Unified Discrete Diffusion for Categorical Data

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Abstract

Discrete diffusion models have attracted significant attention for their application to naturally discrete data, such as language and graphs. While discrete-time discrete diffusion has been established for some time, it was only recently that Campbell et al. [2022] introduced the first framework for continuous-time discrete diffusion. However, their training and backward sampling processes significantly differ from those of the discrete-time version, requiring nontrivial approximations for tractability. In this paper, we first introduce a series of generalizations and simplifications of the variational lower bound (VLB) that facilitate more accurate and easier optimization both discrete- and continuous-time discrete diffusion. We further establish a unification of discrete- and continuous-time discrete diffusion through shared forward process and backward parameterization. Thanks to this unification, the continuous-time diffusion can now utilize the exact and efficient backward process developed for the discrete-time case, avoiding the need for costly and inexact approximations. Similarly, the discrete-time diffusion now also employ the MCMC corrector, which was previously exclusive to the continuous-time case. Extensive experiments and ablations demonstrate the significant improvement.

1 Introduction

Deep generative models have taken the world by storm, capturing complex data distributions and producing realistic data, from human-like text [Brown et al., 2020, Li et al., 2022, OpenAI, 2023] and natural looking images [Dhariwal and Nichol, 2021, Ramesh et al., 2022, Zhang et al., 2023] to novel compounds like molecules and drugs [Kang and Cho, 2018, Li et al., 2021] and video synthesis [Ho et al., 2022]. Denoising diffusion models [Ho et al., 2020b], a powerful class of generative models, are trained through a forward diffusion process that gradually adds noise to the training samples, and a backward process that denoises these diffusion trajectories. New data are then generated by sampling from the noise distribution and employing the trained model for recursive denoising.

Discrete diffusion for categorical data has two modeling paradigms: discrete-time and continuous-time. The former discretizes time such that backward denoising is learned only at pre-specified time points. This limits generation, which can only "jump back" through fixed points. In contrast, continuous-time case allows a path through any point in range, and often yields higher sample quality. Current literature on discrete-time discrete diffusion is relatively established, while only recently Campbell et al. [2022] introduced the first continuous-time discrete diffusion framework. While groundbreaking, their their loss requires multiple network evaluations during training. Moreover, the exact sampling through their learned backward process is extremely tedious for multi-dimensional variables. Due to mathematically complicated and computationally demanding formulations, Campbell et al. [2022] propose nontrivial approximations for tractability with unknown potential errors.

In this paper, we introduce a series of improvements to two critical aspects of discrete diffusion: loss computation and backward sampling, for both discrete- and continuous-time scenarios. What is more,

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分类数据的统一离散扩散

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抽象的

离散扩散模型因其在自然离散数据（例如语言和图形）中的应用而引起了极大的关注。虽然离散时间离散扩散已经建立了一段时间，但直到最近 Campbell 等人才提出。 [2022]引入了第一个连续时间离散扩散框架。然而，它们的训练和向后采样过程与离散时间版本的训练和向后采样过程显着不同，需要非平凡的近似以实现可处理性。在本文中，我们首先介绍了变分下界（VLB）的一系列概括和简化，有助于更准确、更轻松地优化离散时间和连续时间离散扩散。我们通过共享前向过程和后向参数化进一步建立离散时间和连续时间离散扩散的统一。由于这种统一，连续时间扩散现在可以利用为离散时间情况开发的精确且高效的后向过程，从而避免了昂贵且不精确的近似的需要。类似地，离散时间扩散现在也采用 MCMC 校正器，这以前是连续时间情况所独有的。大量的实验和消融证明了显着的改进。

1 简介

深度生成模型席卷了整个世界，捕获复杂的数据分布并生成真实的数据，从类人文本 [Brown 等人，2020，Li 等人，2022，OpenAI，2023] 和自然图像 [Dhariwal 和 Nichol，2021，Ramesh 等人，2022，Zhang 等人，2023] 到分子和药物等新颖化合物 [Kang 和 Cho， 2018，Li et al.，2021]和视频合成[Ho et al.，2022]。去噪扩散模型 [Ho et al., 2020b] 是一类功能强大的生成模型，通过逐渐向训练样本添加噪声的前向扩散过程和对这些扩散轨迹进行去噪的后向过程进行训练。然后通过从噪声分布中采样并使用经过训练的模型进行递归去噪来生成新数据。

分类数据的离散扩散有两种建模范例：离散时间和连续时间。前者将时间离散化，使得仅在预先指定的时间点学习后向去噪。这限制了生成，只能通过固定点"跳回"。相反，连续时间情况允许通过范围内任何点的路径，并且通常会产生更高的样本质量。当前关于离散时间离散扩散的文献相对成熟，而直到最近 Campbell 等人才提出了离散时间离散扩散的研究。 [2022]引入了第一个连续时间离散扩散框架。虽然具有开创性，但它们的损失需要在训练期间进行多次网络评估。此外，对于多维变量来说，通过学习的向后过程进行精确采样是极其乏味的。由于数学公式复杂且计算要求高，Campbell 等人。 [2022]提出了具有未知潜在错误的易处理性的非平凡近似。

在本文中，我们针对离散时间和连续时间场景介绍了对离散扩散的两个关键方面的一系列改进：损失计算和后向采样。更，

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for the first time, we show that the discrete-time and continuous-time discrete diffusion can be unified together such that they share exactly the same forward diffusion and backward sampling process. We elaborate on the details and their benefits as follows:

- **Loss simplifcation and generalization**: We derive significantly simplified yet exact VLB calculations for both discrete&continuous-time discrete diffusion, taking into account the properties of categorical data. Our simplified formulations allow both forward and backward probabilities to accommodate any noise distribution element-wisely, which is particularly attractive for multi-element objects where each element can exhibit a different noise distribution.
- **Efficient backward sampling**: We present a closed-form backward probability $p_\theta(\boldsymbol{x}_s|\boldsymbol{x}_t)$ for all $s < t$ with reconstruction-based model parameterization, paving the way for VLB simplification and accelerated backward sampling, initially derived in the discrete-time case. Furthermore, we demonstrate that this improved backward process can be applied to continuous-time case, eliminating the need for the costly and inexact sampling approximations in Campbell et al. [2022].
- **Unification:** We demonstrate that continuous-time diffusion can share the exact same forward process as discrete-time diffusion for *any* noise distribution. This unified forward process, along with shared backward parameterization, leads to a unification of both forward and backward processes for discrete-& continuous-time cases. This offers mutual benefits: discrete-time diffusion can utilize continuous time's MCMC corrector to enhance performance, while continuous-time diffusion can employ fast and exact backward sampling derived in discrete-time case.

We present both discrete-time (§2) and continuous-time (§3) discrete diffusion in a self-contained manner, accompanied by easy-to-use open-source code at `https://github.com/LingxiaoShawn/USD3`. We explicitly mark our novel contributions in (sub)sections with summarization.

2 Discrete-time Discrete Diffusion

Notation: Let $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x}_0)$ be the random variable of observed data with underlying distribution $p_{\text{data}}(\mathbf{x}_0)$. Let $\mathbf{x}_t \sim q(\mathbf{x}_t)$ be the latent variable at time t of a single-element object, like a pixel of an image or a node/edge of a graph, with maximum time T. Let $\mathbf{x}_{t|s} \sim q(\mathbf{x}_t|\mathbf{x}_s)$ be the conditional random variable. We model the *forward diffusion* process independently for each element of the object, while the *backward denoising* process is modeled jointly for all elements of the object. For simplicity and clarity of presentation, we first assume that the object only has 1 element and extend to multi-element object later. Let $\mathbf{x}_0^{1:D}$ denote the object with D elements, and $\mathbf{x}_t^i$ be the i-th element of latent object at time t. We assume all random variables take categorical values from $\{1, 2, ..., K\}$. Let $\boldsymbol{e}_k \in \{0,1\}^K$ be the one-hot encoding of category k. For a random variable $\mathbf{x}$, we use $\boldsymbol{x}$ denoting its one-hot encoded sample where $\boldsymbol{x} \in \{\boldsymbol{e}_1, ..., \boldsymbol{e}_K\}$. Also, we interchangeably use $q(\mathbf{x}_t|\mathbf{x}_s)$, $q(\mathbf{x}_t = \boldsymbol{x}_t|\mathbf{x}_s = \boldsymbol{x}_s)$, and $q_{t|s}(\boldsymbol{x}_t|\boldsymbol{x}_s)$ when no ambiguity is introduced. Let $\langle \cdot, \cdot \rangle$ denote inner product. All vectors are column-wise vectors.

2.1 Graphical Model View of Diffusion Models

Diffusion models can be represented by latent variable graphical models (see Appx. Fig. 1). We can write the joint probability as $p_\theta(\mathbf{x}_{0:T}) := p_\theta(\mathbf{x}_0, \mathbf{x}_1, ..., \mathbf{x}_T) = p_\theta(\mathbf{x}_T)\prod_{t=1}^T p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ using the Markov condition. Parameters θ are learned by maximizing the loglikehood of the observed variable $\mathbf{x}_0$: $\log p_\theta(\mathbf{x}_0) = \log \int p_\theta(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T}$. However the marginalization is intractable, and instead the following variational lower bound (VLB) is used.

$$\log p_\theta(\mathbf{x}_0) = \log \int q(\mathbf{x}_{1:T}|\mathbf{x}_0)\frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} d\mathbf{x}_{1:T} \geq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\big[\log p_\theta(\mathbf{x}_{0:T}) - \log q(\mathbf{x}_{1:T}|\mathbf{x}_0)\big]. \quad (1)$$

The above inequality holds for any conditional probability $q(\mathbf{x}_{1:T}|\mathbf{x}_0)$ and finding the best $q(\mathbf{x}_{1:T}|\mathbf{x}_0)$ to tighten the bound is the inference problem in graphical models (i.e. E step in EM algorithm). Exact inference is intractable, thus $q(\mathbf{x}_{1:T}|\mathbf{x}_0)$ in diffusion models is fixed or chosen specifically to simplify the learning objective. To simplify Eq. (1), it is important to assume $q(\mathbf{x}_{1:T}|\mathbf{x}_0)$ is *decomposable*. The typical assumption is $q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})$ [Ho et al., 2020a], which we also adopt. Others that have been explored include $q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0)$ [Song et al., 2021a].

Assuming $q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})$, Eq. (1) can be simplified as the following

$$\underbrace{\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)}\big[\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)\big]}_{-\mathcal{L}_1(\theta)} - \underbrace{D_{\mathrm{KL}}\big(q(\mathbf{x}_T|\mathbf{x}_0)||p_\theta(\mathbf{x}_T)\big)}_{\mathcal{L}_{\text{prior}}} - \sum_{t=2}^{T}\underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}\big[D_{\mathrm{KL}}\big(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)||p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)\big)\big]}_{\mathcal{L}_t(\theta)} \quad (2)$$

我们首次证明离散时间和连续时间离散扩散可以统一在一起，以便它们共享完全相同的前向扩散和后向采样过程。我们详细阐述了细节及其好处如下：

- 损失简化和泛化：考虑到分类数据的属性，我们针对离散和连续时间离散扩散得出了显着简化但精确的 VLB 计算。我们的简化公式允许前向和后向概率以元素方式适应任何噪声分布，这对于每个元素可以表现出不同噪声分布的多元素对象特别有吸引力。

- 高效的后向采样：我们通过基于重构的模型参数化为所有 $s<t$ 提供了一个封闭形式的后向概率 $p_\theta(\boldsymbol{x}_s|\boldsymbol{x}_t)$，为 VLB 简化和加速后向采样铺平了道路，最初是在离散时间情况下推导的。此外，我们证明了这种改进的后向过程可以应用于连续时间情况，从而消除了 Ca mpbell 等人对昂贵且不精确的采样近似的需要。 [2022]。
- 统一：我们证明连续时间扩散可以与 *any* 噪声分布的离散时间扩散共享完全相同的前向过程。这种统一的前向过程以及共享的后向参数化导致了离散和连续时间情况下前向和后向过程的统一。这提供了互惠互利：离散时间扩散可以利用连续时间的 MCMC 校正器来增强性能，而连续时间扩散可以采用在离散时间情况下导出的快速且精确的后向采样。

我们以独立的方式展示了离散时间（§2）和连续时间（§3）离散扩散，并附有易于使用的开源代码（https://github.com/LingxiaoShawn/USD3）。我们在（子）部分中明确标记我们的新颖贡献并进行总结。

2 离散时间离散扩散

表示法：令 $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x}_0)$ 为具有底层分布 $p_{\text{data}}(\mathbf{x}_0)$ 的观测数据的随机变量。令 $\mathbf{x}_t \sim q(\mathbf{x}_t)$ 为单元素对象（例如图像的像素或图形的节点/边）在时间 t 的潜在变量，最大时间为 T。令 $\mathbf{x}_{t|s} \sim q(\mathbf{x}_t|\mathbf{x}_s)$ 为条件随机变量。我们为对象的每个元素独立地建模 *forward diffusion* 过程，而为对象的所有元素联合建模 *backward denoising* 过程。为了表达的简单和清晰，我们首先假设该对象只有 1 个元素，然后扩展到多元素对象。令 $\mathbf{x}_0^{1:D}$ 表示具有 D 个元素的对象，而 $\mathbf{x}_t^i$ 是时间 t 时潜在对象的第 i 个元素。我们假设所有随机变量均采用 $\{1,2,...,K\}$ 中的分类值。令 $\boldsymbol{e}_k \in \{0,1\}^K$ 为类别 k 的 one-hot 编码。对于随机变量 $\mathbf{x}$，我们使用 $\boldsymbol{x}$ 表示其独热编码样本，其中 $\boldsymbol{x} \in \{\boldsymbol{e}_1,...,\boldsymbol{e}_K\}$。此外，当不产生歧义时，我们可以互换使用 $q(\mathbf{x}_t|\mathbf{x}_s)$、$q(\mathbf{x}_t=\boldsymbol{x}_t|\mathbf{x}_s=\boldsymbol{x}_s)$ 和 $q_{t|s}(\boldsymbol{x}_t|\boldsymbol{x}_s)$。让 $\langle\cdot,\cdot\rangle$ 表示内积。所有向量都是列向量。

2.1 扩散模型的图形模型视图

扩散模型可以用潜变量图形模型表示（参见附录图1）。我们可以使用马尔可夫条件将联合概率写为 $p_\theta(\mathbf{x}_{0:T}) := p_\theta(\mathbf{x}_0,\mathbf{x}_1,...,\mathbf{x}_T) = p_\theta(\mathbf{x}_T)\prod_{t=1}^T p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$。参数 θ 是通过最大化观察变量 $\mathbf{x}_0$ 的对数似然来学习的： $\log p_\theta(\mathbf{x}_0) = \log\int p_\theta(\mathbf{x}_{0:T})d\mathbf{x}_{1:T}$。然而，边缘化是棘手的，而是使用以下变分下界（VLB）。

$$\log p_\theta(\mathbf{x}_0) = \log\int q(\mathbf{x}_{1:T}|\mathbf{x}_0)\frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}d\mathbf{x}_{1:T} \geq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\big[\log p_\theta(\mathbf{x}_{0:T}) - \log q(\mathbf{x}_{1:T}|\mathbf{x}_0)\big]. \quad (1)$$

上述不等式对于任何条件概率 $q(\mathbf{x}_{1:T}|\mathbf{x}_0)$ 都成立，找到收紧界限的最佳 $q(\mathbf{x}_{1:T}|\mathbf{x}_0)$ 是图模型中的推理问题（即 EM 算法中的 E 步骤）。精确的推理是棘手的，因此扩散模型中的 $q(\mathbf{x}_{1:T}|\mathbf{x}_0)$ 是固定的或专门选择的以简化学习目标。简化方程。 (1)，假设 $q(\mathbf{x}_{1:T}|\mathbf{x}_0)$ 是 *decomposable* 很重要。典型的假设是 $q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})$ [Ho et al., 2020a]，我们也采用该假设。其他已探索过的包括 $q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1},\mathbf{x}_0)$ [Song et al., 2021a]。

假设$q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})$，等式。 (1) 可以简化为如下

$$\underbrace{\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)}\big[\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)\big]}_{-\mathcal{L}_1(\theta)} - \underbrace{D_{\text{KL}}\big(q(\mathbf{x}_T|\mathbf{x}_0)||p_\theta(\mathbf{x}_T)\big)}_{\mathcal{L}_{\text{prior}}} - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}\big[D_{\text{KL}}\big(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)||p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)\big)\big]}_{\mathcal{L}_t(\theta)} \quad (2)$$

Where $\mathcal{L}_{\text{prior}} \approx 0$, since $p_\theta(\mathbf{x}_T) \approx q(\mathbf{x}_T|\mathbf{x}_0)$ is designed as a fixed noise distribution that is easy to sample from. (See Appx. §A.1 for derivation.) To compute Eq. (2), we need to formalize distributions (i) $q(\mathbf{x}_t|\mathbf{x}_0)$ and (ii) $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$, as well as (iii) the parameterization of $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$. We specify these respectively in §2.2.1, §2.2.2, and §2.2.3.

2.2 Components of Discrete-time Discrete Diffusion

After reviewing the forward process for discrete diffusion (§2.2.1), we contribute a series of analytical simplifications for various components (§2.2.2, §2.2.3) of the VLB, providing exact closed-form formulation in §2.2.4, as well as an approximated loss for easier optimization in §2.2.5. We present fast backward sampling in Appx.§A.8, and give the extension to multi-element case in Appx. §A.9.

2.2.1 The Forward Diffusion Process

We assume each discrete random variable $\mathbf{x}_t$ has a categorical distribution, i.e. $\mathbf{x}_t \sim \text{Cat}(\mathbf{x}_t; \boldsymbol{p})$ with $\boldsymbol{p} \in [0,1]^K$ and $\mathbf{1}^\top \boldsymbol{p} = 1$. One can verify that $p(\mathbf{x}_t = \boldsymbol{x}_t) = \boldsymbol{x}_t^\top \boldsymbol{p}$, or simply $p(\mathbf{x}_t) = \boldsymbol{x}_t^\top \boldsymbol{p}$. As shown in [Hoogeboom et al., 2021, Austin et al., 2021], the forward process with discrete variables $q(\mathbf{x}_t|\mathbf{x}_{t-1})$ at t step can be represented as a transition matrix $Q_t \in [0,1]^{K\times K}$ such that $[Q_t]_{ij} = q(\mathbf{x}_t = \boldsymbol{e}_j|\mathbf{x}_{t-1} = \boldsymbol{e}_i)$. Then, we can write the distribution explicitly as

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \text{Cat}(\mathbf{x}_t; Q_t^\top \boldsymbol{x}_{t-1}) \ . \tag{3}$$

Given transition matrices $Q_1, ..., Q_T$, the t-step marginal distribution conditioning on $s, \forall t > s$ is

$$q(\mathbf{x}_t|\mathbf{x}_s) = \text{Cat}(\mathbf{x}_t; \overline{Q}_{t|s}^\top \boldsymbol{x}_s), \text{ with } \overline{Q}_{t|s} = Q_{s+1}...Q_t \ . \tag{4}$$

The s-step posterior distribution conditioning on $\mathbf{x}_0$ and t-step can be derived as

$$q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) = \frac{q(\mathbf{x}_t|\mathbf{x}_s)q(\mathbf{x}_s|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} = \text{Cat}(\mathbf{x}_s; \frac{\overline{Q}_{t|s}\boldsymbol{x}_t \odot \overline{Q}_s^\top \boldsymbol{x}_0}{\boldsymbol{x}_t^\top \overline{Q}_t^\top \boldsymbol{x}_0}), \ \forall t > s \ . \tag{5}$$

The above formulations are valid (derivation in Appx. §A.3) for *any* transition matrices $Q_1, ..., Q_T$. However, to achieve uninformative noise $\mathbf{x}_T$, $\{Q_i\}_{i=1}^T$ should be chosen such that every row of $\overline{Q}_t = \overline{Q}_{t|0}$ converge to the same known stationary distribution when t becomes large enough (at T). Let the known stationary distribution be $\mathbf{m}_0 \sim \text{Cat}(\mathbf{m}_0; \boldsymbol{m})$. Then, the constraint can be stated as

$$\lim_{t\to T} \overline{Q}_t = \mathbf{1}\boldsymbol{m}^\top \ . \tag{6}$$

In addition, this paper focuses on *nominal data* (see Appx. §A.2) where categories are unordered and only equality comparison is defined. Hence, no ordering prior *except checking equality* should be used to define Q_t[2]. To achieve the desired convergence on nominal data while keeping the flexibility of choosing any categorical stationary distribution $\mathbf{m}_0 \sim \text{Cat}(\mathbf{m}_0; \boldsymbol{m})$, we define Q_t as

$$Q_t = \alpha_t I + (1-\alpha_t)\mathbf{1}\boldsymbol{m}^\top \ , \tag{7}$$

where $\alpha_t \in [0,1]$. This results in the accumulated transition matrix $\overline{Q}_{t|s}$ being equal to

$$\overline{Q}_{t|s} = \overline{\alpha}_{t|s} I + (1-\overline{\alpha}_{t|s})\mathbf{1}\boldsymbol{m}^\top, \ \forall t > s \ , \tag{8}$$

where $\overline{\alpha}_{t|s} = \prod_{i=s+1}^t \alpha_i$. Note that $\overline{\alpha}_t = \overline{\alpha}_{t|0} = \overline{\alpha}_{t|s}\overline{\alpha}_s$. We can achieve the uninformative noise with satisfying the constraint Eq. (6) by picking α_t such that $\lim_{t\to T} \overline{\alpha}_t = 0$.

2.2.2 Analytical Form of $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$

The formulation in Eq. (7) can be used to simplify $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$. We provide a general formulation of $q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$ for any s, t with $0 < s < t \leq T$, which will be useful for unifying with continuous-time diffusion. One can recover $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ by setting s as $t-1$.

Proposition 2.1 (Zheng et al. [2023]). *For both discrete- and continuous-time discrete diffusion, we can write the conditional distribution as*

$$q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) = \begin{cases} \mathit{Cat}\Big(\mathbf{x}_s; \ (1-\lambda_{t|s}) \cdot \boldsymbol{x}_t + \lambda_{t|s} \cdot \boldsymbol{m}\Big) & \textit{when } \boldsymbol{x}_t = \boldsymbol{x}_0 \\ \mathit{Cat}\Big(\mathbf{x}_s; \ (1-\mu_{t|s}) \cdot \boldsymbol{x}_0 + \mu_{t|s}\overline{\alpha}_{t|s} \cdot \boldsymbol{x}_t + \mu_{t|s}(1-\overline{\alpha}_{t|s}) \cdot \boldsymbol{m}\Big) & \textit{when } \boldsymbol{x}_t \neq \boldsymbol{x}_0 \end{cases} \tag{9}$$

where $\lambda_{t|s} := \frac{(1-\overline{\alpha}_s)(1-\overline{\alpha}_{t|s})\langle \boldsymbol{m}, \boldsymbol{x}_t\rangle}{\overline{\alpha}_t + (1-\overline{\alpha}_t)\langle \boldsymbol{m}, \boldsymbol{x}_t\rangle}$, $\mu_{t|s} := \frac{1-\overline{\alpha}_s}{1-\overline{\alpha}_t}$.

We remark that the above formulation is originally presented in Zheng et al. [2023] with $s = t-1$. Appx. §A.3 gives the detailed derivation of the probability's formulation.

[2] Mathematically, this means $\forall k, j$, $q(\mathbf{x}_t|\mathbf{x}_{t-1} = \boldsymbol{e}_j, \mathbf{x}_t \neq \{\boldsymbol{e}_j, \boldsymbol{e}_k\}) = q(\mathbf{x}_t|\mathbf{x}_{t-1} = \boldsymbol{e}_k, \mathbf{x}_t \neq \{\boldsymbol{e}_j, \boldsymbol{e}_k\})$.

其中 $\mathcal{L}_{\text{prior}} \approx 0$，因为 $p_\theta(\mathbf{x}_T) \approx q(\mathbf{x}_T|\mathbf{x}_0)$ 被设计为易于采样的固定噪声分布。（推导参见附录§A.1。）(2)，我们需要形式化分布 (i) $q(\mathbf{x}_t|\mathbf{x}_0)$ 和 (ii) $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$，以及 (iii) $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ 的参数化。我们分别在§2.2.1、§2.2.2 和§2.2.3 中指定这些内容。

2.2 离散时间离散扩散的组成部分

在回顾了离散扩散的正向过程（第 2.2.1 节）之后，我们对 VLB 的各个组件（第 2.2.2 节、第 2.2.3 节）进行了一系列分析简化，在第 2.2.4 节中提供了精确的封闭形式公式，并在第 2.2.5 节中提供了近似损失以便于优化。我们在 Appx.§A.8 中提出了快速向后采样，并在 Appx 中给出了多元素情况的扩展。§A.9。

2.2.1 前向扩散过程

我们假设每个离散随机变量 $\mathbf{x}_t$ 具有分类分布，即 $\mathbf{x}_t \sim$ Cat($\mathbf{x}_t$; $\boldsymbol{p}$) 与 $\boldsymbol{p} \in [0,1]^K$ 和 $\mathbf{1}^\top \boldsymbol{p} = 1$。人们可以验证 $p(\mathbf{x}_t = \boldsymbol{x}_t) = \boldsymbol{x}_t^\top \boldsymbol{p}$，或者简单地验证 $p(\mathbf{x}_t) = \boldsymbol{x}_t^\top \boldsymbol{p}$。如 [Hoogeboom et al., 2021, Austin et al., 2021] 所示，在 t 步骤处使用离散变量 $q(\mathbf{x}_t|\mathbf{x}_{t-1})$ 的前向过程可以表示为转换矩阵 $Q_t \in [0,1]^{K\times K}$，使得 $[Q_t]_{ij} = q(\mathbf{x}_t = \boldsymbol{e}_j|\mathbf{x}_{t-1} = \boldsymbol{e}_i)$。然后，我们可以将分布显式写为

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \text{Cat}(\mathbf{x}_t; Q_t^\top \boldsymbol{x}_{t-1}) \ . \tag{3}$$

给定转移矩阵 $Q_1, ..., Q_T$，$s, \forall t > s$ 上的 t 步边际分布条件为

$$q(\mathbf{x}_t|\mathbf{x}_s) = \text{Cat}(\mathbf{x}_t; \overline{Q}_{t|s}^\top \boldsymbol{x}_s), \text{ with } \overline{Q}_{t|s} = Q_{s+1} ... Q_t \ . \tag{4}$$

$\mathbf{x}_0$ 和 t 步上的 s 步后验分布条件可以导出为

$$q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) = \frac{q(\mathbf{x}_t|\mathbf{x}_s)q(\mathbf{x}_s|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} = \text{Cat}(\mathbf{x}_s; \frac{\overline{Q}_{t|s}\boldsymbol{x}_t \odot \overline{Q}_s^\top \boldsymbol{x}_0}{\boldsymbol{x}_t^\top \overline{Q}_t^\top \boldsymbol{x}_0}), \ \forall t > s \ . \tag{5}$$

上述公式对于 *any* 转移矩阵 $Q_1, ..., Q_T$ 是有效的（附录 §A.3 中的推导）。然而，为了实现无信息噪声 $\mathbf{x}_T$，应选择 $\{Q_i\}_{i=1}^T$，以便当 t 变得足够大（在 T 处）时，$Q_t = \overline{Q}_{t|0}$ 的每一行收敛到相同的已知平稳分布。令已知的平稳分布为 $\mathbf{m}_0 \sim$ Cat($\mathbf{m}_0$; $\boldsymbol{m}$)。那么，约束可以表示为

$$\lim_{t\to T} \overline{Q}_t = \mathbf{1}\boldsymbol{m}^\top \ . \tag{6}$$

此外，本文重点介绍*nominal data* (参见Appx。§A.2)，其中类别是无序的，并且仅定义了相等比较。因此，不应使用先验排序 *except checking equality* 来定义 Q_t[2]。为了在名义数据上实现所需的收敛，同时保持选择任何分类平稳分布 $\mathbf{m}_0 \sim$ Cat($\mathbf{m}_0$; $\boldsymbol{m}$) 的灵活性，我们将 Q_t 定义为

$$Q_t = \alpha_t I + (1-\alpha_t)\mathbf{1}\boldsymbol{m}^\top \ , \tag{7}$$

其中 $\alpha_t \in [0,1]$。这导致累积的转移矩阵 $Q_{t|s}$ 等于

$$\overline{Q}_{t|s} = \overline{\alpha}_{t|s} I + (1-\overline{\alpha}_{t|s})\mathbf{1}\boldsymbol{m}^\top, \ \forall t > s \ , \tag{8}$$

其中$\overline{\alpha_{t|s}} = \prod_{i=s+1}^t \alpha_i$。请注意$\alpha_t \equiv \overline{\alpha}_{t|0} = \overline{\alpha}_{t|s}\overline{\alpha}_s$。我们可以通过满足约束方程来获得无信息的噪声。(6) 通过选择 α_t 使得 $\lim_{t\to T} \overline{\alpha}_t =$ 为 0。

2.2.2 $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$的解析形式

方程中的公式。(7) 可以用来化简$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$。我们为任何具有 $0 < s < t \leq T$ 的 s, t 提供了 $q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$ 的通用公式，这对于与连续时间扩散统一非常有用。可以通过将 s 设置为 $t-1$ 来恢复 $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$。

命题 2.1（Zheng 等人[2023]）。*For both discrete- and continuous-time discrete diffusion, we can write the conditional distribution as*

$$q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) = \begin{cases} Cat\Big(\mathbf{x}_s;\ (1-\lambda_{t|s})\cdot \boldsymbol{x}_t + \lambda_{t|s}\cdot \boldsymbol{m}\Big) & \text{when } \boldsymbol{x}_t = \boldsymbol{x}_0 \\ Cat\Big(\mathbf{x}_s;\ (1-\mu_{t|s})\cdot \boldsymbol{x}_0 + \mu_{t|s}\overline{\alpha}_{t|s}\cdot \boldsymbol{x}_t + \mu_{t|s}(1-\overline{\alpha}_{t|s})\cdot \boldsymbol{m}\Big) & \text{when } \boldsymbol{x}_t \neq \boldsymbol{x}_0 \end{cases} \tag{9}$$

where $\lambda_{t|s} := \frac{(1-\overline{\alpha}_s)(1-\overline{\alpha}_{t|s})\langle \boldsymbol{m}, \boldsymbol{x}_t\rangle}{\overline{\alpha}_t + (1-\overline{\alpha}_t)\langle \boldsymbol{m}, \boldsymbol{x}_t\rangle}$, $\mu_{t|s} := \frac{1-\overline{\alpha}_s}{1-\overline{\alpha}_t}$.

我们注意到，上述公式最初是在 Cheng 等人中提出的。[2023] 与 $s = t-1$. 附录。§A.3 给出了概率公式的详细推导。

[2] Mathematically, this means $\forall k, j$, $q(\mathbf{x}_t|\mathbf{x}_{t-1} = \boldsymbol{e}_j, \mathbf{x}_t \neq \{\boldsymbol{e}_j, \boldsymbol{e}_k\}) = q(\mathbf{x}_t|\mathbf{x}_{t-1} = \boldsymbol{e}_k, \mathbf{x}_t \neq \{\boldsymbol{e}_j, \boldsymbol{e}_k\})$.

2.2.3 Contrib. 1: Analytical Form of $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$

The literature has explored three different parameterizations of $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$: (1) parameterizing $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ directly; (2) parameterizing $p_\theta(\mathbf{x}_0|\mathbf{x}_t)$ with f_t^θ such that $p_\theta(\mathbf{x}_0|\mathbf{x}_t) = \text{Cat}(\mathbf{x}_0; f_t^\theta(\mathbf{x}_t))$ and letting $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = q(\mathbf{x}_{t-1}|\mathbf{x}_t, f_t^\theta(\mathbf{x}_t))$; and (3) parameterizing $p_\theta(\mathbf{x}_0|\mathbf{x}_t)$ with f_t^θ and then marginalizing $q(\mathbf{x}_{t-1}, \mathbf{x}_0|\mathbf{x}_t)$ such that $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \sum_{\mathbf{x}_0} q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)p_\theta(\mathbf{x}_0|\mathbf{x}_t)$.

Method (1) does not reuse any distribution in forward process, and hence is less effective. Method (2) has been widely used for continuous diffusion models as in Ho et al. [2020a] and Song et al. [2021a], and some discrete diffusion models like in Hoogeboom et al. [2021] and Zheng et al. [2023]. It avoids marginalization and works effectively for continuous diffusion. However for discrete diffusion, as shown in Eq. (9), sample $\mathbf{x}_0$ determines which categorical distribution should be used, which cannot be determined without the true $\mathbf{x}_0$. Some heuristics have been proposed in Zheng et al. [2023], however those can have a large gap to the true $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$, leading to an inaccurate sampling process.

Method (3) has been used in Austin et al. [2021] by directly marginalizing out $\mathbf{x}_0$, which introduces additional computational cost in both loss function computation and sampling process as the formulation of $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ has not been simplified to a closed-form distribution. In this paper, we show that method (3) parameterization can be simplified to a clean formulation of categorical distribution.

Proposition 2.2. *The parameterization of $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ can be simplified for any $0 < s < t \leq T$ as*

$$p_\theta(\mathbf{x}_s|\mathbf{x}_t) = \mathit{Cat}\Big(\mathbf{x}_s; (1-\mu_{t|s}) \cdot f_t^\theta(\boldsymbol{x}_t) + (\mu_{t|s}\overline{\alpha}_{t|s} + \gamma_{t|s}^\theta) \cdot \boldsymbol{x}_t + (\mu_{t|s}(1-\overline{\alpha}_{t|s}) - \gamma_{t|s}^\theta) \cdot \boldsymbol{m}\Big) \quad (10)$$

where $\gamma_{t|s}^\theta$ is affected by $f_t^\theta(\mathbf{x}_t)$, $\gamma_{t|s}^\theta := (\mu_{t|s} - \lambda_{t|s} - \mu_{t|s}\overline{\alpha}_{t|s})\langle f_t^\theta(\boldsymbol{x}_t), \boldsymbol{x}_t\rangle$.

The detailed proof is in Appx. §A.4. Notice that $f_t^\theta(\boldsymbol{x}_t)$ is a parameterized neural network with $\mathbf{1}^T f_t^\theta(\boldsymbol{x}_t) = 1$. As we show next, the above formulation simplifies the negative VLB loss computation greatly (§2.2.4), further motivates an approximated loss that is much easier to optimize (§2.2.5), and accelerates the sampling process through reparameterization (§A.8).

2.2.4 Loss Function

With $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ in Eq. (9) and $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ in Eq. (10), the $\mathcal{L}_t(\theta)$ in Eq. (2) can be written as

$$\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}\Big[\delta_{\mathbf{x}_t,\mathbf{x}_0} D_{\text{KL}}\big(q(\mathbf{x}_{t-1}|\mathbf{x}_t = \mathbf{x}_0)\|p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)\big) + (1-\delta_{\mathbf{x}_t,\mathbf{x}_0}) D_{\text{KL}}\big(q(\mathbf{x}_{t-1}|\mathbf{x}_t \neq \mathbf{x}_0)\|p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)\big)\Big] \ , \quad (11)$$

where $\delta_{\mathbf{x}_t,\mathbf{x}_0}$ denotes the Kronecker delta of $\mathbf{x}_t$ and $\mathbf{x}_0$. $q(\mathbf{x}_{t-1}|\mathbf{x}_t = \mathbf{x}_0)$ and $q(\mathbf{x}_{t-1}|\mathbf{x}_t \neq \mathbf{x}_0)$ represent the first and second categorical distribution in Eq. (9), respectively.

Apart from negative VLB, another commonly employed auxiliary loss is the cross-entropy (CE) loss between $q(\mathbf{x}_t|\mathbf{x}_0)$ and $p_\theta(\mathbf{x}_0|\mathbf{x}_t)$, which measures the reconstruction quality.

$$\mathcal{L}_t^{CE}(\theta) := \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}[-\log p_\theta(\mathbf{x}_0|\mathbf{x}_t)] \quad (12)$$

Eq. (12) and Eq. (11) share the same global minima with $p_\theta(\mathbf{x}_0|\mathbf{x}_t)$ being the true posterior $q(\mathbf{x}_0|\mathbf{x}_t)$. However they have different optimization landscape; and thus under limited data and network capacity, which loss is be easier to minimize is unknown [Tewari and Bartlett, 2007, Demirkaya et al., 2020].

2.2.5 Contrib. 2: Further Loss Simplification for Easier Optimization

While Eq. (11) is the *exact* negative VLB loss, in practice we find it harder to minimize than $\mathcal{L}_t^{CE}$. In this section, we first derive a much simpler, approximated loss by observing a relation between $q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$ and $p_\theta(\mathbf{x}_t|\mathbf{x}_0)$. We then show that coefficient simplification and $\mathcal{L}_t^{CE}$ are *both* valuable for optimizing a general negative VLB where only partial time steps are observed. We combine all designs to derive the final approximated loss, denoted as $\tilde{\mathcal{L}}_t$.

Proposition 2.3. *For any $0 < s < t \leq T$, with $\mathbf{x}_0$ known, $\triangle \boldsymbol{p}_\theta(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$ is defined as*

$$p_\theta(\mathbf{x}_s|\mathbf{x}_t) - q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) = (1-\mu_{t|s})[f_t^\theta(\boldsymbol{x}_t) - \boldsymbol{x}_0 + \phi_{t|s}\langle f_t^\theta(\boldsymbol{x}_t) - \boldsymbol{x}_0, \boldsymbol{x}_t\rangle(\boldsymbol{x}_t - \boldsymbol{m})] \ , \quad (13)$$

where $\phi_{t|s} := \frac{(1-\bar{\alpha}_s)\bar{\alpha}_{t|s}}{\bar{\alpha}_t + (1-\bar{\alpha}_t)\langle \mathbf{x}_t, \boldsymbol{m}\rangle}$.

Proposition 2.3 shows that the distribution difference between $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ and $q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$ has a closed-form formulation. (See Appx. §A.5 for the proof.) With it, we can apply the Taylor expansion (up to second order) to approximate the KL divergence directly (See Appx. §A.6.)

$$D_{\text{KL}}\big(q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)\|p_\theta(\mathbf{x}_s|\mathbf{x}_t)\big) \approx \sum_{\mathbf{x}_s} \frac{|\triangle \boldsymbol{p}_\theta(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)|^2}{q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)} \quad (14)$$

2.2.3 贡献。 1：$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$的解析形式

文献探索了$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$的三种不同参数化：（1）直接参数化$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$；(2) 用 f_t^θ 参数化 $p_\theta(\mathbf{x}_0|\mathbf{x}_t)$，使得 $p_\theta(\mathbf{x}_0|\mathbf{x}_t) = \text{Cat}(\mathbf{x}_0; f_t^\theta(\mathbf{x}_t))$ 并令 $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = q(\mathbf{x}_{t-1}|\mathbf{x}_t, f_t^\theta(\mathbf{x}_t))$；(3) 用 f_t^θ 参数化 $p_\theta(\mathbf{x}_0|\mathbf{x}_t)$，然后边缘化 $q(\mathbf{x}_{t-1}, \mathbf{x}_0|\mathbf{x}_t)$，使得 $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \sum_{\mathbf{x}_0} q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)p_\theta(\mathbf{x}_0|\mathbf{x}_t)$。

方法（1）在前向过程中不重用任何分布，因此效率较低。方法（2）已广泛用于连续扩散模型，如 Ho 等人。 [2020a] 和宋等人。 [2021a]，以及一些离散扩散模型，如 Hoogeboom 等人。 [2021]和郑等人。 [2023]。它避免了边缘化并有效地实现了持续扩散。然而对于离散扩散，如方程所示。 (9)，样本$\mathbf{x}_0$决定应该使用哪种分类分布，如果没有真实的$\mathbf{x}_0$就无法确定。郑等人提出了一些启发式方法。 [2023]，然而，这些可能与真实的 $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$ 有很大差距，导致采样过程不准确。

Austin 等人使用了方法（3）。 [2021]通过直接边缘化$\mathbf{x}_0$，这在损失函数计算和采样过程中引入了额外的计算成本，因为$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$的公式尚未简化为封闭形式分布。在本文中，我们证明方法（3）参数化可以简化为分类分布的清晰公式。

命题2.2。 *The parameterization of* $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ *can be simplified for any* $0 < s < t \leq T$ *as*

$$p_\theta(\mathbf{x}_s|\mathbf{x}_t) = Cat\Big(\mathbf{x}_s; (1-\mu_{t|s}) \cdot f_t^\theta(\boldsymbol{x}_t) + (\mu_{t|s}\overline{\alpha}_{t|s} + \gamma_{t|s}^\theta) \cdot \boldsymbol{x}_t + (\mu_{t|s}(1-\overline{\alpha}_{t|s}) - \gamma_{t|s}^\theta) \cdot \boldsymbol{m}\Big) \tag{10}$$

where $\gamma_{t|s}^\theta$ *is affected by* $f_t^\theta(\mathbf{x}_t)$, $\gamma_{t|s}^\theta := (\mu_{t|s} - \lambda_{t|s} - \mu_{t|s}\overline{\alpha}_{t|s})\langle f_t^\theta(\boldsymbol{x}_t), \boldsymbol{x}_t\rangle$.

详细证明见Appx. §A.4。请注意，$f_t^\theta(\boldsymbol{x}_t)$ 是一个参数化神经网络，$\mathbf{1}^T f_t^\theta(\boldsymbol{x}_t) =$ 为 1。正如我们接下来展示的，上述公式极大地简化了负 VLB 损失计算（第 2.2.4 节），进一步激发了更容易优化的近似损失（第 2.2.5 节），并通过重新参数化加速了采样过程（第 A.8 节）。

2.2.4 损失函数

方程中的 $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ (9) 和等式中的$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ (10) 式中的$\mathcal{L}_t(\theta)$ (2) 可以写成

$$\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}\Big[\delta_{\mathbf{x}_t,\mathbf{x}_0} D_{\mathrm{KL}}\big(q(\mathbf{x}_{t-1}|\mathbf{x}_t = \mathbf{x}_0)\|p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)\big) + (1-\delta_{\mathbf{x}_t,\mathbf{x}_0}) D_{\mathrm{KL}}\big(q(\mathbf{x}_{t-1}|\mathbf{x}_t \neq \mathbf{x}_0)\|p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)\big)\Big], \tag{11}$$

其中 $\delta_{\mathbf{x}_t,\mathbf{x}_0}$ 表示 $\mathbf{x}_t$ 和 $\mathbf{x}_0$ 的克罗内克增量。 $q(\mathbf{x}_{t-1}|\mathbf{x}_t = \mathbf{x}_0)$和$q(\mathbf{x}_{t-1}|\mathbf{x}_t \neq \mathbf{x}_0)$代表等式中的第一和第二分类分布。 (9) 分别。

除了负VLB之外，另一个常用的辅助损失是$q(\mathbf{x}_t|\mathbf{x}_0)$和$p_\theta(\mathbf{x}_0|\mathbf{x}_t)$之间的交叉熵（CE）损失，它衡量重建质量。

$$\mathcal{L}_t^{CE}(\theta) := \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}[-\log p_\theta(\mathbf{x}_0|\mathbf{x}_t)] \tag{12}$$

等式。 (12) 和等式。 (11) 与作为真实后验值 $q(\mathbf{x}_0|\mathbf{x}_t)$ 的 $p_\theta(\mathbf{x}_0|\mathbf{x}_t)$ 共享相同的全局最小值。然而，它们有不同的优化前景；因此，在有限的数据和网络容量下，哪种损失更容易最小化是未知的[Tewari and Bartlett, 2007, Demirkaya et al., 2020]。

2.2.5 贡献。 2：进一步简化损失以更轻松地优化

而方程。 (11) 是 *exact* 负 VLB 损失，实际上我们发现它比 $\mathcal{L}_t^{CE}$ 更难最小化。在本节中，我们首先通过观察 $q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$ 和 $p_\theta(\mathbf{x}_t|\mathbf{x}_0)$ 之间的关系得出一个更简单的近似损失。然后我们证明系数简化和 $\mathcal{L}_t^{CE}$ 对于优化仅观察到部分时间步长的一般负 VLB 很有价值。我们结合所有设计来得出最终的近似损失，表示为$\tilde{\mathcal{L}}_t$。

命题2.3。 *For any* $0 < s < t \leq T$*, with* $\mathbf{x}_0$ *known,* $\triangle \boldsymbol{p}_\theta(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$ *is defined as*

$$p_\theta(\mathbf{x}_s|\mathbf{x}_t) - q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) = (1-\mu_{t|s})[f_t^\theta(\boldsymbol{x}_t) - \boldsymbol{x}_0 + \phi_{t|s}\langle f_t^\theta(\boldsymbol{x}_t) - \boldsymbol{x}_0, \boldsymbol{x}_t\rangle(\boldsymbol{x}_t - \boldsymbol{m})], \tag{13}$$

where $\phi_{t|s} := \frac{(1-\bar{\alpha}_s)\bar{\alpha}_{t|s}}{\bar{\alpha}_t + (1-\bar{\alpha}_t)\langle \mathbf{x}_t, \boldsymbol{m}\rangle}$.

命题 2.3 表明 $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ 和 $q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$ 之间的分布差异具有闭合形式的公式。（参见附录 § A.5 的证明。)有了它，我们可以应用泰勒展开式（高达二阶）来直接近似 KL 散度（参见附录 §A.6。）

$$D_{\mathrm{KL}}\big(q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)\|p_\theta(\mathbf{x}_s|\mathbf{x}_t)\big) \approx \sum_{\mathbf{x}_s} \frac{|\triangle \boldsymbol{p}_\theta(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)|^2}{q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)} \tag{14}$$

The above formulation along with prop.2.3 are valid for *any* $0 < s < t \leq T$. We next show that minimizing divergence between q and p_θ at any s and t is also valid, as it is inside a general negative VLB with partial time steps. The initial version of the VLB is derived under the assumption that observations are made at every time step. Its backward denoising process is designed to advance by a single time step during each generation step for best generation quality. Let us consider a more general case where only partial time steps are observed in the forward process, then, minimizing its negative VLB can help improve generation quality with fewer steps. Assuming only $\mathbf{x}_s$ and $\mathbf{x}_t$ are observed, where $0 < s < t \leq T$, a derivation analogous to that of Eq. (2) can show that

$$\log p_\theta(\mathbf{x}_0) \geq -D_{\mathrm{KL}}[q(\mathbf{x}_t|\mathbf{x}_0)\|p_\theta(\mathbf{x}_t)] + \mathbb{E}_{q(\mathbf{x}_s|\mathbf{x}_0)}[\log p_\theta(\mathbf{x}_0|\mathbf{x}_s)] - D_{\mathrm{KL}}[q(\mathbf{x}_s|\mathbf{x}_t,\mathbf{x}_0)\|p_\theta(\mathbf{x}_s|\mathbf{x}_t)], \tag{15}$$

where the first divergence term between the prior and posterior quantifies the quality of the backward denoising process from T to t. The second term represents the CE loss, $\mathcal{L}_s^{\mathrm{CE}}$, which influences the generation quality from time s to time 0. The final term is given by Eq. (14), and contributes to the generation process from t to s. (See Appx. §A.7 for proof.)

This generalized formulation of the VLB highlights the significance of the CE loss and the alignment between $q(\mathbf{x}_s|\mathbf{x}_s,\mathbf{x}_0)$ and $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ at any observed times s and t. While CE loss does not have a coefficient that depends on noise schedules and time, changing s and t or noise schedule (which determines $\bar{\alpha}_t$) during training will greatly impact the scale of the term in Eq. (14). By rendering the loss scaling term independent of time and the noise schedule, the minimization of this adjusted loss concurrently leads to the minimization of the original loss in Eq. (14) for any given s and t. Hence, by removing the sensitive scale $\frac{(1-\mu_{t|s})^2}{q(\mathbf{x}_s|\mathbf{x}_t,\mathbf{x}_0)}$ in Eq. (14), we reformulate the loss as

$$\mathcal{L}_t^2 := \|f_t^\theta(\boldsymbol{x}_t) - \boldsymbol{x}_0 + \phi_{t|s}\langle f_t^\theta(\boldsymbol{x}_t) - \boldsymbol{x}_0, \boldsymbol{x}_t\rangle(\boldsymbol{x}_t - \boldsymbol{m})\|_2^2, \tag{16}$$

where we can further clip $\phi_{t|s}$ to $\min(1, \phi_{t|s})$ for minimal scaling influence. It is important to note that while we have modified the coefficient to be invariant to the noise schedule and time s to effectively minimize Eq. (14) at any t and s, a similar approach to coefficient revision has been previously explored in Ho et al. [2020a], Karras et al. [2022], primarily to facilitate an easier optimization by achieving a balance of terms in the loss function.

Overall, we have the final approximated loss as $\tilde{\mathcal{L}}_t(\theta) = \mathcal{L}_t^2(\theta) + \mathcal{L}_t^{CE}(\theta)$ .We find that in practice this loss is much easier to optimize than the original exact negative VLB for more complicated distributions, leading to faster convergence and improved generation quality.

3 Continuous-time Discrete Diffusion

Despite being simple, discrete-time diffusion limits the generation process as we can only "jump back" through fixed time points. Recent works generalize continuous-state diffusion models to continuous-time [Song et al., 2021b]. This generalization enables great flexibility in backward generation as one can "jump back" through any time in $[0, T]$ with improved sample quality.

Nevertheless, generalizing discrete-state diffusion model from discrete-time to continuous-time is nontrivial, as the score-matching based technique [Song et al., 2021b] in continuous-state models requires the score function $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ to be available. This function, however, is evidently non-existent for discrete distributions. Recently, Campbell et al. [2022] presented the first continuous-time diffusion model for discrete data. It formulates the forward process through a Continuous Time Markov Chain (CTMC) and aims at learning a reverse CTMC that matches the marginal distribution with the forward CTMC at any time t. While being theoretically solid, the formulation in Campbell et al. [2022] has two problems: **(1)** the negative VLB loss of matching the forward and backward CTMCs is analytically complicated and hard to implement; and **(2)** the exact sampling through the learned backward CTMC is unrealistic. Campbell et al. [2022] propose approximate solutions, which however, trade off computational tractability with unknown errors and sample quality. SDDM Sun et al. [2023] takes a different approach with ratio matching Hyvärinen [2007], Lyu [2009], which is the generalization of score matching to discrete data. However, it needs a specific model architecture. Lou et al. [2024] advances ratio matching, yielding a loss that requires only a single model pass. Interestingly, this loss is similar to ours in formulation, despite originating from a different angle.

In this paper, we build on the CTMC formulation in Campbell et al. [2022], and show that the loss can be analytically simplified for nominal data. This simplification also inspires an improved MCMC corrector with closed-form formulation. For problem **(2)**, we argue that the difficulty arises from directly using the learned backward CTMC's transition rate. Instead, realizing that the reverse transition rate is computed based on the learned $p_\theta(\mathbf{x}_0|\mathbf{x}_t)$, we propose to compute $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ through

上述公式以及 prop.2.3 对于 *any* $0 < s < t \leq T$ 有效。接下来我们证明，在任何 s 和 t 处最小化 q 和 p_θ 之间的分歧也是有效的，因为它位于具有部分时间步长的一般负 VLB 内。 VLB 的初始版本是在每个时间步进行观测的假设下得出的。其后向降噪过程被设计为在每个生成步骤中前进一个时间步，以获得最佳生成质量。让我们考虑一个更一般的情况，其中在前向过程中仅观察到部分时间步，然后，最小化其负 VLB 可以帮助以更少的步数提高生成质量。假设仅观察到 $\mathbf{x}_s$ 和 $\mathbf{x}_t$，其中 $0 < s < t \leq T$，类似于等式的推导。 (2) 可以证明

$$\text{日志} p_\theta(\mathbf{x}_0) \geq -D_{\mathrm{KL}}[q(\mathbf{x}_t|\mathbf{x}_0)\|p_\theta(\mathbf{x}_t)] + \mathbb{E}_{q(\mathbf{x}_s|\mathbf{x}_0)}[\text{日志} p_\theta(\mathbf{x}_0|\mathbf{x}_s)] - D_{\mathrm{KL}}[q(\mathbf{x}_s|\mathbf{x}_t,\mathbf{x}_0)\|p_\theta(\mathbf{x}_s|\mathbf{x}_t)], \quad (15)$$

其中先验和后验之间的第一个散度项量化了从 T 到 t 的后向去噪过程的质量。第二项表示 CE 损失 $\mathcal{L}_s^{\mathrm{CE}}$，它影响从时间 s 到时间 0 的生成质量。最后一项由等式 1 给出。 (14)，并有助于从t到s的生成过程。 （证明参见附录§A.7。）

VLB 的这种广义公式强调了 CE 损失的重要性以及在任何观察时间 s 和 t 处 $q(\mathbf{x}_s|\mathbf{x}_s,\mathbf{x}_0)$ 和 $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ 之间的对齐。虽然CE损失没有依赖于噪声调度和时间的系数，但在训练期间改变s和t或噪声调度（决定$\bar{\alpha}_t$）将极大地影响等式（1）中的项的规模。 （14）。通过使损失缩放项独立于时间和噪声调度，该调整损失的最小化同时导致等式1中原始损失的最小化。 (14) 对于任何给定的 s 和 t。因此，通过去除等式中的敏感标度$\frac{(1-\mu_{t|s})^2}{q(\mathbf{x}_s|\mathbf{x}_t,\mathbf{x}_0)}$。 (14)，我们将损失重新表述为

$$\mathcal{L}_t^2 := \|f_t^\theta(\boldsymbol{x}_t) - \boldsymbol{x}_0 + \phi_{t|s}\langle f_t^\theta(\boldsymbol{x}_t) - \boldsymbol{x}_0, \boldsymbol{x}_t\rangle(\boldsymbol{x}_t - \boldsymbol{m})\|_2^2, \quad (16)$$

我们可以进一步将 $\phi_{t|s}$ 裁剪为 $\min(1, \phi_{t|s})$ 以最小化缩放影响。值得注意的是，虽然我们已经将系数修改为对噪声调度和时间 s 不变，以有效地最小化方程（1）。 (14) 在任何 t 和 s 处，Ho 等人之前已经探索了类似的系数修正方法。 [2020a]，卡拉斯等人。 [2022]，主要是为了通过实现损失函数中的项平衡来促进更容易的优化。

总的来说，我们最终的近似损失为$\tilde{\mathcal{L}}_t(\theta) = \mathcal{L}_t^2(\theta) + \mathcal{L}_t^{CE}(\theta)$. 我们发现，在实践中，对于更复杂的分布，这种损失比原始的精确负 VLB 更容易优化，从而导致更快的收敛并提高生成质量。

3 连续时间离散扩散

尽管很简单，但离散时间扩散限制了生成过程，因为我们只能“跳回”固定时间点。最近的工作将连续状态扩散模型推广到连续时间 [Song et al., 2021b]。这种泛化为后向生成提供了极大的灵活性，因为人们可以“跳回”$[0,T]$中的任何时间，并提高样本质量。

然而，将离散状态扩散模型从离散时间推广到连续时间并非易事，因为连续状态模型中基于分数匹配的技术 [Song et al., 2021b] 需要分数函数 $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ 可用。然而，对于离散分布来说，这个函数显然不存在。最近，坎贝尔等人。 [2022]提出了第一个离散数据的连续时间扩散模型。它通过连续时间马尔可夫链（CTMC）制定前向过程，旨在学习在任意时刻t将边际分布与前向CTMC相匹配的反向CTMC。虽然理论上是可靠的，但坎贝尔等人的公式。 [2022]有两个问题：（1）匹配前向和后向 CTMC 的负 VLB 损失分析复杂且难以实现；(2) 通过学习的后向CTMC进行精确采样是不现实的。坎贝尔等人。 [2022]提出了近似解决方案，然而，该解决方案在计算易处理性与未知错误和样本质量之间进行了权衡。 SDDM 孙等人。 [2023] 采用了不同的比率匹配方法 Hyvärinen [2007]、Lyu [2009]，这是分数匹配到离散数据的推广。然而，它需要特定的模型架构。卢等人。 [2024] 改进了比率匹配，产生仅需要单个模型传递的损失。有趣的是，这种损失在表述上与我们的类似，尽管源自不同的角度。

在本文中，我们以 Campbell 等人的 CTMC 公式为基础。 [2022]，并表明对于标称数据可以对损失进行分析简化。这种简化也激发了采用封闭式公式改进 MCMC 校正器的灵感。对于问题（2），我们认为困难来自于直接使用学习到的后向 CTMC 的转移率。相反，意识到反向转移率是根据学习的 $p_\theta(\mathbf{x}_0|\mathbf{x}_t)$ 计算的，我们建议通过以下方式计算 $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$

$p_\theta(\mathbf{x}_0|\mathbf{x}_t)$ without using the transition rate matrix. This avoids the approximation error of sampling directly from the backward CTMC and greatly simplifies the generation process.

Remarkably, with the new formulation of generation, we show that the continuous-time and discrete-time diffusion models can be unified together, with exactly the same forward diffusion and now also backward generation process. Moreover, we demonstrate that this unification offers mutual benefits: the continuous-time diffusion can leverage the swift and precise sampling formulation derived from the discrete-time case (as detailed in §A.8), while the discrete-time diffusion can utilize the MCMC corrector from the continuous-time scenario (see Appx. §A.17 and §A.19.6).

3.1 Background: Continuous-Time Markov Chain

CTMC generalizes Markov chain from discrete- to continuous-time via the Markov property: $\mathbf{x}_{t_1} \perp\!\!\!\perp \mathbf{x}_{t_3} \mid \mathbf{x}_{t_2}, \forall t_1 < t_2 < t_3$. Anderson [2012] provides an introduction to time-homogeneous CTMC. It can be derived from discrete-time Markov chain by increasing the number of time stamps N to infinite while keeping the total time T fixed. Specifically, we can define $\triangle t = \frac{T}{N}$, $t_i = i\triangle t$, and a discrete-time Markov chain characterized by transition probability $q(\mathbf{x}_{t_i}|\mathbf{x}_{t_{i-1}})$ and transition matrix Q_{t_i} with $[Q_{t_i}]_{jk} = q(\mathbf{x}_{t_i} = \boldsymbol{e}_k|\mathbf{x}_{t_{i-1}} = \boldsymbol{e}_j)$. By setting N to infinite, the transition probability $q(\mathbf{x}_{t_i}|\mathbf{x}_{t_{i-1}})$ converges to 0, hence is not suitable for describing CTMC. Instead, CTMC is fully characterized by its *transition rate* $r_t(\boldsymbol{y}|\boldsymbol{x})$, s.t.

$$r_t(\boldsymbol{y}|\boldsymbol{x}) = \lim_{N\to\infty} \frac{q(\mathbf{x}_{t_i} = \boldsymbol{y}|\mathbf{x}_{t_{i-1}} = \boldsymbol{x}) - \delta_{\boldsymbol{x},\boldsymbol{y}}}{t_i - t_{i-1}} = \lim_{\triangle t\to 0} \frac{q_{t|t-\triangle t}(\boldsymbol{y}|\boldsymbol{x}) - \delta_{\boldsymbol{x},\boldsymbol{y}}}{\triangle t} . \tag{17}$$

As the name suggests, $r_t(\boldsymbol{y}|\boldsymbol{x})$ measures the change rate of the transition probability of moving from state $\boldsymbol{x}$ to state $\boldsymbol{y}$ at time t in the direction of the process. The corresponding *transition rate matrix* R_t with $[R_t]_{ij} = r_t(\boldsymbol{e}_j|\boldsymbol{e}_i)$ fully determines the underlying stochastic process. A CTMC's transition probabilities satisfy the Kolmogorov equations [Kolmogoroff, 1931] (see Appx. §A.10), which have unique solution. As Rindos et al. [1995] stated, when R_{t_1} and R_{t_2} commute (i.e. $R_{t_1}R_{t_2} = R_{t_2}R_{t_1}$) for any t_1, t_2, the transition probability matrix can be written as $\overline{Q}_{t|s} = \exp\left(\int_s^t R_a da\right)$, where $\exp(M) := \sum_{k=0}^{\infty} \frac{M^k}{k!}$. The commutative property of R_t can be achieved by choosing $R_t = \beta(t)R_b$ where $R_b \in \mathbb{R}^{K\times K}$ is a time-independent base rate matrix.

3.2 Contrib. 3: Simplified Forward & Backward CTMCs that Connect to Discrete-Time Case

Forward CTMC. Two properties are needed for modeling the forward process: P1) the process can converge to an easy-to-sample stationary distribution at final time T; and P2) the conditional marginal distribution $q(\mathbf{x}_t|\mathbf{x}_0)$ can be obtained analytically for efficient training. As given above, P2) can be achieved by choosing commutative transition rate matrices with $R_t = \beta(t)R_b$.

We next show that property P1), i.e. $\lim_{t\to T} \overline{Q}_{t|0} = \overline{Q}_{T|0} = \mathbf{1}\boldsymbol{m}^\top$ for some stationary distribution $\boldsymbol{m}$, can be achieved by choosing $R_b = \mathbf{1}\boldsymbol{m}^\top - I$, which is a valid transition rate matrix with the property $(-R_b)^2 = (-R_b)$ (see derivation in Appx. §A.11). Then we have

$$\overline{Q}_{t|s} = \exp(\overline{\beta}_{t|s}R_b) = e^{-\overline{\beta}_{t|s}}I + (1 - e^{-\overline{\beta}_{t|s}})\mathbf{1}\boldsymbol{m}^\top. \tag{18}$$

★ **Unified forward process.** Eq. (18) has the same formulation as the transition matrix of the discrete-time case in Eq. (8), if we set $\overline{\alpha}_{t|s} = \exp(-\overline{\beta}_{t|s}) = \exp(-\int_s^t \beta(a)da)$. Thus, *this formulation unifies the forward processes of adding noise for both discrete- and continuous-time discrete diffusion*. With $\lim_{t\to T} \overline{\alpha}_{t|0} = 0$, or equivalently $\lim_{t\to T} \int_0^T \beta(a)da = \infty$, we achieve the goal $\overline{Q}_{T|0} = \mathbf{1}\boldsymbol{m}^\top$. We use $\overline{\alpha}_{t|s}$ directly in the following sections, i.e. Eq. (8). To summarize, for the forward CTMC

$$R_t = \beta(t)(\mathbf{1}\boldsymbol{m}^\top - I) \text{ , and } \quad \overline{Q}_{t|s} = \overline{\alpha}_{t|s}I + (1 - \overline{\alpha}_{t|s})\mathbf{1}\boldsymbol{m}^\top . \tag{19}$$

We can further get vector-form forward rate

$$r_t(\boldsymbol{x}|\cdot) = R_t\boldsymbol{x} = \beta(t)\big(\langle \boldsymbol{x}, \boldsymbol{m}\rangle\mathbf{1} - \boldsymbol{x}\big), \quad r_t(\cdot|\boldsymbol{x}) = R_t^\top\boldsymbol{x} = \beta(t)\big(\boldsymbol{m} - \boldsymbol{x}\big) . \tag{20}$$

Backward CTMC. To generate samples from the target distribution, we have to reverse the process of the forward CTMC. Let $\widehat{r}_t$ be the transition rate of the backward CTMC with corresponding matrix $\widehat{R}_t$. When the forward and backward CTMCs are matched exactly, theoretically the forward and

$p_\theta(\mathbf{x}_{0|\mathbf{x}_t})$ 不使用转移率矩阵。这避免了直接从后向CTMC采样的近似误差，大大简化了生成过程。

Remarkably，通过新的生成公式，我们表明连续时间和离散时间扩散模型可以统一在一起，具有完全相同的前向扩散和现在的后向生成过程。此外，我们证明这种统一提供了互惠互利：连续时间扩散可以利用从离散时间情况得出的快速而精确的采样公式（如§A.8中详述），而离散时间扩散可以利用连续时间场景中的MCMC校正器（参见附录§A.17和§A.19.6）。

3.1 背景：连续时间马尔可夫链

CTMC 通过马尔可夫属性将马尔可夫链从离散时间推广到连续时间：$\mathbf{x}_{t_1} \perp\!\!\!\perp$、$\mathbf{x}_{t_3} \mid \mathbf{x}_{t_2}$、$\forall t_1 < t_2 < t_3$。Anderson [2012] 介绍了时间同质 CTMC。它可以通过将时间戳 N 的数量增加到无限同时保持总时间 T 固定，从离散时间马尔可夫链导出。具体来说，我们可以定义 $\triangle t = \frac{T}{N}$、$t_i = i\triangle t$ 和一个以转移概率 $q(\mathbf{x}_{t_i}|\mathbf{x}_{t_{i-1}})$ 和转移矩阵 Q_{t_i} 为特征的离散时间马尔可夫链 $[Q_{t_i}]_{jk} = q(\mathbf{x}_{t_i} = \boldsymbol{e}_k|\mathbf{x}_{t_{i-1}} = \boldsymbol{e}_j)$。通过将$N$设置为无穷大，转移概率$q(\mathbf{x}_{t_i}|\mathbf{x}_{t_{i-1}})$收敛于0，因此不适合描述CTMC。相反，CTMC 的特点是其 *transition rate* $r_t(\boldsymbol{y}|\boldsymbol{x})$、s.t.

$$r_t(\boldsymbol{y}|\boldsymbol{x}) = \lim_{N\to\infty} \frac{q(\mathbf{x}_{t_i} = \boldsymbol{y}|\mathbf{x}_{t_{i-1}} = \boldsymbol{x}) - \delta_{\boldsymbol{x},\boldsymbol{y}}}{t_i - t_{i-1}} = \lim_{\triangle t\to 0} \frac{q_{t|t-\triangle t}(\boldsymbol{y}|\boldsymbol{x}) - \delta_{\boldsymbol{x},\boldsymbol{y}}}{\triangle t} \; . \tag{17}$$

顾名思义，$r_t(\boldsymbol{y}|\boldsymbol{x})$ 衡量在时间 t 时从状态 $\boldsymbol{x}$ 转移到状态 $\boldsymbol{y}$ 的转移概率在过程方向上的变化率。对应的 *transition rate matrix* R_t 与 $[R_t]_{ij} = r_t(\boldsymbol{e}_j|\boldsymbol{e}_i)$ 完全确定了底层随机过程。CTMC 的转移概率满足 Kolmogorov 方程 [Kolmogoroff, 1931]（参见附录 §A.10），该方程具有唯一解。正如 Rindos 等人。[1995] 指出，当 R_{t_1} 和 R_{t_2} 对任何 t_1, t_2 交换（即 $R_{t_1}R_{t_2} = R_{t_2}R_{t_1}$）时，转移概率矩阵可以写为 $Q_{t|s} = \exp\left(\int_s^t R_a da\right)$，其中 $\exp(M) := \sum_{k=0}^{\infty} \frac{M^k}{k!}$。$R_t$ 的交换律可以通过选择 $R_t = \beta(t)R_b$ 来实现，其中 $R_b \in \mathbb{R}^{K\times K}$ 是与时间无关的基本速率矩阵。

3.2 贡献。3：连接到离散时间案例的简化前向和后向 CTMC

向前CTMC。建模前向过程需要两个属性：P1) 该过程可以在最终时间 T 收敛到易于采样的平稳分布；P2) 可以通过分析获得条件边际分布 $q(\mathbf{x}_t|\mathbf{x}_0)$，以进行有效的训练。如上所述，P2)可以通过选择具有$R_t = \beta(t)R_b$的交换转移率矩阵来实现。

接下来我们展示属性 P1)，即某些平稳分布 $\boldsymbol{m}$ 的 $\lim_{t\to T} \overline{Q}_{t|0} = \overline{Q}_{T|0} = \mathbf{1}\boldsymbol{m}^\top$，可以通过选择 $R_b = \mathbf{1}\boldsymbol{m}^\top - I$ 来实现，它是具有属性 $(-R_b)^2 = (-R_b)$ (的有效转移率矩阵，请参阅 Appx 中的推导。§A.11)。然后我们有

$$\overline{Q}_{t|s} = \exp(\overline{\beta}_{t|s}R_b) = e^{-\overline{\beta}_{t|s}}I + (1 - e^{-\overline{\beta}_{t|s}})\mathbf{1}\boldsymbol{m}^\top. \tag{18}$$

★ 统一转发流程。等式。(18) 与式 (18) 中离散时间情况的转移矩阵具有相同的公式。(8)，如果我们设置 $\alpha_{t|s} = \exp(-\overline{\beta}_{t|s}) = \exp(-\int_s^t \beta(a)da)$。因此，*this formulation unifies the forward processes of adding noise for both discrete- and continuous-time discrete diffusion*。当 $\lim_{t\to T} \overline{\alpha}_{t|0} = 0$ 或等价的 $\lim_{t\to T} \int_0^T \beta(a)da = \infty$ 时，我们实现了目标 $Q_{T|0} = \mathbf{1}\boldsymbol{m}^\top$。我们在以下部分中直接使用 $\alpha_{t|s}$，即等式：（8）。总结一下，对于未来的CTMC

$$R_t = \beta(t)(\mathbf{1}\boldsymbol{m}^\top - I) \text{ , and } \quad \overline{Q}_{t|s} = \overline{\alpha}_{t|s}I + (1 - \overline{\alpha}_{t|s})\mathbf{1}\boldsymbol{m}^\top \; . \tag{19}$$

我们可以进一步得到向量形式的转发速率

$$r_t(\boldsymbol{x}|\cdot) = R_t\boldsymbol{x} = \beta(t)\big(\langle\boldsymbol{x}, \boldsymbol{m}\rangle\mathbf{1} - \boldsymbol{x}\big), \quad r_t(\cdot|\boldsymbol{x}) = R_t^\top\boldsymbol{x} = \beta(t)\big(\boldsymbol{m} - \boldsymbol{x}\big) \; . \tag{20}$$

落后的CTMC。为了从目标分布生成样本，我们必须反转前向 CTMC 的过程。令$\hat{r}_t$为具有相应矩阵$\widehat{R}_t$的后向CTMC的转移率。当前向和后向CTMC精确匹配时，理论上前向和后向CTMC backward CTMCs have the following relationship (see Campbell et al. [2022]'s Proposition 1):

$$\widehat{r}_t(\boldsymbol{x}|\boldsymbol{y}) = r_t(\boldsymbol{y}|\boldsymbol{x})\frac{q_t(\boldsymbol{x})}{q_t(\boldsymbol{y})}, \quad \forall \boldsymbol{x} \neq \boldsymbol{y} \ . \tag{21}$$

However, the marginal distributions $q_t(\boldsymbol{x})$ and $q_t(\boldsymbol{y})$ are intractable analytically, hence we cannot derive the backward CTMC directly from Eq. (21). Instead, Campbell et al. [2022] parameterize the transition rate $\widehat{r}_t^\theta$, by observing that $\frac{q_t(\boldsymbol{x})}{q_t(\boldsymbol{y})} = \sum_{\boldsymbol{x}_0} \frac{q_{t|0}(\boldsymbol{x}|\boldsymbol{x}_0)}{q_{t|0}(\boldsymbol{y}|\boldsymbol{x}_0)} q_{0|t}(\boldsymbol{x}_0|\boldsymbol{y})$[3], as follows

$$\widehat{r}_t^\theta(\boldsymbol{x}|\boldsymbol{y}) = r_t(\boldsymbol{y}|\boldsymbol{x}) \sum_{\boldsymbol{x}_0} \frac{q_{t|0}(\boldsymbol{x}|\boldsymbol{x}_0)}{q_{t|0}(\boldsymbol{y}|\boldsymbol{x}_0)} p^\theta_{0|t}(\boldsymbol{x}_0|\boldsymbol{y}). \tag{22}$$

Then, $\widehat{r}_t^\theta$ is learned with θ to minimize the continuous-time negative VLB introduced next.

Negative VLB. Similar to the discrete-time case, the backward CTMC can be learned by maximizing the VLB for data log-likelihood. Computing VLB for CTMC is nontrivial, and fortunately Campbell et al. [2022] has derived (see their Proposition 2) that the negative VLB can be formulated as

$$T\,\mathbb{E}_{\substack{t\sim \text{Uni}(0,T)\\ \boldsymbol{x}\sim q(\mathbf{x}_t|\mathbf{x}_0)}} \Big[\sum_{\boldsymbol{z}\neq \boldsymbol{x}} \widehat{r}_t^\theta(\boldsymbol{z}|\boldsymbol{x}) - \sum_{\boldsymbol{z}\neq\boldsymbol{x}} r_t(\boldsymbol{z}|\boldsymbol{x}) \log \widehat{r}_t^\theta(\boldsymbol{x}|\boldsymbol{z}) \Big] \ . \tag{23}$$

However, Campbell et al. [2022]'s original design did not simplify the negative VLB with the parameterization of $\widehat{r}_t^\theta$ in Eq. (22), making the implementation nontrivial and inefficient. In this section, we show that their formulation can be greatly simplified to a closed-form evaluation.

To simplify Eq. (23), we introduce $g_t^\theta(\boldsymbol{x}|\boldsymbol{y})$ such that $\widehat{r}_t^\theta(\boldsymbol{x}|\boldsymbol{y}) = r_t(\boldsymbol{y}|\boldsymbol{x}) g_t^\theta(\boldsymbol{x}|\boldsymbol{y})$, with

$$g_t^\theta(\boldsymbol{x}|\boldsymbol{y}) := \sum_{\boldsymbol{x}_0} \frac{q_{t|0}(\boldsymbol{x}|\boldsymbol{x}_0)}{q_{t|0}(\boldsymbol{y}|\boldsymbol{x}_0)} p^\theta_{0|t}(\boldsymbol{x}_0|\boldsymbol{y}) \approx \frac{q_t(\boldsymbol{x})}{q_t(\boldsymbol{y})} \ , \tag{24}$$

which is the estimator of the marginal probability ratio.

Proposition 3.1. *The vector form parameterization of $g_t^\theta(\boldsymbol{x}|\boldsymbol{y})$ can be simplified analytically as:*

$$g_t^\theta(\cdot|\boldsymbol{y}) = \Big[\big(1 - \frac{\overline{\alpha}_{t|0} \langle f_t^\theta(\boldsymbol{y}), \boldsymbol{y}\rangle}{\overline{\alpha}_{t|0} + (1-\overline{\alpha}_{t|0})\langle \boldsymbol{y}, \boldsymbol{m}\rangle}\big)\boldsymbol{m} + \frac{\overline{\alpha}_{t|0}}{1-\overline{\alpha}_{t|0}} f_t^\theta(\boldsymbol{y}) \Big] \odot \frac{\mathbf{1}-\boldsymbol{y}}{\langle \boldsymbol{y}, \boldsymbol{m}\rangle} + \boldsymbol{y} \tag{25}$$

The proof is in Appx. §A.12, with its extension to multi-element $g_t^{\theta,d}(\cdot|\boldsymbol{y}^{1:D})$ in Eq. (75).

3.3 Contrib. 4: Simplification of Continuous-time Negative VLB

As the derivation is much harder in multi-element case, we work on it directly and single-element can be induced as a special case. Given $\boldsymbol{x}^{1:D}$, let $\boldsymbol{x}^{\backslash d}$ represent $\boldsymbol{x}^{1:D\backslash d}$, i.e. the object without d-th element. As we assume the forward processes are independent for different elements, r_t^d represents the transition rate of the forward CTMC process at the d-th element.

Proposition 3.2. *The negative VLB in Eq. (23) in multi-element case can be simplified as*

$$T\,\mathbb{E}_{\substack{t\sim Uni(0,T)\\ \boldsymbol{x}^{1:D}\sim q_{t|0}(\cdot|\boldsymbol{x}_0^{1:D})\\ \boldsymbol{z}^{1:D}\sim S_t(\cdot|\boldsymbol{x}^{1:D})}} \Big[\sum_{d=1}^D r_t^d(\boldsymbol{x}^d|\cdot)^\top g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D}) - \frac{1}{\mathcal{M}_{S_t}(\boldsymbol{z}^{1:D}|\boldsymbol{x}_0^{1:D})} \sum_{d=1}^D \frac{\mathbf{1}^\top [q_{t|0}(\cdot|\boldsymbol{x}_0^d) \odot r_t^d(\boldsymbol{z}^d|\cdot) \odot \log g_t^{\theta,d}(\cdot|\boldsymbol{z}^{1:D})]}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)} \Big] \tag{26}$$

where $S_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D})$ is any unnormalized distribution (see Eq. (82) in Appx.) of sampling the auxiliary variable $\boldsymbol{z}^{1:D}$ from $\boldsymbol{x}^{1:D}$. The auxiliary variable is introduced to avoid multiple passes of the model for computing the second term of Eq. (23). M_{S_t} is a normalization scalar (see Eq. (86) in Appx.) that only depends on $S_t, \boldsymbol{z}^{1:D}$, and $\boldsymbol{x}_0^{1:D}$.

The detailed proof is in Appx. §A.14. This formulation simplifies and generalizes the result in Campbell et al. [2022], such that any S_t can be used for introducing the auxiliary variable $\boldsymbol{z}_{1:D}$. Importantly, Eq. (26) shows that changing S_t only affects a scalar weight. Computing the loss requires two passes of the model for $\boldsymbol{z}^{1:D}$ and $\boldsymbol{x}^{1:D}$ separately. Now we show that it can be further simplified to only a single pass of the model.

Proposition 3.3. *The loss in Eq. (26) can be further simplified as*

$$T\,\mathbb{E}_{\substack{t\sim Uni(0,T)\\ \boldsymbol{x}^{1:D}\sim q_{t|0}(\cdot|\boldsymbol{x}_0^{1:D})}} \Big[\sum_{d=1}^D r_t^d(\boldsymbol{x}^d|\cdot)^\top \Big(g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D}) - \frac{q_{t|0}(\cdot|\boldsymbol{x}_0^d) \odot \log g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D})}{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)} \Big) \Big] \tag{27}$$

[3]As $q_t(\boldsymbol{x}) = \sum_{\boldsymbol{x}_0} q_{t|0}(\boldsymbol{x}|\boldsymbol{x}_0) q_{\text{data}}(\boldsymbol{x}_0)$ and $q_t(\boldsymbol{x}) = \frac{q_{t|0}(\boldsymbol{x}|\tilde{\boldsymbol{x}}_0) q_{\text{data}}(\tilde{\boldsymbol{x}}_0)}{q_{0|t}(\tilde{\boldsymbol{x}}_0|\boldsymbol{x})}$.

落后的 CTMC 具有以下关系（参见 Campbell 等人[2022]的命题 1）：

$$\hat{r}_t(\boldsymbol{x}|\boldsymbol{y}) = r_t(\boldsymbol{y}|\boldsymbol{x})\frac{q_t(\boldsymbol{x})}{q_t(\boldsymbol{y})}, \quad \forall \boldsymbol{x} \neq \boldsymbol{y} \ . \tag{21}$$

然而，边际分布$q_t(\boldsymbol{x})$和$q_t(\boldsymbol{y})$在分析上是难以处理的，因此我们不能直接从方程（1）导出后向CTMC。(21)。相反，坎贝尔等人。 [2022] 通过观察 $\frac{q_t(\boldsymbol{x})}{q_t(\boldsymbol{y})} = \sum_{\boldsymbol{x}_0} \frac{q_{t|0}(\boldsymbol{x}|\boldsymbol{x}_0)}{q_{t|0}(\boldsymbol{y}|\boldsymbol{x}_0)} q_{0|t}(\boldsymbol{x}_0|\boldsymbol{y})^3$ 来参数化转换率 $\hat{r}_t^\theta$，如下所示

$$\hat{r}_t^\theta(\boldsymbol{x}|\boldsymbol{y}) = r_t(\boldsymbol{y}|\boldsymbol{x}) \sum_{\boldsymbol{x}_0} \frac{q_{t|0}(\boldsymbol{x}|\boldsymbol{x}_0)}{q_{t|0}(\boldsymbol{y}|\boldsymbol{x}_0)} p_{0|t}^\theta(\boldsymbol{x}_0|\boldsymbol{y}). \tag{22}$$

然后，用θ学习$\hat{r}_t^\theta$，以最小化接下来引入的连续时间负VLB。

VLB 阴性。与离散时间情况类似，可以通过最大化数据对数似然的 VLB 来学习后向 CTMC。计算 CTMC 的 VLB 并非易事，幸运的是 Campbell 等人。 [2022] 推导出（参见他们的命题 2）负 VLB 可以表示为

$$T\ \mathbb{E}_{\substack{t\sim \mathrm{Uni}(0,T) \\ \boldsymbol{x}\sim q(\mathbf{x}_t|\mathbf{x}_0)}} \Big[\sum_{\boldsymbol{z}\neq \boldsymbol{x}} \hat{r}_t^\theta(\boldsymbol{z}|\boldsymbol{x}) - \sum_{\boldsymbol{z}\neq \boldsymbol{x}} r_t(\boldsymbol{z}|\boldsymbol{x}) \log \hat{r}_t^\theta(\boldsymbol{x}|\boldsymbol{z}) \Big] \ . \tag{23}$$

然而，坎贝尔等人。 [2022]的原始设计并没有通过等式中$\hat{r}_t^\theta$的参数化来简化负VLB。(22), 使得实施变得不平凡且低效。在本节中，我们展示了它们的公式可以大大简化为封闭式评估。

简化方程。(23)，我们引入 $g_t^\theta(\boldsymbol{x}|\boldsymbol{y})$ 使得 $\hat{r}_t^\theta(\boldsymbol{x}|\boldsymbol{y}) = r_t(\boldsymbol{y}|\boldsymbol{x}) g_t^\theta(\boldsymbol{x}|\boldsymbol{y})$，其中

$$g_t^\theta(\boldsymbol{x}|\boldsymbol{y}) := \sum_{\boldsymbol{x}_0} \frac{q_{t|0}(\boldsymbol{x}|\boldsymbol{x}_0)}{q_{t|0}(\boldsymbol{y}|\boldsymbol{x}_0)} p_{0|t}^\theta(\boldsymbol{x}_0|\boldsymbol{y}) \approx \frac{q_t(\boldsymbol{x})}{q_t(\boldsymbol{y})} \ , \tag{24}$$

这是边际概率比的估计。

命题3.1。 *The vector form parameterization of $g_t^\theta(\boldsymbol{x}|\boldsymbol{y})$ can be simplified analytically as:*

$$g_t^\theta(\cdot|\boldsymbol{y}) = \Big[(1 - \frac{\overline{\alpha}_{t|0}\langle f_t^\theta(\boldsymbol{y}), \boldsymbol{y}\rangle}{\overline{\alpha}_{t|0} + (1-\overline{\alpha}_{t|0})\langle \boldsymbol{y}, \boldsymbol{m}\rangle}) \boldsymbol{m} + \frac{\overline{\alpha}_{t|0}}{1-\overline{\alpha}_{t|0}} f_t^\theta(\boldsymbol{y}) \Big] \odot \frac{\mathbf{1}-\boldsymbol{y}}{\langle \boldsymbol{y}, \boldsymbol{m}\rangle} + \boldsymbol{y} \tag{25}$$

证明在Appx中。 §A.12，及其对等式中多元素$g_t^{\theta,d}(\cdot|\boldsymbol{y}^{1:D})$的扩展。(75)。

3.3 贡献。 4：连续时间负V的简化 磅

由于多元素情况下推导比较困难，我们直接对其进行处理，单元素可以作为特例进行归纳。给定 $\boldsymbol{x}^{1:D}$，让 $\boldsymbol{x}^{\setminus d}$ 代表 $\boldsymbol{x}^{1:D\setminus d}$，即没有第 d 个元素的对象。由于我们假设前向过程对于不同的元素是独立的，因此 r_t^d 表示前向 CTMC 过程在第 d 个元素处的转换率。

命题3.2。 *T he negative VLB in Eq. (23) in multi-element case can be simplified as*

$$T\ \mathbb{E}_{\substack{t\sim Uni(0,T) \\ \boldsymbol{x}^{1:D}\sim q_{t|0}(\cdot|\boldsymbol{x}_0^{1:D}) \\ \boldsymbol{z}^{1:D}\sim S_t(\cdot|\boldsymbol{x}^{1:D})}} \Big[\sum_{d=1}^{D} r_t^d(\boldsymbol{x}^d|\cdot)^\top g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D}) - \frac{1}{\mathcal{M}_{S_t}(\boldsymbol{z}^{1:D}|\boldsymbol{x}_0^{1:D})} \sum_{d=1}^{D} \frac{\mathbf{1}^\top[q_{t|0}(\cdot|\boldsymbol{x}_0^d) \odot r_t^d(\boldsymbol{z}^d|\cdot) \odot \log g_t^{\theta,d}(\cdot|\boldsymbol{z}^{1:D})]}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)} \Big] \tag{26}$$

where $S_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D})$ is any unnormalized distribution (see Eq. (82) in Appx.) of sampling the auxiliary variable $\boldsymbol{z}^{1:D}$ from $\boldsymbol{x}^{1:D}$. The auxiliary variable is introduced to avoid multiple passes of the model for computing the second term of Eq. (23). $\mathcal{M}_{S_t}$ is a normalization scalar (see Eq. (86) in Appx.) that only depends on $S_t, \boldsymbol{z}^{1:D}$, and $\boldsymbol{x}^{1:D}{}_0$.

详细证明见Appx. §A.14。该公式简化并概括了 Campbell 等人的结果。 [2022]，使得任何S_t都可以用于引入辅助变量$\boldsymbol{z}_{1:D}$。重要的是，等式。 (26) 表明改变 S_t 仅影响标量权重。计算损失需要分别对 $\boldsymbol{z}^{1:D}$ 和 $\boldsymbol{x}^{1:D}$ 模型进行两次传递。现在我们证明它可以进一步简化为模型的一次传递。

命题3.3。 *The loss in Eq. (26) can be further simplified as*

$$T\ \mathbb{E}_{\substack{t\sim Uni(0,T) \\ \boldsymbol{x}^{1:D}\sim q_{t|0}(\cdot|\boldsymbol{x}_0^{1:D})}} \Big[\sum_{d=1}^{D} r_t^d(\boldsymbol{x}^d|\cdot)^\top \Big(g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D}) - \frac{q_{t|0}(\cdot|\boldsymbol{x}_0^d) \odot \log g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D})}{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)} \Big) \Big] \tag{27}$$

3如 $q_t(\boldsymbol{x}) = \sum_{\boldsymbol{x}_0} q_{t|0}(\boldsymbol{x}|\boldsymbol{x}_0) q_{\text{data}}(\boldsymbol{x}_0)$ 和 $q_t(\boldsymbol{x}) = \frac{q_{t|0}(\boldsymbol{x}|\tilde{\boldsymbol{x}}_0) q_{\text{data}}(\tilde{\boldsymbol{x}}_0)}{q_{0|t}(\tilde{\boldsymbol{x}}_0|\boldsymbol{x})}$。

The detailed proof can be found in Appendix §A.15. Note that this loss requires only a single forward pass of the model, making it extremely efficient. Additionally, this loss aligns with the DWDSE loss (while presented in single-element case) in Lou et al. [2024], which is derived from the different concrete score matching direction. This indicates that the VLB is essentially the same as the ratio matching loss, as also demonstrated in the continuous-state setting, such that the score matching Song et al. [2021b] is shown to be equivalent to VLB computation in Ho et al. [2020a].

3.4 Contrib. 5: Fast Backward Sampling & Unification of Discrete Diffusion

Campbell et al. [2022] proposes to use the learned transition rate $\widehat{r}_t^\theta$ of the backward CTMC to sample reversely for the target distribution $p_{\text{data}}(\mathbf{x}_0)$. However, different from the forward CTMC where all elements are sampled independently, the backward CTMC processes for all elements are coupled together and do not have the closed-form transition probability derived from the learned transition rate. Direct and exact sampling from a CTMC uses the algorithm by Gillespie [1977], which is intractable for multi-element objects. Campbell et al. [2022] proposes to use tau-leaping [Gillespie, 2001] that approximately samples all transitions from time t to $t-\tau$, assuming $R_t^{1:D}$ fixed during the time period. While being tractable, it introduces unknown errors that needs correction process.

We propose to avoid the costly sampling from using the transition rate of backward CTMC. We first observe that the estimated transition rate $\widehat{r}_t^\theta$ is essentially derived from the estimated $p_{0|t}^\theta(\mathbf{x}_0|\mathbf{x}_t)$. Hence using $\widehat{r}_t^\theta$ is equivalent to using $p_{0|t}^\theta(\mathbf{x}_0|\mathbf{x}_t)$ in an indirect way. We realize that the transition probability $q_{s|t}(\mathbf{x}_s|\mathbf{x}_t)$ can be computed easily from $p_{0|t}^\theta(\mathbf{x}_0|\mathbf{x}_t)$ directly, as shown in Eq. (10). With the help of Eq. (10), we can sample $\mathbf{x}_0$ or equivalently $\mathbf{x}_{0|T}$ through sampling $\mathbf{x}_{t_{n-1}|t_n}, ..., \mathbf{x}_{t_1|t_2}, \mathbf{x}_{t_0|t_1}$ sequentially, with any $\{t_i\}_{i=0}^n$ that satisfies $0{=}t_0 < t_1 < ... < t_n{=}T$. Although Eq. (10) is derived for discrete-time case, it applies directly in continuous-time case, thanks to the unification of the forward process for discrete- and continuous-time diffusion (§3.2). Hence, discrete-time and continuous-time diffusion also have the same **unified backward generation process**.

$\star$ **Unified Discrete Diffusion.** All in all, through a series of mathematical simplifications, we have shown that both discrete&continuous-time discrete diffusion share **(1)** the same forward diffusion process; **(2)** the same parameterization for learning $p_{0|t}^\theta$; and **(3)** the same backward denoising process. In light of our reformulations, we propose USD3, a novel Unified and Simplified Discrete Denoising Diffusion model. Notably, USD3 utilizes the same source code for both discrete- and continuous-time, up to a single alteration in the loss function during training, and a shared sample generation process (resp. Algo. 1 and 2 in Appx. §A.16).

4 Experiments

Discrete diffusion has limited , no established guidelines on model training. Many prior work optimize a combined VLB and cross-entropy (CE) loss with a fixed weight, without deeper investigation. In this work we not only improve VLB loss mathematically, but also empirically explore an extensive testbed of training regimes for discrete diffusion—including various loss combinations, both discrete- and continuous-time training, as well as varying model sizes—toward a deeper understanding of which yield tractable optimization and higher generation quality. USD3 is a hybrid of our simplified exact VLB and CE, and USD3* refers to its approximation with further simplifications (§2.2.5 and §3.3). USD3-CE and USD3-VLB are variants, resp. with CE or VLB loss *only*.

Baselines. We compare USD3 and variants to three latest SOTA discrete diffusion models in the literature: D3PM in Austin et al. [2021], RDM in Zhang et al. [2023] (discrete-time), τ-LDR-0 in Campbell et al. [2022] , SDDM in Sun et al. [2023] and SEDD in Lou et al. [2024] (continuous time).

Training Details. Thanks to unification, we can evaluate both discrete- and continuous-time models with both cosine and constant noise schedulers at ease. In USD3, we combine VLB and CE with weight 0.001 for the latter, following prior literature. For USD3*, we combine VLB with CE weight 1.0. As architecture, we parameterize $f_t^\theta(\mathbf{x}_t)$ with a sequence transformer model. By varying its depths and widths, we study the impact of model size. The details are described in Appx. §A.18.4.

4.1 Music Generation

Lakh Pianoroll Datasets. We evaluate monophonic music generation on PIANO, the cleaned Lakh pianoroll dataset [Raffel [2016], Dong et al. [2017]], containing $6,000$ training and 973 evaluation (or test) sequences of 256 notes each. Here we perform conditional generation; given the first 32

详细证明可见附录§A.15。请注意，这种损失仅需要模型的一次前向传递，因此非常高效。此外，该损失与 Lou 等人中的 DWDSE 损失（虽然在单元素情况下呈现）一致。 [2024], 这是由不同的具体分数匹配方向得出的。这表明 VLB 本质上与比率匹配损失相同，这也在连续状态设置中得到证明，使得分数匹配 Song 等人。 [2021b]被证明与Ho等人中的VLB计算等效。 [2020a]。

3.4 贡献。 5：快速向后采样和离散扩散的统一

坎贝尔等人。 [2022]提出使用后向CTMC的学习转移率$\hat{r}_t^\theta$来反向采样目标分布$p_{\text{data}}(\mathbf{x}_0)$。然而，与所有元素独立采样的前向 CTMC 不同，所有元素的后向 CTMC 过程耦合在一起，并且不具有从学习的转移率导出的闭合形式转移概率。 CTMC 的直接精确采样使用 Gillespie [1977] 的算法，这对于多元素对象来说很棘手。坎贝尔等人。 [2022] 建议使用 tau-leaping [Gillespie, 2001] 来近似采样从时间 t 到 $t-\tau$ 的所有转换，假设 $R_t^{1:D}$ 在该时间段内固定。虽然易于处理，但它会引入需要纠正过程的未知错误。

我们建议避免使用向后 CTMC 的转换率来进行昂贵的采样。我们首先观察到估计的转换率 $\hat{r}_t^\theta$ 本质上是从估计的 $p_{0|t}^\theta(\mathbf{x}_0|\mathbf{x}_t)$ 导出的。因此，使用 $\hat{r}_t^\theta$ 相当于间接使用 $p_{0|t}^\theta(\mathbf{x}_0|\mathbf{x}_t)$。我们意识到转移概率 $q_{s|t}(\mathbf{x}_s|\mathbf{x}_t)$ 可以很容易地直接从 $p_{0|t}^\theta(\mathbf{x}_0|\mathbf{x}_t)$ 计算出来，如方程所示。（10）。在方程式的帮助下。 (10)，我们可以通过顺序采样 $\mathbf{x}_{t_{n-1}|t_n},\dots,\mathbf{x}_{t_1|t_2},\mathbf{x}_{t_0|t_1}$ 来采样 $\mathbf{x}_0$ 或等效的 $\mathbf{x}_{0|T}$，其中任何满足 0=$t_0<t_1<\dots<t_n$=T 的 $\{t_i\}_{i=0}^n$。虽然等式。 (10) 是针对离散时间情况导出的，由于离散时间和连续时间扩散的前向过程的统一，它直接适用于连续时间情况（§3.2）。因此，离散时间和连续时间扩散也具有相同的统一后向生成过程。

★ 统一离散扩散。总而言之，通过一系列的数学简化，我们证明了离散扩散和连续时间离散扩散共享（1）相同的前向扩散过程； (2) 学习$p_{0|t}^\theta$的相同参数化； (3)同样的后向去噪过程。根据我们的重新表述，我们提出了 USD3，一种新颖的统一和简化的离散去噪扩散模型。值得注意的是，USD3 在离散时间和连续时间上使用相同的源代码，最多在训练期间对损失函数进行一次更改，并共享样本生成过程（分别为附录§A.16 中的算法 1 和 2）。

4 实验

离散扩散有限，没有既定的模型训练指南。许多先前的工作以固定权重优化组合的 VLB 和交叉熵 (CE) 损失，而没有进行更深入的研究。在这项工作中，我们不仅在数学上改进了 VLB 损失，而且还根据经验探索了离散扩散训练机制的广泛测试平台（包括各种损失组合、离散时间和连续时间训练以及不同的模型大小），以更深入地了解哪些会产生易于处理的优化和更高的生成质量。 USD3 是我们简化的精确 VLB 和 CE 的混合体，USD3* 指的是其进一步简化的近似值（§2.2.5 和 §3.3）。 USD3-CE 和 USD3-VLB 分别是变体。具有 CE 或 VLB 丢失 *only*。

基线。我们将 USD3 及其变体与文献中三个最新的 SOTA 离散扩散模型进行比较：Austin 等人的 D3PM。 [2021]，Zhang 等人的 RDM。 [2023]（离散时间），Campbell 等人中的 τ-LDR-0。 [2022]，Sun 等人中的 SDDM。 [2023] 和 Lou 等人的 SEDD。 [2024]（连续时间）。

培训细节。由于统一，我们可以使用余弦和恒定噪声调度器轻松评估离散和连续时间模型。在 USD3 中，我们遵循先前文献，将 VLB 和 CE 结合起来，后者的权重为 0.001。对于 USD3*，我们将 VLB 与 CE 权重 1.0 结合起来。作为架构，我们使用序列转换器模型参数化 $f_t^\theta(\mathbf{x}_t)$。通过改变其深度和宽度，我们研究了模型尺寸的影响。详细信息在 Appx 中描述。 §A.18.4。

4.1 音乐生成

十万个钢琴卷轴数据集。我们在 PIANO 上评估单音音乐的生成，PIANO 是经过清理的 Lakh 钢琴卷轴数据集 [Raffel [2016]，Dong 等人。 [2017]]，包含 6, 000 个训练序列和 973 个评估（或测试）序列，每个序列有 256 个音符。这里我们进行条件生成；给定前 32 个

notes, the models are required to generate the rest 224 notes. Interestingly, we find that some (124) evaluation sequences share the *same* first 32 notes as one or more training sequences. Such repetition gives us a unique opportunity to investigate what-we-call "parroting"—the phenomenon that models are simply re-playing the exact training data during generation. To that end we create PIANO-P, a subset consisting only of these 124 evaluation sequences paired with their first-32-note-matching training sequences. Details are in Appx. §A.18.1.

Metrics. We use a number of new metrics for PIANO: 2- and 3-gram (1) Hellinger Distance (↓) and (2) Proportion of Outliers (↓), in addition to previously used 1-gram counterparts based only on *single* note appearances in Campbell et al. [2022]; (3) Diverse Edit Distance (↑), which accounts for the creativity/novelty across multiple generated samples. On PIANO-P, we also report (4) Train-to-Test Ratios (↑) for {1,2,3}-gram Hellinger as well as Proportion of Outliers, which compare the weighted distance of a generated sample to its matching training vs. test sequence that share the same first 32 notes. Such ratios quantify the extent of "parroting"; a smaller ratio indicates a higher (lower) degree of memorization (generalization). Calculation details of all metrics are given in Appx. §A.18.3.

Results. Table 1 shows that USD3 and its variants perform better than the baseline methods across metrics. An exception is D3PM's Diverse Edit Distance, which trades-off high novelty with low generation quality. USD3-CE, while performing well w.r.t. 1-gram metrics, does not compete with USD3-VLB and USD3 w.r.t. {2,3}-gram metrics, which indicates that *CE only* loss captures the least sequential information. Models with combined losses, USD3 and USD3*, achieve higher Train-to-Test ratio than pure CE or VLB based ones, suggesting that the combination of two losses can alleviate overfitting, i.e. "parroting" the training data. We find continuous-time USD3 to perform better than discrete-time counterparts models because its ability of denoising through any timestep.

Table 1: Conditional music gen. quality (3 samples avg.) on PIANO w.r.t. n-gram Hellinger, n-gram Prop. of Out., and Diverse Edit Dist. Also, Train-to-Test Ratio for on PIANO-P quantifies "parroting". **First** & **Second** shown in color (Detailed version is in Table 4).

	Method	n-gram Hellinger(↓) 1gram	2gram	3gram	n-gram Prop. of Out.(↓) 1gram	2gram	3gram	Div. Edit Dist. (↑)	3g-Prop. Ratio(↑)
Discrete-time	D3PM	0.398	0.530	0.591	0.120	0.253	0.379	**0.295**	2.221
	RDM	0.412	0.596	0.620	0.140	0.246	0.386	0.289	2.521
	USD3-CE	0.375	0.483	0.574	0.107	0.209	0.303	0.047	2.888
	USD3-VLB	0.379	0.464	**0.542**	0.117	0.184	0.273	**0.082**	2.863
	USD3	0.377	0.469	0.552	**0.107**	0.186	0.286	0.064	**3.083**
	USD3*	0.375	0.470	0.555	0.110	0.191	0.283	0.066	2.959
Continuous-time	SDDM	0.375	0.485	0.577	0.110	0.205	0.340	0.060	2.901
	τ-LDR-0	0.379	0.481	0.571	0.114	0.207	0.320	0.050	2.965
	SEDD	0.381	0.471	0.547	**0.105**	**0.182**	0.271	0.061	3.042
	USD3-CE	**0.373**	0.483	0.577	0.116	0.222	0.346	0.043	2.637
	USD3-VLB	0.376	**0.462**	0.549	0.120	0.186	0.295	0.061	2.904
	USD3	**0.374**	**0.462**	**0.543**	0.112	**0.184**	**0.270**	0.066	**3.083**
	USD3*	0.374	0.462	0.549	0.114	0.184	**0.271**	0.052	2.923

Table 2: Image generation quality w.r.t. IS and FID over 50,000 unconditional generated CIFAR10 samples. **First** & **Second** shown in color.

	Method	IS (↑)	FID (↓)
Discrete-time	D3PM	8.13	18.08
	RDM	7.16	24.23
	USD3-CE	9.02	12.64
	USD3*	**9.27**	**12.07**
	USD3	8.85	13.25
Cont.-time	SDDM	8.72	14.17
	τ-LDR	8.37	17.61
	SEDD	8.51	17.01
	USD3-CE	**9.23**	**11.97**
	USD3*	8.83	14.03
	USD3	8.97	13.09
	VQGAN Recons.	10.42	7.68

4.2 Image Generation

VQCIFAR10 Dataset. CIFAR10 contains $50,000$ training images in continuous values, which we convert to vectors from 64×512-dimensional quantization hash-code space with a pre-trained VQGAN Esser et al. [2020]. This conversion allows us to (1) evaluate our methods on *nominal* data by breaking the neighboring orders in the image space, and (2) select a closed-form stationary distribution $\boldsymbol{m}$. Details on VQCIFAR10 dataset are given in Appx. §A.18.2.

Metrics. After feeding the generated samples through the VQGAN decoder to obtain representations from the discretized space, we measure the Inception Score (IS) (↑) and Frechet Inception Distance (FID) (↓) against the original CIFAR10 dataset. Note that our training set, which are discretized images from VQGAN, achieves IS=10.42 and FID=7.68, which are optimistic limits for generation.

Results. Table 2 shows that our proposed approaches outperform existing baselines. Discrete-time D3PM and RDM fall short for the harder image generation task, whereas our simplified discrete-time loss boosts quality significantly. In continuous-time, τ-LDR with a similar loss to USD3 falls short due to its complicated generation process, whereas SDDM is most competitive among the baselines, although requires substantial compute resources while being limited to a specialized model architecture. USD3-CE achieves competitive IS and FID scores, which validates our finding (recall Eq. (15)) that CE loss is essential for diffusion loss minimization.

音符，模型需要生成其余 224 个音符。有趣的是，我们发现一些（124）评估序列共享 *same* 前 32 个音符作为一个或多个训练序列。这种重复为我们提供了一个独特的机会来研究我们所说的“鹦鹉学舌”——模型在生成过程中简单地重放精确的训练数据的现象。为此，我们创建了 PIANO-P，一个仅由这 124 个评估序列及其前 32 个音符匹配训练序列组成的子集。详细信息参见附录。 §A.18.1。

指标。我们对 PIANO 使用了许多新指标：2-gram 和 3-gram (1) 海林格距离 (↓) 和 (2) 异常值比例 (↓)，以及之前仅基于 Campbell 等人中的 *single* 音符出现而使用的 1-gram 对应指标。 [2022]; (3) 多样化编辑距离 (↑)，它解释了多个生成样本的创造力/新颖性。在 PIANO-P 上，我们还报告了 {1,2,3}-gram Hellinger 的 (4) 训练测试比 (↑) 以及异常值比例，它将生成的样本的加权距离与其匹配的训练与测试序列（共享相同的前 32 个音符）进行比较。这些比率量化了“鹦鹉学舌”的程度；比率越小，表示记忆（泛化）程度越高（越低）。所有指标的计算详细信息在 Appx 中给出。 §A.18.3。

结果。表 1 显示 USD3 及其变体在各个指标上的表现均优于基准方法。一个例外是 D3PM 的多样化编辑距离，它在高新颖性和低生成质量之间进行了权衡。 USD3-CE，同时表现良好。 1 克指标，不与 USD3-VLB 和 USD3 w.r.t 竞争。 {2,3}-gram 指标，表明 *CE only* 损失捕获最少的顺序信息。具有组合损失的模型 USD3 和 USD3* 比纯粹基于 CE 或 VLB 的模型实现了更高的训练测试比，这表明两种损失的组合可以减轻过度拟合，即“鹦鹉学舌”训练数据。我们发现连续时间 USD3 比离散时间对应模型表现更好，因为它能够通过任何时间步进行去噪。

表 1：条件音乐生成。 PIANO w.r.t. 的质量（平均 3 个样本） n-gram Hellinger、n-gram Prop. of Out. 和 Diverse Edit Dist.此外，PIANO-P 的训练测试比量化了“鹦鹉学舌”。第一和第二以颜色显示（详细版本见表 4）。

	Method	n-gram Hellinger(↓) 1gram	2gram	3gram	n-gram Prop. of Out.(↓) 1gram	2gram	3gram	Div. Edit Dist. (↑)	3g-Prop. Ratio(↑)
Discrete-time	D3PM	0.398	0.530	0.591	0.120	0.253	0.379	**0.295**	2.221
	RDM	0.412	0.596	0.620	0.140	0.246	0.386	0.289	2.521
	USD3-CE	0.375	0.483	0.574	0.107	0.209	0.303	0.047	2.888
	USD3-VLB	0.379	0.464	**0.542**	0.117	0.184	0.273	**0.082**	2.863
	USD3	0.377	0.469	0.552	**0.107**	0.186	0.286	0.064	**3.083**
	USD3*	0.375	0.470	0.555	0.110	0.191	0.283	0.066	2.959
Continuous-time	SDDM	0.375	0.485	0.577	0.110	0.205	0.340	0.060	2.901
	τ-LDR-0	0.379	0.481	0.571	0.114	0.207	0.320	0.050	2.965
	SEDD	0.381	0.471	0.547	**0.105**	**0.182**	0.271	0.061	3.042
	USD3-CE	**0.373**	0.483	0.577	0.116	0.222	0.346	0.043	2.637
	USD3-VLB	0.376	**0.462**	0.549	0.120	0.186	0.295	0.061	2.904
	USD3	**0.374**	**0.462**	**0.543**	0.112	**0.184**	**0.270**	0.066	**3.083**
	USD3*	0.374	0.462	0.549	0.114	0.184	**0.271**	0.052	2.923

表 2：图像生成质量IS 和 FID 超过 50,000 个无条件生成的 CIFAR10 样本。第一和第二以颜色显示。

	Method	IS (↑)	FID (↓)
Discrete-time	D3PM	8.13	18.08
	RDM	7.16	24.23
	USD3-CE	9.02	12.64
	USD3*	**9.27**	**12.07**
	USD3	8.85	13.25
Continuous-time	SDDM	8.72	14.17
	τ-LDR	8.37	17.61
	SEDD	8.51	17.01
	USD3-CE	**9.23**	**11.97**
	USD3*	8.83	14.03
	USD3	8.97	13.09
	VQGAN Recons.	10.42	7.68

4.2 图像生成

VQCIFAR10 数据集。 CIFAR10 包含 50, 000 个连续值训练图像，我们使用预训练的 VQGAN Esser 等人将其转换为 64×512 维量化哈希码空间的向量。 [2020]。这种转换使我们能够（1）通过破坏图像空间中的相邻顺序来评估我们在 *nominal* 数据上的方法，以及（2）选择封闭形式的平稳分布 $\boldsymbol{m}$。 Appx 中给出了 VQCIFAR10 数据集的详细信息。 §A.18.2。

指标。将生成的样本输入 VQGAN 解码器以获取离散空间的表示后，我们根据原始 CIFAR10 数据集测量初始分数 (IS) (↑) 和 Frechet 初始距离 (FID) (↓)。请注意，我们的训练集是来自 VQGAN 的离散图像，达到了 IS=10.42 和 FID=7.68，这是生成的乐观限制。

结果。表 2 显示我们提出的方法优于现有基线。离散时间 D3PM 和 RDM 无法满足较难的图像生成任务，而我们简化的离散时间损失却显着提高了质量。在连续时间内，损失与 USD3 类似的 τ-LDR 由于其复杂的生成过程而表现不佳，而 SDDM 在基线中最具竞争力，尽管需要大量计算资源，同时仅限于专门的模型架构。 USD3-CE 实现了具有竞争力的 IS 和 FID 分数，这验证了我们的发现（回想式（15）），即 CE 损失对于扩散损失最小化至关重要。

Additional results. We provide memory and runtimes of models in Table 7 in Appx. §A.19.4, generated images in Fig.2 in Appx. §A.19.5, and ablations of MCMC sampling in Appx. §A.19.6.

5 Conclusion

We introduce two fundamental contributions for both discrete-time and continuous-time diffusion for categorical data. First, we presented extensive *mathematical simplifications for the loss functions*, including exact closed-form derivations as well as novel easy-to-optimize approximations. Second, we established a *mathematical unification* of discrete-time and continuous-time discrete diffusion, enabling mutual benefits. Our proposed approach USD3 for discrete diffusion achieved significant improvements on established datasets across a suite of generation quality metrics.

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额外的结果。我们在 Appx 中的表 7 中提供了模型的内存和运行时间。 §A.19.4，生成的图像如图 Appx 中的图 2 所示。 §A.19.5，以及附录中 MCMC 采样的消融。 §A.19.6。

5 结论

我们介绍了分类数据的离散时间和连续时间扩散的两个基本贡献。首先，我们提出了广泛的 *mathematical simplifications for the loss functions*，包括精确的封闭形式推导以及新颖的易于优化的近似值。其次，我们建立了离散时间和连续时间离散扩散的*mathematical unification*，实现互惠互利。我们提出的离散扩散方法 USD3 在一系列发电质量指标的既定数据集上取得了显着改进。

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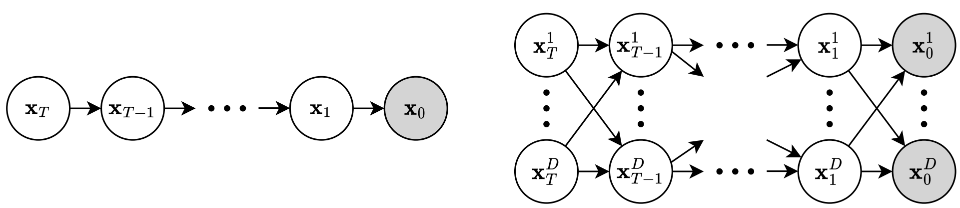
A Appendix

A.1 Variational Lower Bound Derivation

$$
\begin{aligned}
\log \int p_\theta(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T} &\geq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\big[\log p_\theta(\mathbf{x}_{0:T}) - \log q(\mathbf{x}_{1:T}|\mathbf{x}_0)\big] \\
&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\big[\log p_\theta(\mathbf{x}_T) + \sum_{t=1}^{T} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})}\big] \\
&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\big[\log p_\theta(\mathbf{x}_T) + \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \sum_{t=2}^{T} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0)}\big] \\
&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\big[\log p_\theta(\mathbf{x}_T) + \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \sum_{t=2}^{T} \log \left(\frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \cdot \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} \right)\big] \\
&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\big[\log p_\theta(\mathbf{x}_T) + \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{q(\mathbf{x}_T|\mathbf{x}_0)} + \sum_{t=2}^{T} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}\big] \\
&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\big[\log p_\theta(\mathbf{x}_0|\mathbf{x}_1) + \log \frac{p_\theta(\mathbf{x}_T)}{q(\mathbf{x}_T|\mathbf{x}_0)} - \sum_{t=2}^{T} \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}\big] \\
&= \underbrace{\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)}\big[\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)\big]}_{-\mathcal{L}_1(\theta)} - \sum_{t=2}^{T} \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}\big[D_{\mathrm{KL}}\big(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)\big)\big]}_{\mathcal{L}_t(\theta)} - \text{const.}
\end{aligned} \tag{28}
$$



(a) Latent graphical model for diffusion

(b) Generalize (a) to multi-element object

Figure 1: The graphical model diagram for both element-wise diffusion and multi-element diffusion.

A.2 Definition of nominal data

Nominal data, or categorical data, refers to data that can be categorized but not ranked or ordered. In mathematics, nominal data can be represented as a set of discrete categories or distinct labels without inherent ordering. As there is no ordering inside, operations of comparing categories is meaningless except checking equality. In the real world, most categories of data are nominal in nature, such as nationality, blood type, colors, among others.

A.3 Derivation of $q(x_{t-1}|x_t, x_0)$

First, let us define $\overline{Q}_{t|s} = Q_{s+1}...Q_t$. Note that $\overline{Q}_{t|0} = \overline{Q}_t$ and $\overline{Q}_{t|t-1} = Q_t$. Accordingly, we can derive the following two equalities.

$$
q(\mathbf{x}_t|\mathbf{x}_s) = \mathrm{Cat}(\mathbf{x}_t; \overline{Q}_{t|s}^\top \mathbf{x}_s) \tag{29}
$$

$$
\begin{aligned}
q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) &= \frac{q(\mathbf{x}_t|\mathbf{x}_s) q(\mathbf{x}_s|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} = \frac{\mathrm{Cat}(\mathbf{x}_t; \overline{Q}_{t|s}^\top \mathbf{x}_s) \mathrm{Cat}(\mathbf{x}_s; \overline{Q}_s^\top \mathbf{x}_0)}{\mathrm{Cat}(\mathbf{x}_t; \overline{Q}_t^\top \mathbf{x}_0)} \\
&= \frac{\mathbf{x}_s^\top \overline{Q}_{t|s} \mathbf{x}_t \cdot \mathbf{x}_s^\top \overline{Q}_s^\top \mathbf{x}_0}{\mathbf{x}_t^\top \overline{Q}_t^\top \mathbf{x}_0} = \mathbf{x}_s^\top \frac{\overline{Q}_{t|s} \mathbf{x}_t \odot \overline{Q}_s^\top \mathbf{x}_0}{\mathbf{x}_t^\top \overline{Q}_t^\top \mathbf{x}_0} = \mathrm{Cat}(\mathbf{x}_s; \frac{\overline{Q}_{t|s} \mathbf{x}_t \odot \overline{Q}_s^\top \mathbf{x}_0}{\mathbf{x}_t^\top \overline{Q}_t^\top \mathbf{x}_0})
\end{aligned} \tag{30}
$$

附录

A.1 变分下界推导

$$
\begin{aligned}
&\log \int p_\theta(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T} \geq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\big[\log p_\theta(\mathbf{x}_{0:T}) - \log q(\mathbf{x}_{1:T}|\mathbf{x}_0)\big] \\
&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\big[\log p_\theta(\mathbf{x}_T) + \sum_{t=1}^{T} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})}\big] \\
&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\big[\log p_\theta(\mathbf{x}_T) + \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \sum_{t=2}^{T} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0)}\big] \\
&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\big[\log p_\theta(\mathbf{x}_T) + \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \sum_{t=2}^{T} \log \left(\frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \cdot \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} \right)\big] \\
&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\big[\log p_\theta(\mathbf{x}_T) + \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{q(\mathbf{x}_T|\mathbf{x}_0)} + \sum_{t=2}^{T} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}\big] \\
&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\big[\log p_\theta(\mathbf{x}_0|\mathbf{x}_1) + \log \frac{p_\theta(\mathbf{x}_T)}{q(\mathbf{x}_T|\mathbf{x}_0)} - \sum_{t=2}^{T} \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}\big] \\
&= \underbrace{\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)}\big[\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)\big]}_{-\mathcal{L}_1(\theta)} - \sum_{t=2}^{T} \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}\big[D_{\mathrm{KL}}\big(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)\big)\big]}_{\mathcal{L}_t(\theta)} - \text{const.}
\end{aligned}
\tag{28}
$$

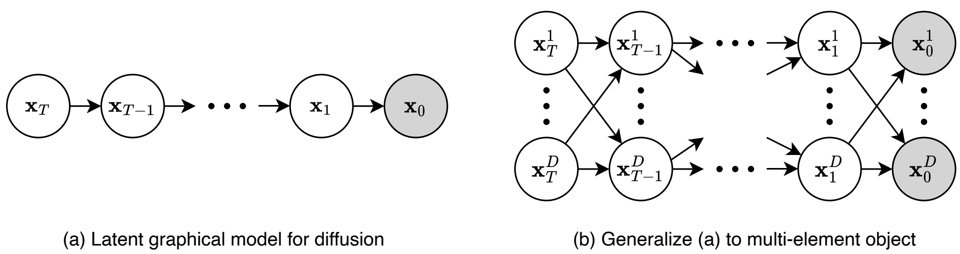


图 1：逐元素扩散和多元素扩散的图形模型图。

A.2 标称数据的定义

名义数据或分类数据是指可以分类但不能排名或排序的数据。在数学中，名义数据可以表示为一组离散类别或不同标签，没有固有的顺序。由于内部没有排序，所以除了检查相等性之外，比较类别的操作没有任何意义。在现实世界中，大多数类别的数据本质上都是名义上的，例如国籍、血型、肤色等。

A.3 $q(x_{t-1}|x_t, x_0)$的推导

首先，让我们定义$\overline{Q}_{t|s} = Q_{s+1}...Q_t$。请注意 $\overline{Q}_{t|0} = \overline{Q}_t$ 和 $\overline{Q}_{t|t-1} = Q_t$。据此，我们可以推导出以下两个等式。

$$
q(\mathbf{x}_t|\mathbf{x}_s) = \mathrm{Cat}(\mathbf{x}_t; \overline{Q}_{t|s}^\top \mathbf{x}_s) \tag{29}
$$

$$
\begin{aligned}
q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) &= \frac{q(\mathbf{x}_t|\mathbf{x}_s)q(\mathbf{x}_s|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} = \frac{\mathrm{Cat}(\mathbf{x}_t; \overline{Q}_{t|s}^\top \mathbf{x}_s)\mathrm{Cat}(\mathbf{x}_s; \overline{Q}_s^\top \mathbf{x}_0)}{\mathrm{Cat}(\mathbf{x}_t; \overline{Q}_t^\top \mathbf{x}_0)} \\
&= \frac{\mathbf{x}_s^\top \overline{Q}_{t|s} \mathbf{x}_t \cdot \mathbf{x}_s^\top \overline{Q}_s^\top \mathbf{x}_0}{\mathbf{x}_t^\top \overline{Q}_t^\top \mathbf{x}_0} = \mathbf{x}_s^\top \frac{\overline{Q}_{t|s} \mathbf{x}_t \odot \overline{Q}_s^\top \mathbf{x}_0}{\mathbf{x}_t^\top \overline{Q}_t^\top \mathbf{x}_0} = \mathrm{Cat}(\mathbf{x}_s; \frac{\overline{Q}_{t|s} \mathbf{x}_t \odot \overline{Q}_s^\top \mathbf{x}_0}{\mathbf{x}_t^\top \overline{Q}_t^\top \mathbf{x}_0})
\end{aligned}
\tag{30}
$$

Using the formulation of Q_t in Eq. (7), $\overline{Q}_{t|s}$ can be written as

$$\overline{Q}_{t|s} = \overline{\alpha}_{t|s} I + (1 - \overline{\alpha}_{t|s}) \mathbf{1} \boldsymbol{m}^\top \tag{31}$$

where $\overline{\alpha}_{t|s} = \prod_{i=s+1}^{t} \alpha_i$. Note that $\overline{\alpha}_t = \overline{\alpha}_{t|s} \overline{\alpha}_s$.

Next, we can simplify Eq. (5) using the above formulations as

$$\begin{aligned} \frac{\overline{Q}_{t|s} \mathbf{x}_t \odot \overline{Q}_s^\top \mathbf{x}_0}{\mathbf{x}_t^\top \overline{Q}_t^\top \mathbf{x}_0} &= \frac{\big(\overline{\alpha}_{t|s} \mathbf{x}_t + (1 - \overline{\alpha}_{t|s}) \langle \boldsymbol{m}, \mathbf{x}_t \rangle \mathbf{1}\big) \odot \big(\overline{\alpha}_s \mathbf{x}_0 + (1 - \overline{\alpha}_s) \boldsymbol{m}\big)}{\overline{\alpha}_t \langle \mathbf{x}_t, \mathbf{x}_0 \rangle + (1 - \overline{\alpha}_t) \langle \boldsymbol{m}, \mathbf{x}_t \rangle} \\ &= \frac{\overline{\alpha}_t \mathbf{x}_t \odot \mathbf{x}_0 + (\overline{\alpha}_s - \overline{\alpha}_t) \langle \boldsymbol{m}, \mathbf{x}_t \rangle \mathbf{x}_0 + (\overline{\alpha}_{t|s} - \overline{\alpha}_t) \mathbf{x}_t \odot \boldsymbol{m} + (1 - \overline{\alpha}_s)(1 - \overline{\alpha}_{t|s}) \langle \boldsymbol{m}, \mathbf{x}_t \rangle \boldsymbol{m}}{\overline{\alpha}_t \langle \mathbf{x}_t, \mathbf{x}_0 \rangle + (1 - \overline{\alpha}_t) \langle \boldsymbol{m}, \mathbf{x}_t \rangle} \end{aligned} \tag{32}$$

We can simplify the above equation further by considering two cases: (1) $\mathbf{x}_t = \mathbf{x}_0$ and (2) $\mathbf{x}_t \neq \mathbf{x}_0$.

(Case 1) When $\mathbf{x}_t = \mathbf{x}_0$, using the fact that both $\mathbf{x}_t$ and $\mathbf{x}_0$ are one-hot encoded, we observe

$$\begin{aligned} \text{Eq. (32)} &= \frac{\overline{\alpha}_t \mathbf{x}_t + (\overline{\alpha}_s - \overline{\alpha}_t) \langle \boldsymbol{m}, \mathbf{x}_t \rangle \mathbf{x}_t + (\overline{\alpha}_{t|s} - \overline{\alpha}_t) \langle \boldsymbol{m}, \mathbf{x}_t \rangle \mathbf{x}_t + (1 - \overline{\alpha}_s)(1 - \overline{\alpha}_{t|s}) \langle \boldsymbol{m}, \mathbf{x}_t \rangle \boldsymbol{m}}{\overline{\alpha}_t + (1 - \overline{\alpha}_t) \langle \boldsymbol{m}, \mathbf{x}_t \rangle} \\ &= \frac{\overline{\alpha}_t + (\overline{\alpha}_{t|s} + \overline{\alpha}_s - 2\overline{\alpha}_t) \langle \boldsymbol{m}, \mathbf{x}_t \rangle}{\overline{\alpha}_t + (1 - \overline{\alpha}_t) \langle \boldsymbol{m}, \mathbf{x}_t \rangle} \mathbf{x}_t + \frac{(1 - \overline{\alpha}_s)(1 - \overline{\alpha}_{t|s}) \langle \boldsymbol{m}, \mathbf{x}_t \rangle}{\overline{\alpha}_t + (1 - \overline{\alpha}_t) \langle \boldsymbol{m}, \mathbf{x}_t \rangle} \boldsymbol{m} \\ &= (1 - \lambda_{t|s}) \cdot \mathbf{x}_t + \lambda_{t|s} \cdot \boldsymbol{m} \ , \end{aligned} \tag{33}$$

where $\lambda_{t|s}$ is defined as

$$\lambda_{t|s} = \frac{(1 - \overline{\alpha}_s)(1 - \overline{\alpha}_{t|s}) \langle \boldsymbol{m}, \mathbf{x}_t \rangle}{\overline{\alpha}_t + (1 - \overline{\alpha}_t) \langle \boldsymbol{m}, \mathbf{x}_t \rangle} \ . \tag{34}$$

(Case 2) When $\mathbf{x}_t \neq \mathbf{x}_0$, we can similarly derive

$$\begin{aligned} \text{Eq. (32)} &= \frac{(\overline{\alpha}_s - \overline{\alpha}_t) \langle \boldsymbol{m}, \mathbf{x}_t \rangle \mathbf{x}_0 + (\overline{\alpha}_{t|s} - \overline{\alpha}_t) \langle \boldsymbol{m}, \mathbf{x}_t \rangle \mathbf{x}_t + (1 - \overline{\alpha}_s)(1 - \overline{\alpha}_{t|s}) \langle \boldsymbol{m}, \mathbf{x}_t \rangle \boldsymbol{m}}{(1 - \overline{\alpha}_t) \langle \boldsymbol{m}, \mathbf{x}_t \rangle} \\ &= \frac{\overline{\alpha}_s - \overline{\alpha}_t}{1 - \overline{\alpha}_t} \mathbf{x}_0 + \frac{1 - \overline{\alpha}_s}{1 - \overline{\alpha}_t} \overline{\alpha}_{t|s} \mathbf{x}_t + \frac{1 - \overline{\alpha}_s}{1 - \overline{\alpha}_t} (1 - \overline{\alpha}_{t|s}) \boldsymbol{m} \\ &= (1 - \mu_{t|s}) \cdot \mathbf{x}_0 + \mu_{t|s} \overline{\alpha}_{t|s} \cdot \mathbf{x}_t + \mu_{t|s} (1 - \overline{\alpha}_{t|s}) \cdot \boldsymbol{m} \ , \end{aligned} \tag{35}$$

where $\mu_{t|s}$ is defined as

$$\mu_{t|s} = \frac{1 - \overline{\alpha}_s}{1 - \overline{\alpha}_t} \tag{36}$$

Combining the results from both cases together, we can write $q(\mathbf{x}_s | \mathbf{x}_t, \mathbf{x}_0)$ in the following form.

$$q(\mathbf{x}_s | \mathbf{x}_t, \mathbf{x}_0) = \begin{cases} \text{Cat}\Big(\mathbf{x}_s; \ (1 - \lambda_{t|s}) \cdot \mathbf{x}_t + \lambda_{t|s} \cdot \boldsymbol{m}\Big) & \text{when } \mathbf{x}_t = \mathbf{x}_0 \\ \text{Cat}\Big(\mathbf{x}_s; \ (1 - \mu_{t|s}) \cdot \mathbf{x}_0 + \mu_{t|s} \overline{\alpha}_{t|s} \cdot \mathbf{x}_t + \mu_{t|s} (1 - \overline{\alpha}_{t|s}) \cdot \boldsymbol{m}\Big) & \text{when } \mathbf{x}_t \neq \mathbf{x}_0 \end{cases} \tag{37}$$

A.4 Parameterization of $p_\theta(x_s | x_t)$

We first provide the formulation as follows.

$$p_\theta(\mathbf{x}_s | \mathbf{x}_t) = \sum_{\mathbf{x}_0} q(\mathbf{x}_s | \mathbf{x}_t, \mathbf{x}_0) p_\theta(\mathbf{x}_0 | \mathbf{x}_t) \tag{38}$$

Using $q(\mathbf{x}_s | \mathbf{x}_t, \mathbf{x}_0)$ in Eq. (9) and $p_\theta(\mathbf{x}_0 | \mathbf{x}_t) = \text{Cat}(\mathbf{x}_0; f_t^\theta(\mathbf{x}_t))$, we can expand it as

使用方程中的 Q_t 公式。(7)、$Q_{t|s}$可写为

$$\overline{Q}_{t|s} = \overline{\alpha}_{t|s} I + (1-\overline{\alpha}_{t|s})\mathbf{1}\boldsymbol{m}^\top \tag{31}$$

其中$\alpha_{\overline{t|s}} = \prod_{i=s+1}^{t} \alpha_i$。请注意$\alpha_t \equiv \overline{\alpha}_{t|s}\overline{\alpha}_s$。

接下来，我们可以简化方程。(5) 将上述公式代为

$$\begin{aligned}\frac{\overline{Q}_{t|s}\mathbf{x}_t \odot \overline{Q}_s^\top \mathbf{x}_0}{\mathbf{x}_t^\top \overline{Q}_t^\top \mathbf{x}_0} &= \frac{\big(\overline{\alpha}_{t|s}\mathbf{x}_t + (1-\overline{\alpha}_{t|s})\langle \boldsymbol{m}, \mathbf{x}_t\rangle \mathbf{1}\big) \odot \big(\overline{\alpha}_s \mathbf{x}_0 + (1-\overline{\alpha}_s)\boldsymbol{m}\big)}{\overline{\alpha}_t \langle \mathbf{x}_t, \mathbf{x}_0\rangle + (1-\overline{\alpha}_t)\langle \boldsymbol{m}, \mathbf{x}_t\rangle} \\ &= \frac{\overline{\alpha}_t \mathbf{x}_t \odot \mathbf{x}_0 + (\overline{\alpha}_s - \overline{\alpha}_t)\langle \boldsymbol{m}, \mathbf{x}_t\rangle \mathbf{x}_0 + (\overline{\alpha}_{t|s} - \overline{\alpha}_t)\mathbf{x}_t \odot \boldsymbol{m} + (1-\overline{\alpha}_s)(1-\overline{\alpha}_{t|s})\langle \boldsymbol{m}, \mathbf{x}_t\rangle \boldsymbol{m}}{\overline{\alpha}_t \langle \mathbf{x}_t, \mathbf{x}_0\rangle + (1-\overline{\alpha}_t)\langle \boldsymbol{m}, \mathbf{x}_t\rangle}\end{aligned} \tag{32}$$

我们可以通过考虑两种情况进一步简化上面的方程：（1）$\mathbf{x}_t = \mathbf{x}_0$和（2）$\mathbf{x}_t \neq \mathbf{x}_0$。（情况1）当 $\mathbf{x}_t = \mathbf{x}_0$ 时，利用 $\mathbf{x}_t$ 和 $\mathbf{x}_0$ 都是 one-hot 编码这一事实，我们观察到

等式。(32)

$$\begin{aligned} &= \frac{\overline{\alpha}_t \mathbf{x}_t + (\overline{\alpha}_s - \overline{\alpha}_t)\langle \boldsymbol{m}, \mathbf{x}_t\rangle \mathbf{x}_t + (\overline{\alpha}_{t|s} - \overline{\alpha}_t)\langle \boldsymbol{m}, \mathbf{x}_t\rangle \mathbf{x}_t + (1-\overline{\alpha}_s)(1-\overline{\alpha}_{t|s})\langle \boldsymbol{m}, \mathbf{x}_t\rangle \boldsymbol{m}}{\overline{\alpha}_t + (1-\overline{\alpha}_t)\langle \boldsymbol{m}, \mathbf{x}_t\rangle} \\ &= \frac{\overline{\alpha}_t + (\overline{\alpha}_{t|s} + \overline{\alpha}_s - 2\overline{\alpha}_t)\langle \boldsymbol{m}, \mathbf{x}_t\rangle}{\overline{\alpha}_t + (1-\overline{\alpha}_t)\langle \boldsymbol{m}, \mathbf{x}_t\rangle}\mathbf{x}_t + \frac{(1-\overline{\alpha}_s)(1-\overline{\alpha}_{t|s})\langle \boldsymbol{m}, \mathbf{x}_t\rangle}{\overline{\alpha}_t + (1-\overline{\alpha}_t)\langle \boldsymbol{m}, \mathbf{x}_t\rangle}\boldsymbol{m} \\ &= (1-\lambda_{t|s})\cdot \mathbf{x}_t + \lambda_{t|s}\cdot \boldsymbol{m}\ , \end{aligned} \tag{33}$$

其中 $\lambda_{t|s}$ 定义为

$$\lambda_{t|s} = \frac{(1-\overline{\alpha}_s)(1-\overline{\alpha}_{t|s})\langle \boldsymbol{m}, \mathbf{x}_t\rangle}{\overline{\alpha}_t + (1-\overline{\alpha}_t)\langle \boldsymbol{m}, \mathbf{x}_t\rangle}\ . \tag{34}$$

(情况2) 当$\mathbf{x}_t \neq \mathbf{x}_0$时，我们可以类似地推导出

$$\begin{aligned}\text{Eq. (32)} &= \frac{(\overline{\alpha}_s - \overline{\alpha}_t)\langle \boldsymbol{m}, \mathbf{x}_t\rangle \mathbf{x}_0 + (\overline{\alpha}_{t|s} - \overline{\alpha}_t)\langle \boldsymbol{m}, \mathbf{x}_t\rangle \mathbf{x}_t + (1-\overline{\alpha}_s)(1-\overline{\alpha}_{t|s})\langle \boldsymbol{m}, \mathbf{x}_t\rangle \boldsymbol{m}}{(1-\overline{\alpha}_t)\langle \boldsymbol{m}, \mathbf{x}_t\rangle} \\ &= \frac{\overline{\alpha}_s - \overline{\alpha}_t}{1-\overline{\alpha}_t}\mathbf{x}_0 + \frac{1-\overline{\alpha}_s}{1-\overline{\alpha}_t}\overline{\alpha}_{t|s}\mathbf{x}_t + \frac{1-\overline{\alpha}_s}{1-\overline{\alpha}_t}(1-\overline{\alpha}_{t|s})\boldsymbol{m} \\ &= (1-\mu_{t|s})\cdot \mathbf{x}_0 + \mu_{t|s}\overline{\alpha}_{t|s}\cdot \mathbf{x}_t + \mu_{t|s}(1-\overline{\alpha}_{t|s})\cdot \boldsymbol{m}\ , \end{aligned} \tag{35}$$

其中 $\mu_{t|s}$ 定义为

$$\mu_{t|s} = \frac{1-\overline{\alpha}_s}{1-\overline{\alpha}_t} \tag{36}$$

将两种情况的结果结合在一起，我们可以将 $q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$ 写成以下形式。

$$q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) = \begin{cases} \text{Cat}\Big(\mathbf{x}_s;\ (1-\lambda_{t|s})\cdot \mathbf{x}_t + \lambda_{t|s}\cdot \boldsymbol{m}\Big) & \text{when } \mathbf{x}_t = \mathbf{x}_0 \\ \text{Cat}\Big(\mathbf{x}_s;\ (1-\mu_{t|s})\cdot \mathbf{x}_0 + \mu_{t|s}\overline{\alpha}_{t|s}\cdot \mathbf{x}_t + \mu_{t|s}(1-\overline{\alpha}_{t|s})\cdot \boldsymbol{m}\Big) & \text{when } \mathbf{x}_t \neq \mathbf{x}_0 \end{cases} \tag{37}$$

A.4 $p_\theta(x_s|x_t)$ 的参数化

我们首先提供如下公式。

$$p_\theta(\mathbf{x}_s|\mathbf{x}_t) = \sum_{\mathbf{x}_0} q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)p_\theta(\mathbf{x}_0|\mathbf{x}_t) \tag{38}$$

在等式中使用 $q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$ (9) 和 $p_\theta(\mathbf{x}_0|\mathbf{x}_t) = \text{Cat}(\mathbf{x}_0;\ f_t^\theta(\mathbf{x}_t))$，我们可以将其展开为

$$
\begin{aligned}
p_\theta(\mathbf{x}_s|\mathbf{x}_t) &= q(\mathbf{x}_s|\mathbf{x}_t,\mathbf{x}_t)p_\theta(\mathbf{x}_t|\mathbf{x}_t) + \sum_{\mathbf{x}\neq\mathbf{x}_t} q(\mathbf{x}_s|\mathbf{x}_t,\mathbf{x})p_\theta(\mathbf{x}|\mathbf{x}_t) \\
=&\mathbf{x}_s^\top\Big((1-\lambda_{t|s})\mathbf{x}_t + \lambda_{t|s}\boldsymbol{m}\Big)\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t) \\
&+ \sum_{\mathbf{x}\neq\mathbf{x}_t} \mathbf{x}_s^T\Big((1-\mu_{t|s})\mathbf{x} + \mu_{t|s}\overline{\alpha}_{t|s}\mathbf{x}_t + \mu_{t|s}(1-\overline{\alpha}_{t|s})\boldsymbol{m}\Big)\mathbf{x}^\top f_t^\theta(\mathbf{x}_t) \\
=&\mathbf{x}_s^\top\Big[((1-\lambda_{t|s})\mathbf{x}_t + \lambda_{t|s}\boldsymbol{m})\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t) + (1-\mu_{t|s})(\sum_{\mathbf{x}\neq\mathbf{x}_t}\mathbf{x}\mathbf{x}^\top)f_t^\theta(\mathbf{x}_t) \\
&+ (\mu_{t|s}\overline{\alpha}_{t|s}\mathbf{x}_t + \mu_{t|s}(1-\overline{\alpha}_{t|s})\boldsymbol{m})(\sum_{\mathbf{x}\neq\mathbf{x}_t}\mathbf{x})^\top f_t^\theta(\mathbf{x}_t)\Big] \\
=&\mathbf{x}_s^\top\Big[((1-\lambda_{t|s})\mathbf{x}_t + \lambda_{t|s}\boldsymbol{m})\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t) + (1-\mu_{t|s})(I-\mathbf{x}_t\mathbf{x}_t^\top)f_t^\theta(\mathbf{x}_t) \\
&+ (\mu_{t|s}\overline{\alpha}_{t|s}\mathbf{x}_t + \mu_{t|s}(1-\overline{\alpha}_{t|s})\boldsymbol{m})(\mathbf{1}-\mathbf{x}_t)^\top f_t^\theta(\mathbf{x}_t)\Big] \\
=&\mathbf{x}_s^\top\Big[(1-\lambda_{t|s})\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t)\mathbf{x}_t + \lambda_{t|s}\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t)\boldsymbol{m} + (1-\mu_{t|s})(f_t^\theta(\mathbf{x}_t) - \mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t)\mathbf{x}_t) \\
&+ (\mu_{t|s}\overline{\alpha}_{t|s}\mathbf{x}_t + \mu_{t|s}(1-\overline{\alpha}_{t|s})\boldsymbol{m})(1-\mathbf{x}_t^T f_t^\theta(\mathbf{x}_t))\Big] \\
=&\mathbf{x}_s^\top\Big[(1-\mu_{t|s})\cdot f_t^\theta(\mathbf{x}_t) + \Big((\mu_{t|s}-\lambda_{t|s}-\mu_{t|s}\overline{\alpha}_{t|s})\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t) + \mu_{t|s}\overline{\alpha}_{t|s}\Big)\cdot\mathbf{x}_t \\
&+ \Big(-(\mu_{t|s}-\lambda_{t|s}-\mu_{t|s}\overline{\alpha}_{t|s})\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t) + \mu_{t|s}(1-\overline{\alpha}_{t|s})\Big)\cdot\boldsymbol{m}\Big] \\
=&\quad \text{Cat}\Big(\mathbf{x}_s;\ (1-\mu_{t|s})\cdot f_t^\theta(\mathbf{x}_t) + (\mu_{t|s}\overline{\alpha}_{t|s} + \gamma_{t|s}^\theta)\cdot\mathbf{x}_t + (\mu_{t|s}(1-\overline{\alpha}_{t|s}) - \gamma_{t|s}^\theta)\cdot\boldsymbol{m}\Big)
\end{aligned} \tag{39}
$$

A.5 Proof of Proposition 3

As $q(\mathbf{x}_s|\mathbf{x}_t,\mathbf{x}_0)$ has two different categorical distributions based on whether $\mathbf{x}_t$ is equal to $\mathbf{x}_0$, we prove the Proposition 3 based on two cases.

Case 1: $\mathbf{x}_t = \mathbf{x}_0$. In this case, let $h_t^\theta := f_t^\theta(\boldsymbol{x}_t) - \boldsymbol{x}_0$, now replace f_t^θ with g_t^θ in Eq. (2.2) and Eq. (10), we get

$$
\gamma_{t|s}^\theta = (\mu_{t|s}-\lambda_{t|s}-\mu_{t|s}\overline{\alpha}_{t|s})\langle h_t^\theta + \boldsymbol{x}_0, \boldsymbol{x}_t\rangle = (\mu_{t|s}-\lambda_{t|s}-\mu_{t|s}\overline{\alpha}_{t|s})\langle h_t^\theta, \boldsymbol{x}_t\rangle + (\mu_{t|s}-\lambda_{t|s}-\mu_{t|s}\overline{\alpha}_{t|s}) \tag{40}
$$

Taking the formulation into Eq. (10) and cancel out terms, we can simplify $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ as

$$
\begin{aligned}
p_\theta(\mathbf{x}_s|\mathbf{x}_t) &= (1-\mu_{t|s})h_t^\theta + (1-\lambda_{t|s})\boldsymbol{x}_t + \lambda_{t|s}\boldsymbol{m} + (\mu_{t|s}-\lambda_{t|s}-\mu_{t|s}\overline{\alpha}_{t|s})\langle h_t^\theta, \boldsymbol{x}_t\rangle(\boldsymbol{x}_t - \boldsymbol{m}) \\
&= q(\mathbf{x}_s|\mathbf{x}_t,\mathbf{x}_0) + (1-\mu_{t|s})\Big[h_t^\theta + \frac{\mu_{t|s}-\lambda_{t|s}-\mu_{t|s}\overline{\alpha}_{t|s}}{1-\mu_{t|s}}\langle h_t^\theta, \boldsymbol{x}_t\rangle(\boldsymbol{x}_t-\boldsymbol{m})\Big] \\
&= q(\mathbf{x}_s|\mathbf{x}_t,\mathbf{x}_0) + (1-\mu_{t|s})\Big[h_t^\theta + \phi_{t|s}\langle h_t^\theta, \boldsymbol{x}_t\rangle(\boldsymbol{x}_t-\boldsymbol{m})\Big]
\end{aligned} \tag{41}
$$

Case 2: $\mathbf{x}_t \neq \mathbf{x}_0$**.** Based on Eq. (9) and Eq. (10), it's easy to find that

$$
\begin{aligned}
p_\theta(\mathbf{x}_s|\mathbf{x}_t) &= (1-\mu_{t|s})(f_t^\theta(\boldsymbol{x}_t) - \boldsymbol{x}_0) + \gamma_{t|s}^\theta(\boldsymbol{x}_t-\boldsymbol{m}) + (1-\mu_{t|s})\boldsymbol{x}_0 + \mu_{t|s}\bar{\alpha}_{t|s}\boldsymbol{x}_t + \mu_{t|s}(1-\bar{\alpha}_{t|s})\boldsymbol{m} \\
&= (1-\mu_{t|s})(f_t^\theta(\boldsymbol{x}_t) - \boldsymbol{x}_0) + \gamma_{t|s}^\theta(\boldsymbol{x}_t-\boldsymbol{m}) + q(\mathbf{x}_s|\mathbf{x}_t,\mathbf{x}_0) \\
&= q(\mathbf{x}_s|\mathbf{x}_t,\mathbf{x}_0) + (1-\mu_{t|s})\Big[f_t^\theta(\boldsymbol{x}_t) - \boldsymbol{x}_0 + \frac{\gamma_{t|s}^\theta}{(1-\mu_{t|s})}(\boldsymbol{x}_t-\boldsymbol{m})\Big]
\end{aligned} \tag{42}
$$

Taking the definition of $\gamma_{t|s}^\theta$ in Eq. (2.2) and using the fact $< \boldsymbol{x}_0, \boldsymbol{x}_t >= 0$ we get the formulation in proposition 3.

For both cases, we proved that the formulation in proposition 3 is correct.

$$
\begin{aligned}
p_\theta(\mathbf{x}_s|\mathbf{x}_t) = & q(\mathbf{x}_s|\mathbf{x}_t,\mathbf{x}_t)p_\theta(\mathbf{x}_t|\mathbf{x}_t) + \sum_{\mathbf{x}\neq\mathbf{x}_t} q(\mathbf{x}_s|\mathbf{x}_t,\mathbf{x})p_\theta(\mathbf{x}|\mathbf{x}_t) \\
=&\mathbf{x}_s^\top\Big((1-\lambda_{t|s})\mathbf{x}_t+\lambda_{t|s}\boldsymbol{m}\Big)\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t) \\
&+\sum_{\mathbf{x}\neq\mathbf{x}_t}\mathbf{x}_s^T\Big((1-\mu_{t|s})\mathbf{x}+\mu_{t|s}\overline{\alpha}_{t|s}\mathbf{x}_t+\mu_{t|s}(1-\overline{\alpha}_{t|s})\boldsymbol{m}\Big)\mathbf{x}^\top f_t^\theta(\mathbf{x}_t) \\
=&\mathbf{x}_s^\top\Big[\big((1-\lambda_{t|s})\mathbf{x}_t+\lambda_{t|s}\boldsymbol{m}\big)\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t)+(1-\mu_{t|s})(\sum_{\mathbf{x}\neq\mathbf{x}_t}\mathbf{x}\mathbf{x}^\top)f_t^\theta(\mathbf{x}_t) \\
&+(\mu_{t|s}\overline{\alpha}_{t|s}\mathbf{x}_t+\mu_{t|s}(1-\overline{\alpha}_{t|s})\boldsymbol{m})(\sum_{\mathbf{x}\neq\mathbf{x}_t}\mathbf{x})^\top f_t^\theta(\mathbf{x}_t)\Big] \\
=&\mathbf{x}_s^\top\Big[\big((1-\lambda_{t|s})\mathbf{x}_t+\lambda_{t|s}\boldsymbol{m}\big)\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t)+(1-\mu_{t|s})(I-\mathbf{x}_t\mathbf{x}_t^\top)f_t^\theta(\mathbf{x}_t) \\
&+(\mu_{t|s}\overline{\alpha}_{t|s}\mathbf{x}_t+\mu_{t|s}(1-\overline{\alpha}_{t|s})\boldsymbol{m})(\mathbf{1}-\mathbf{x}_t)^\top f_t^\theta(\mathbf{x}_t)\Big] \\
=&\mathbf{x}_s^\top\Big[(1-\lambda_{t|s})\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t)\mathbf{x}_t+\lambda_{t|s}\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t)\boldsymbol{m}+(1-\mu_{t|s})(f_t^\theta(\mathbf{x}_t)-\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t)\mathbf{x}_t) \\
&+(\mu_{t|s}\overline{\alpha}_{t|s}\mathbf{x}_t+\mu_{t|s}(1-\overline{\alpha}_{t|s})\boldsymbol{m})(1-\mathbf{x}_t^T f_t^\theta(\mathbf{x}_t))\Big] \\
=&\mathbf{x}_s^\top\Big[(1-\mu_{t|s})\cdot f_t^\theta(\mathbf{x}_t)+\Big((\mu_{t|s}-\lambda_{t|s}-\mu_{t|s}\overline{\alpha}_{t|s})\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t)+\mu_{t|s}\overline{\alpha}_{t|s}\Big)\cdot\mathbf{x}_t \\
&+\Big(-(\mu_{t|s}-\lambda_{t|s}-\mu_{t|s}\overline{\alpha}_{t|s})\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t)+\mu_{t|s}(1-\overline{\alpha}_{t|s})\Big)\cdot\boldsymbol{m}\Big] \\
=&\quad \mathrm{Cat}\Big(\mathbf{x}_s;\ (1-\mu_{t|s})\cdot f_t^\theta(\mathbf{x}_t)+(\mu_{t|s}\overline{\alpha}_{t|s}+\gamma_{t|s}^\theta)\cdot\mathbf{x}_t+(\mu_{t|s}(1-\overline{\alpha}_{t|s})-\gamma_{t|s}^\theta)\cdot\boldsymbol{m}\Big)
\end{aligned} \tag{39}
$$

A.5 命题3的证明

由于根据$\mathbf{x}_t$是否等于$\mathbf{x}_0$，$q(\mathbf{x}_s|\mathbf{x}_t,\mathbf{x}_0)$具有两种不同的分类分布，因此我们基于两种情况证明命题3。

情况1：$\mathbf{x}_t = \mathbf{x}_0$。在这种情况下，让 $h_t^\theta := f_t^\theta(\boldsymbol{x}_t) - \boldsymbol{x}_0$，现在将等式中的 f_t^θ 替换为 g_t^θ。(2.2) 和等式。（10），我们得到

$$
\gamma_{t|s}^\theta = (\mu_{t|s}-\lambda_{t|s}-\mu_{t|s}\overline{\alpha}_{t|s})\langle h_t^\theta+\boldsymbol{x}_0,\boldsymbol{x}_t\rangle = (\mu_{t|s}-\lambda_{t|s}-\mu_{t|s}\overline{\alpha}_{t|s})\langle h_t^\theta,\boldsymbol{x}_t\rangle + (\mu_{t|s}-\lambda_{t|s}-\mu_{t|s}\overline{\alpha}_{t|s}) \tag{40}
$$

采取公式 化成方程。(10) 并取消项，我们可以简化 $p_\theta(\mathbf{x}$ $_s|\mathbf{x}_t)$ 作为

$$
\begin{aligned}
p_\theta(\mathbf{x}_s|\mathbf{x}_t) &= (1-\mu_{t|s})h_t^\theta+(1-\lambda_{t|s})\boldsymbol{x}_t+\lambda_{t|s}\boldsymbol{m}+(\mu_{t|s}-\lambda_{t|s}-\mu_{t|s}\overline{\alpha}_{t|s})\langle h_t^\theta,\boldsymbol{x}_t\rangle(\boldsymbol{x}_t-\boldsymbol{m}) \\
&= q(\mathbf{x}_s|\mathbf{x}_t,\mathbf{x}_0)+(1-\mu_{t|s})\Big[h_t^\theta+\frac{\mu_{t|s}-\lambda_{t|s}-\mu_{t|s}\overline{\alpha}_{t|s}}{1-\mu_{t|s}}\langle h_t^\theta,\boldsymbol{x}_t\rangle(\boldsymbol{x}_t-\boldsymbol{m})\Big] \\
&= q(\mathbf{x}_s|\mathbf{x}_t,\mathbf{x}_0)+(1-\mu_{t|s})\Big[h_t^\theta+\phi_{t|s}\langle h_t^\theta,\boldsymbol{x}_t\rangle(\boldsymbol{x}_t-\boldsymbol{m})\Big]
\end{aligned} \tag{41}
$$

情况2：$\mathbf{x}_t \neq \mathbf{x}_0$。基于等式。(9) 和等式。(10) 容易发现

$$
\begin{aligned}
p_\theta(\mathbf{x}_s|\mathbf{x}_t) &= (1-\mu_{t|s})(f_t^\theta(\boldsymbol{x}_t)-\boldsymbol{x}_0)+\gamma_{t|s}^\theta(\boldsymbol{x}_t-\boldsymbol{m})+(1-\mu_{t|s})\boldsymbol{x}_0+\mu_{t|s}\bar{\alpha}_{t|s}\boldsymbol{x}_t+\mu_{t|s}(1-\bar{\alpha}_{t|s})\boldsymbol{m} \\
&= (1-\mu_{t|s})(f_t^\theta(\boldsymbol{x}_t)-\boldsymbol{x}_0)+\gamma_{t|s}^\theta(\boldsymbol{x}_t-\boldsymbol{m})+q(\mathbf{x}_s|\mathbf{x}_t,\mathbf{x}_0) \\
&= q(\mathbf{x}_s|\mathbf{x}_t,\mathbf{x}_0)+(1-\mu_{t|s})\Big[f_t^\theta(\boldsymbol{x}_t)-\boldsymbol{x}_0+\frac{\gamma_{t|s}^\theta}{(1-\mu_{t|s})}(\boldsymbol{x}_t-\boldsymbol{m})\Big]
\end{aligned} \tag{42}
$$

采用等式中 $\gamma_{t|s}^\theta$ 的定义。(2.2) 并使用事实 $<\boldsymbol{x}_0,\boldsymbol{x}_t>=0$ 我们得到命题 3 中的公式。

对于这两种情况，我们证明了命题 3 中的表述是正确的。

A.6 Proof of KL divergence approximation

Assume that $p(\boldsymbol{x}) = q(\boldsymbol{x}) + \triangle p(\boldsymbol{x})$, and both $p(\boldsymbol{x})$ and $q(\boldsymbol{x})$ are valid categorical distributions. Then

$$D_{\mathrm{KL}}\big(q(\boldsymbol{x})\|p(\boldsymbol{x})\big) = -\sum_{\boldsymbol{x}} q(\boldsymbol{x}) \log \frac{p(\boldsymbol{x})}{q(\boldsymbol{x})} = -\sum_{\boldsymbol{x}} q(\boldsymbol{x}) \log(1 + \frac{\triangle p(\boldsymbol{x})}{q(\boldsymbol{x})}) \tag{43}$$

By Taylor expansion, $\log(1+x) \approx x - \frac{x^2}{2}$, we ignore larger than 2 order terms. Then apply this we get

$$\begin{aligned} D_{\mathrm{KL}}\big(q(\boldsymbol{x})\|p(\boldsymbol{x})\big) &= -\sum_{\boldsymbol{x}} q(\boldsymbol{x})[\frac{\triangle p(\boldsymbol{x})}{q(\boldsymbol{x})} - (\frac{\triangle p(\boldsymbol{x})}{q(\boldsymbol{x})})^2] \\ &= -\sum_{\boldsymbol{x}} \triangle p(\boldsymbol{x}) + \sum_{\boldsymbol{x}} \frac{\triangle p(\boldsymbol{x})^2}{q(\boldsymbol{x})} \\ &= \sum_{\boldsymbol{x}} \frac{\triangle p(\boldsymbol{x})^2}{q(\boldsymbol{x})} \end{aligned} \tag{44}$$

Change probabilities q and p to $q_(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$ and $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ we get the targeted formulation.

A.7 VLB with partial time steps

Assume that we only observe $\mathbf{x}_t$, $\mathbf{x}_s$, and $\mathbf{x}_0$, and assume that we have learned the prior $p_\theta(\mathbf{x}_t)$.

$$\begin{aligned} \log p_\theta(\mathbf{x}_0) &= \log \mathbb{E}_{q(\mathbf{x}_t,\mathbf{x}_s|\mathbf{x}_0)} \frac{p_\theta(\mathbf{x}_t, \mathbf{x}_s, \mathbf{x}_0)}{q(\mathbf{x}_t, \mathbf{x}_s|\mathbf{x}_0)} \geq \mathbb{E}_{q(\mathbf{x}_t,\mathbf{x}_s|\mathbf{x}_0)} \log \frac{p_\theta(\mathbf{x}_t, \mathbf{x}_s, \mathbf{x}_0)}{q(\mathbf{x}_t, \mathbf{x}_s|\mathbf{x}_0)} \\ &= \mathbb{E}_{q(\mathbf{x}_t,\mathbf{x}_s|\mathbf{x}_0)}[\log \frac{p_\theta(\mathbf{x}_t)p_\theta(\mathbf{x}_s|\mathbf{x}_t)p_\theta(\mathbf{x}_0|\mathbf{x}_s)}{q(\mathbf{x}_t|\mathbf{x}_0)q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)}] \\ &= \mathbb{E}_{q(\mathbf{x}_t,\mathbf{x}_s|\mathbf{x}_0)}\Big[\log \frac{p_\theta(\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_0)} + \log p_\theta(\mathbf{x}_t) + \log \frac{p_\theta(\mathbf{x}_s|\mathbf{x}_t)}{q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)}\Big] \\ &= -D_{\mathrm{KL}}[q(\mathbf{x}_t|\mathbf{x}_0)\|p_\theta(\mathbf{x}_t)] + \mathbb{E}_{q(\mathbf{x}_s|\mathbf{x}_0)} \log p_\theta(\mathbf{x}_0|\mathbf{x}_s) - D_{\mathrm{KL}}[q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)\|p_\theta(\mathbf{x}_s|\mathbf{x}_t)] \end{aligned} \tag{45}$$

A.8 Reparameterization Form for Sampling

In practice, we need to sample $\mathbf{x}_{t|0} \sim q(\mathbf{x}_t|\mathbf{x}_0)$ for training and $\mathbf{x}_{s|t} \sim p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ for generation (backward denoising). In what follows, we provide the reparameterization form for these to facilitate fast sampling. Given $q(\mathbf{x}_t|\mathbf{x}_0) = \mathrm{Cat}(\mathbf{x}_t; \overline{\alpha}_t\mathbf{x}_0 + (1-\overline{\alpha}_t)\boldsymbol{m})$ and $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ as in Eq. (10), we can rewrite the corresponding variables as

$$\mathbf{x}_{t|0} = \delta_{1,\mathbf{b}_t}\mathbf{x}_0 + (1-\delta_{1,\mathbf{b}_t})\mathbf{m}_0, \text{ where } \mathbf{b}_t \sim \mathrm{Bernoulli}(\overline{\alpha}_t) \tag{46}$$

$$\mathbf{x}_{s|t} = \delta_{1,\mathbf{b}_{s|t}}\tilde{\mathbf{x}}_{0|t} \;+\; \delta_{2,\mathbf{b}_{s|t}}\mathbf{x}_t \;+\; \delta_{3,\mathbf{b}_{s|t}}\mathbf{m}_0, \tag{47}$$

where $\tilde{\mathbf{x}}_{0|t} \sim \mathrm{Cat}(f_t^\theta(\mathbf{x}_t))$ and $\mathbf{b}_{s|t} \sim \mathrm{Cat}(\cdot; [1-\mu_{t|s}, \mu_{t|s}\overline{\alpha}_{t|s} + \gamma_{t|s}^\theta, \ \mu_{t|s}(1-\overline{\alpha}_{t|s}) - \gamma_{t|s}^\theta])$.

Eq. (46) and Eq. (47) essentially show that the sampling process can be divided into two steps: first, sample the branch indicator $\mathbf{b}_t$ (or $\mathbf{b}_{s|t}$), and then sample from the categorical distribution of that branch, i.e. $\mathbf{x}_t$, $\mathbf{m}_0$, or $\tilde{\mathbf{x}}_{0|t}$. Moreover, the three terms in Eq. (47) highlight that the denoising step of generating $\mathbf{x}_s$ from $\mathbf{x}_t$ essentially draws samples via three levers: (1) use the predicted sample $\mathbf{x}_0$ from the trained network $f_t^\theta(\mathbf{x}_t)$ directly, (2) keep it unchanged as $\mathbf{x}_t$, or (3) roll it back to noise $\mathbf{m}_0$, offering an intuitive understanding.

A.9 Discrete-Time Multi-element Object Extension

We first extend $q(\mathbf{x}_t|\mathbf{x}_0)$, $q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$, and $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ to multi-element object, and then present the negative VLB loss extension. First, for the forward process conditional on $\mathbf{x}_0$, each element $\mathbf{x}_t^d$ of the multi-element object $\mathbf{x}_t^{1:D}$ has its own diffusion process without interactions with others. Formally,

$$q(\mathbf{x}_t^{1:D}|\mathbf{x}_0^{1:D}) = \prod_{d=1}^{D} q(\mathbf{x}_t^d|\mathbf{x}_0^d) \quad \text{, and} \quad q(\mathbf{x}_s^{1:D}|\mathbf{x}_t^{1:D}, \mathbf{x}_0^{1:D}) = \prod_{d=1}^{D} q(\mathbf{x}_s^d|\mathbf{x}_t^d, \mathbf{x}_0^d) \ . \tag{48}$$

A.6 KL散度近似的证明

假设 $p(\boldsymbol{x}) = q(\boldsymbol{x}) + \triangle p(\boldsymbol{x})$、$p(\boldsymbol{x})$ 和 $q(\boldsymbol{x})$ 都是有效的分类分布。然后

$$D_{\mathrm{KL}}\big(q(\boldsymbol{x})\|p(\boldsymbol{x})\big) = -\sum_{\boldsymbol{x}} q(\boldsymbol{x}) \log \frac{p(\boldsymbol{x})}{q(\boldsymbol{x})} = -\sum_{\boldsymbol{x}} q(\boldsymbol{x}) \log(1 + \frac{\triangle p(\boldsymbol{x})}{q(\boldsymbol{x})}) \tag{43}$$

通过泰勒展开式 $\log(1 + x) \approx x - \frac{x^2}{2}$，我们忽略大于 2 阶的项。然后应用这个我们得到

$$\begin{aligned} D_{\mathrm{KL}}\big(q(\boldsymbol{x})\|p(\boldsymbol{x})\big) &= -\sum_{\boldsymbol{x}} q(\boldsymbol{x})[\frac{\triangle p(\boldsymbol{x})}{q(\boldsymbol{x})} - (\frac{\triangle p(\boldsymbol{x})}{q(\boldsymbol{x})})^2] \\ &= -\sum_{\boldsymbol{x}} \triangle p(\boldsymbol{x}) + \sum_{\boldsymbol{x}} \frac{\triangle p(\boldsymbol{x})^2}{q(\boldsymbol{x})} \\ &= \sum_{\boldsymbol{x}} \frac{\triangle p(\boldsymbol{x})^2}{q(\boldsymbol{x})} \end{aligned} \tag{44}$$

改变概率　　lities q 和 p 到 $q_(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$ 和 $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ 我们得到了目标　　配方。

A.7 具有部分时间步长的 VLB

假设我们只观察$\mathbf{x}_t$、$\mathbf{x}_s$和$\mathbf{x}_0$，并假设我们已经学习了先验的$p_\theta(\mathbf{x}_t)$。

$$\begin{aligned} \log p_\theta(\mathbf{x}_0) &= \log \mathbb{E}_{q(\mathbf{x}_t,\mathbf{x}_s|\mathbf{x}_0)} \frac{p_\theta(\mathbf{x}_t,\mathbf{x}_s,\mathbf{x}_0)}{q(\mathbf{x}_t,\mathbf{x}_s|\mathbf{x}_0)} \geq \mathbb{E}_{q(\mathbf{x}_t,\mathbf{x}_s|\mathbf{x}_0)} \log \frac{p_\theta(\mathbf{x}_t,\mathbf{x}_s,\mathbf{x}_0)}{q(\mathbf{x}_t,\mathbf{x}_s|\mathbf{x}_0)} \\ &= \mathbb{E}_{q(\mathbf{x}_t,\mathbf{x}_s|\mathbf{x}_0)}[\log \frac{p_\theta(\mathbf{x}_t)p_\theta(\mathbf{x}_s|\mathbf{x}_t)p_\theta(\mathbf{x}_0|\mathbf{x}_s)}{q(\mathbf{x}_t|\mathbf{x}_0)q(\mathbf{x}_s|\mathbf{x}_t,\mathbf{x}_0)}] \\ &= \mathbb{E}_{q(\mathbf{x}_t,\mathbf{x}_s|\mathbf{x}_0)}\Big[\log \frac{p_\theta(\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_0)} + \log p_\theta(\mathbf{x}_t) + \log \frac{p_\theta(\mathbf{x}_s|\mathbf{x}_t)}{q(\mathbf{x}_s|\mathbf{x}_t,\mathbf{x}_0)}\Big] \\ &= -D_{\mathrm{KL}}[q(\mathbf{x}_t|\mathbf{x}_0)\|p_\theta(\mathbf{x}_t)] + \mathbb{E}_{q(\mathbf{x}_s|\mathbf{x}_0)} \log p_\theta(\mathbf{x}_0|\mathbf{x}_s) - D_{\mathrm{KL}}[q(\mathbf{x}_s|\mathbf{x}_t,\mathbf{x}_0)\|p_\theta(\mathbf{x}_s|\mathbf{x}_t)] \end{aligned} \tag{45}$$

A.8 抽样重新参数化表格

在实践中，我们需要对 $\mathbf{x}_{t|0} \sim q(\mathbf{x}_t|\mathbf{x}_0)$ 进行采样以进行训练，对 $\mathbf{x}_{s|t} \sim p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ 进行采样以进行生成（向后去噪）。接下来，我们提供了这些的重新参数化形式，以促进快速采样。给定 $q(\mathbf{x}_t|\mathbf{x}_0) = \mathrm{Cat}(\mathbf{x}_t; \alpha_t\mathbf{x}_0 + (1-\overline{\alpha}_t)\boldsymbol{m})$ 和 $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$，如等式：(10) 则我们可以将相应的变量改写为

$$\mathbf{x}_{t|0} = \delta_{1,\mathbf{b}_t}\mathbf{x}_0 + (1-\delta_{1,\mathbf{b}_t})\mathbf{m}_0, \text{ where } \mathbf{b}_t \sim \text{Bernoulli}(\overline{\alpha}_t) \tag{46}$$

$$\mathbf{x}_{s|t} = \delta_{1,\mathbf{b}_{s|t}}\tilde{\mathbf{x}}_{0|t} \;+\; \delta_{2,\mathbf{b}_{s|t}}\mathbf{x}_t \;+\; \delta_{3,\mathbf{b}_{s|t}}\mathbf{m}_0, \tag{47}$$

其中 $\tilde{\mathbf{x}}_{0|t} \sim \mathrm{Cat}(f_t^\theta(\mathbf{x}_t))$ 和 $\mathbf{b}_{s|t} \sim \mathrm{Cat}(\cdot; [1-\mu_{t|s}, \mu_{t|s}\overline{\alpha}_{t|s} + \gamma_{t|s}^\theta,\ \mu_{t|s}(1-\overline{\alpha}_{t|s}) - \gamma_{t|s}^\theta])$。

等式。(46) 和等式。(47)本质上表明采样过程可以分为两个步骤：首先，对分支指示符$\mathbf{b}_t$ (或$\mathbf{b}_{s|t}$)进行采样，然后从该分支的分类分布中进行采样，即$\mathbf{x}_t$、$\mathbf{m}_0$或$\tilde{\mathbf{x}}_{0|t}$。此外，方程中的三项。(47) 强调从 $\mathbf{x}_t$ 生成 $\mathbf{x}_s$ 的去噪步骤本质上是通过三个杠杆提取样本：(1) 直接使用来自训练网络 $f_t^\theta(\mathbf{x}_t)$ 的预测样本 $\mathbf{x}_0$，(2) 保持其不变为 $\mathbf{x}_t$，或 (3) 将其回滚到噪声 $\mathbf{m}_0$，提供直观的理解。

A.9 离散时间多元素对象扩展

我们首先将 $q(\mathbf{x}_t|\mathbf{x}_0)$、$q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$ 和 $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ 扩展到多元素对象，然后提出负 VLB 损失扩展。首先，对于以$\mathbf{x}_0$为条件的前向过程，多元素对象$\mathbf{x}_t^{1:D}$的每个元素$\mathbf{x}_t^d$都有自己的扩散过程，而不与其他元素相互作用。正式地，

$$q(\mathbf{x}_t^{1:D}|\mathbf{x}_0^{1:D}) = \prod_{d=1}^{D} q(\mathbf{x}_t^d|\mathbf{x}_0^d) \quad \text{, and} \quad q(\mathbf{x}_s^{1:D}|\mathbf{x}_t^{1:D}, \mathbf{x}_0^{1:D}) = \prod_{d=1}^{D} q(\mathbf{x}_s^d|\mathbf{x}_t^d, \mathbf{x}_0^d)\ . \tag{48}$$

The corresponding forward reparameterization form for generating $\mathbf{x}_{t|0}^{1:D}$ can be updated as

$$\mathbf{x}_{t|0}^{1:D} = \delta_{1,\mathbf{b}_t^{1:D}}^{1:D} \odot \mathbf{x}_0^{1:D} + (1 - \delta_{1,\mathbf{b}_t^{1:D}}^{1:D}) \odot \mathbf{m}_0^{1:D} \quad , \tag{49}$$

where $\delta_{1,\mathbf{b}_t^{1:D}}^{1:D}$ is a D-dimensional vector with d-th element being $\delta_{1,\mathbf{b}_t^d}$ and $\mathbf{b}_t^d \sim \text{Bernoulli}(\overline{\alpha}_t)$.

For the backward process, all elements' transition processes are coupled together, as shown in the graphical model in Appx. Fig. 1(b). We start with the parameterization $p_\theta(\mathbf{x}_0^{1:D}|\mathbf{x}_t^{1:D})$, with the form

$$p_\theta(\mathbf{x}_0^{1:D}|\mathbf{x}_t^{1:D}) = \prod_{d=1}^{D} p_\theta(\mathbf{x}_0^d|\mathbf{x}_t^{1:D}) \quad \text{with} \quad p_\theta(\mathbf{x}_0^d|\mathbf{x}_t^{1:D}) = \text{Cat}\big(\mathbf{x}_0^d|f_t^{\theta,d}(\mathbf{x}_t^{1:D})\big) \tag{50}$$

Then, $p_\theta(\mathbf{x}_s^{1:D}|\mathbf{x}_t^{1:D}) = \prod_{d=1}^{D} p_\theta(\mathbf{x}_s^d|\mathbf{x}_t^{1:D})$, with the component $p_\theta(\mathbf{x}_s^d|\mathbf{x}_t^{1:D})$ has the form

$$\text{Cat}\Big(\mathbf{x}_s^d;\ (1-\mu_{t|s})f_t^{\theta,d}(\mathbf{x}_t^{1:D}) + (\mu_{t|s}\overline{\alpha}_{t|s} + \gamma_{t|s}^{\theta,d})\mathbf{x}_t^d + (\mu_{t|s}(1-\overline{\alpha}_{t|s}) - \gamma_{t|s}^{\theta,d})\boldsymbol{m}^d\Big) , \tag{51}$$

where $\gamma_{t|s}^{\theta,d} = (\mu_{t|s} - \lambda_{t|s}^d - \mu_{t|s}\overline{\alpha}_{t|s})\langle f_t^{\theta,d}(\mathbf{x}_t^{1:D}), \mathbf{x}_t^d\rangle,\ \lambda_{t|s}^d = \dfrac{(1-\overline{\alpha}_s)(1-\overline{\alpha}_{t|s})\langle \boldsymbol{m}^d, \mathbf{x}_t^d\rangle}{\overline{\alpha}_t + (1-\overline{\alpha}_t)\langle \boldsymbol{m}^d, \mathbf{x}_t^d\rangle}$.

Similarly, the reparameterization form can be expressed as

$$\mathbf{x}_{s|t}^{1:D} = \delta_{1,\mathbf{b}_{s|t}^{1:D}}^{1:D} \odot \tilde{\mathbf{x}}_{0|t}^{1:D} \ + \ \delta_{2,\mathbf{b}_{s|t}^{1:D}}^{1:D} \odot \mathbf{x}_t^{1:D} \ + \ \delta_{3,\mathbf{b}_{s|t}^{1:D}}^{1:D} \odot \mathbf{m}_0^{1:D} , \tag{52}$$

where $\tilde{\mathbf{x}}_{0|t}^d \sim \text{Cat}(f_t^{\theta,d}(\mathbf{x}_t^{1:D}))$ and $\mathbf{b}_{s|t}^d \sim \text{Cat}(1-\mu_{t|s},\ \mu_{t|s}\overline{\alpha}_{t|s} + \gamma_{t|s}^{\theta,d},\ \mu_{t|s}(1-\overline{\alpha}_{t|s}) - \gamma_{t|s}^{\theta,d})$.

Finally, the following reformulate the loss functions $\mathcal{L}_t(\theta)$ for multi-element objects, where we drop the constant term for simplicity.

$$\begin{aligned}\mathcal{L}_t^{1:D}(\theta) =& \mathbb{E}_{q(\mathbf{x}_t^{1:D}|\mathbf{x}_0^{1:D})} \sum_{d=1}^{D} \Big[\delta_{\mathbf{x}_t^d,\mathbf{x}_0^d} \mathbb{E}_{q(\mathbf{x}_{t-1}^d|\mathbf{x}_t^d=\mathbf{x}_0^d)}[-\log p_\theta(\mathbf{x}_{t-1}^d|\mathbf{x}_t^{1:D})] + \\ & (1-\delta_{\mathbf{x}_t^d,\mathbf{x}_0^d})\mathbb{E}_{q(\mathbf{x}_{t-1}^d|\mathbf{x}_t^d\neq\mathbf{x}_0^d)}[-\log p_\theta(\mathbf{x}_{t-1}^d|\mathbf{x}_t^{1:D})]\Big]\end{aligned} \tag{53}$$

A.10 CTMC introduction

The CTMC process can go with either increasing t or decreasing t, and we use increasing t as the default direction to introduce CTMC. For any CTMC, its *transition rate matrix* R_t with $[R_t]_{ij} = r_t(\boldsymbol{e}_j|\boldsymbol{e}_i)$ fully determines the underlying stochastic process. CTMC is categorized into time-homogeneous and time-*in*homogeneous based on whether R_t is static with respect to t. In this paper we work with time-inhomogeneous CTMC.

Based on the definition in Eq. (17), we can derive some properties as follows

$$\forall \boldsymbol{x}, r_t(\boldsymbol{x}|\boldsymbol{x}) \leq 0; \qquad \forall \boldsymbol{y} \neq \boldsymbol{x}, r_t(\boldsymbol{y}|\boldsymbol{x}) \geq 0; \qquad \forall \boldsymbol{x}, \sum_{\boldsymbol{y}} r_t(\boldsymbol{y}|\boldsymbol{x}) = 0 \tag{54}$$

$$q_{t|t-\triangle t}(\boldsymbol{y}|\boldsymbol{x}) = \delta_{\boldsymbol{x},\boldsymbol{y}} + r_t(\boldsymbol{y}|\boldsymbol{x})\triangle t + o(\triangle t) \tag{55}$$

Eq. (54) shows the properties of transition rate, which imply $r_t(\boldsymbol{x}|\boldsymbol{x}) = -\sum_{\boldsymbol{y}\neq\boldsymbol{x}} r_t(\boldsymbol{y}|\boldsymbol{x})$. Eq. (55) characterizes the infinitesimal change of transition probability, and can be used to derive the relationship between transition matrix $\overline{Q}_{t|s}$ and transition rate matrix R_t. Formally, a CTMC's transition probabilities satisfies the Kolmogorov forward and backward equations [Kolmogoroff, 1931] :

$$\frac{d}{dt} q_{t|s}(\boldsymbol{y}|\boldsymbol{x}) = \sum_{\mathbf{z}} q_{t|s}(\boldsymbol{z}|\boldsymbol{x}) r_t(\boldsymbol{y}|\boldsymbol{z}) \quad \text{or} \quad \frac{d}{dt}\overline{Q}_{t|s} = \overline{Q}_{t|s} R_t \qquad \text{Kolmogorov Forward} \tag{56}$$

$$\frac{d}{ds} q_{t|s}(\boldsymbol{y}|\boldsymbol{x}) = -\sum_{\boldsymbol{z}} r_s(\boldsymbol{z}|\boldsymbol{x}) q_{t|s}(\boldsymbol{y}|\boldsymbol{z}) \quad \text{or} \quad \frac{d}{ds}\overline{Q}_{t|s} = -R_s \overline{Q}_{t|s} \quad \text{Kolmogorov Backward} \tag{57}$$

The above equations are Ordinary Different Equations (ODEs) and have unique solutions. Rindos et al. [1995] mentioned that when R_{t_1} and R_{t_2} commute (i.e. $R_{t_1}R_{t_2} = R_{t_2}R_{t_1}$) for any t_1, t_2, the transition probability matrix can be written as

$$\overline{Q}_{t|s} = \exp\Big(\int_s^t R_a da\Big) \tag{58}$$

用于生成 $\mathbf{x}_{t|0}^{1:D}$ 的相应前向重新参数化形式可以更新为

$$\mathbf{x}_{t|0}^{1:D}=\delta_{1,\mathbf{b}_t^{1:D}}^{1:D}\odot\mathbf{x}_0^{1:D}+(1-\delta_{1,\mathbf{b}_t^{1:D}}^{1:D})\odot\mathbf{m}_0^{1:D}\quad, \tag{49}$$

其中 $\delta_{1,\mathbf{b}_t^{1:D}}^{1:D}$ 是一个 D 维向量，第 d 个元素为 $\delta_{1,\mathbf{b}_t^d}$ 和 $\mathbf{b}_t^d\sim$ Bernoulli(α_t)。 –

对于后向过程，所有元素的转换过程都耦合在一起，如 Appx 中的图形模型所示。图1(b)。我们从参数化 $p_\theta(\mathbf{x}_0^{1:D}|\mathbf{x}_t^{1:D})$ 开始，其形式为

$$p_\theta(\mathbf{x}_0^{1:D}|\mathbf{x}_t^{1:D})=\prod_{d=1}^{D}p_\theta(\mathbf{x}_0^d|\mathbf{x}_t^{1:D})\quad\text{with}\quad p_\theta(\mathbf{x}_0^d|\mathbf{x}_t^{1:D})=\mathrm{Cat}\big(\mathbf{x}_0^d|f_t^{\theta,d}(\mathbf{x}_t^{1:D})\big) \tag{50}$$

那么，$p_\theta(\mathbf{x}_s^{1:D}|\mathbf{x}_t^{1:D})=\prod_{d=1}^{D}p_\theta(\mathbf{x}_s^d|\mathbf{x}_t^{1:D})$ 以及组件 $p_\theta(\mathbf{x}_s^d|\mathbf{x}_t^{1:D})$ 的形式为

$$\mathrm{Cat}\Big(\mathbf{x}_s^d;\ (1-\mu_{t|s})f_t^{\theta,d}(\mathbf{x}_t^{1:D})+(\mu_{t|s}\overline{\alpha}_{t|s}+\gamma_{t|s}^{\theta,d})\mathbf{x}_t^d+(\mu_{t|s}(1-\overline{\alpha}_{t|s})-\gamma_{t|s}^{\theta,d})\boldsymbol{m}^d\Big)\ , \tag{51}$$

$$\text{where }\gamma_{t|s}^{\theta,d}=(\mu_{t|s}-\lambda_{t|s}^d-\mu_{t|s}\overline{\alpha}_{t|s})\langle f_t^{\theta,d}(\mathbf{x}_t^{1:D}),\mathbf{x}_t^d\rangle,\ \lambda_{t|s}^d=\frac{(1-\overline{\alpha}_s)(1-\overline{\alpha}_{t|s})\langle\boldsymbol{m}^d,\mathbf{x}_t^d\rangle}{\overline{\alpha}_t+(1-\overline{\alpha}_t)\langle\boldsymbol{m}^d,\mathbf{x}_t^d\rangle}\ .$$

类似地，重参数化形式可以表示为

$$\mathbf{x}_{s|t}^{1:D}=\delta_{1,\mathbf{b}_{s|t}^{1:D}}^{1:D}\odot\tilde{\mathbf{x}}_{0|t}^{1:D}\ +\ \delta_{2,\mathbf{b}_{s|t}^{1:D}}^{1:D}\odot\mathbf{x}_t^{1:D}\ +\ \delta_{3,\mathbf{b}_{s|t}^{1:D}}^{1:D}\odot\mathbf{m}_0^{1:D}\ , \tag{52}$$

其中 $\tilde{\mathbf{x}}_{0|t}^d\sim\mathrm{Cat}(f_t^{\theta,d}(\mathbf{x}_t^{1:D}))$ 和 $\mathbf{b}_{s|t}^d\sim\mathrm{Cat}(1-\mu_{t|s},\ \mu_{t|s}\overline{\alpha}_{t|s}+\gamma_{t|s}^{\theta,d},\ \mu_{t|s}(1-\overline{\alpha}_{t|s})-\gamma_{t|s}^{\theta,d})$。

最后，下面重新表述多元素对象的损失函数 $\mathcal{L}_t(\theta)$，为了简单起见，我们删除了常数项。

$$\begin{aligned}\mathcal{L}_t^{1:D}(\theta)=&\mathbb{E}_{q(\mathbf{x}_t^{1:D}|\mathbf{x}_0^{1:D})}\sum_{d=1}^{D}\Big[\delta_{\mathbf{x}_t^d,\mathbf{x}_0^d}\mathbb{E}_{q(\mathbf{x}_{t-1}^d|\mathbf{x}_t^d=\mathbf{x}_0^d)}[-\log p_\theta(\mathbf{x}_{t-1}^d|\mathbf{x}_t^{1:D})]+\\&(1-\delta_{\mathbf{x}_t^d,\mathbf{x}_0^d})\mathbb{E}_{q(\mathbf{x}_{t-1}^d|\mathbf{x}_t^d\neq\mathbf{x}_0^d)}[-\log p_\theta(\mathbf{x}_{t-1}^d|\mathbf{x}_t^{1:D})]\Big]\end{aligned} \tag{53}$$

A.10 CTMC介绍

CTMC 过程可以采用增加 t 或减少 t 的方式进行，我们使用增加 t 作为引入 CTMC 的默认方向。对于任何 CTMC，其 *transition rate matrix* R_t 和 $[R_t]_{ij}=r_t(\boldsymbol{e}_j|\boldsymbol{e}_i)$ 完全决定了底层的随机过程。根据R_t相对于t是否静态，CTMC分为时间齐次和时间*in*齐次。在本文中，我们使用时间不均匀的 CTMC。

基于等式中的定义。 (17)，我们可以推导出一些性质如下

$$\forall\boldsymbol{x},r_t(\boldsymbol{x}|\boldsymbol{x})\leq 0;\qquad\forall\boldsymbol{y}\neq\boldsymbol{x},r_t(\boldsymbol{y}|\boldsymbol{x})\geq 0;\qquad\forall\boldsymbol{x},\sum_{\boldsymbol{y}}r_t(\boldsymbol{y}|\boldsymbol{x})=0 \tag{54}$$

$$q_{t|t-\triangle t}(\boldsymbol{y}|\boldsymbol{x})=\delta_{\boldsymbol{x},\boldsymbol{y}}+r_t(\boldsymbol{y}|\boldsymbol{x})\triangle t+o(\triangle t) \tag{55}$$

等式。 (54)显示了转移率的性质，这意味着$r_t(\boldsymbol{x}|\boldsymbol{x})=-\sum_{\boldsymbol{y}\neq\boldsymbol{x}}r_t(\boldsymbol{y}|\boldsymbol{x})$。等式。式(55)表征了转移概率的无穷小变化，可以用来推导转移矩阵$Q_{t|s}$和转移率矩阵R_t之间的关系。形式上，CTMC 的转移概率满足 Kolmogorov 前向和后向方程 [Kolmogoroff, 1931]:

$$\frac{d}{dt}q_{t|s}(\boldsymbol{y}|\boldsymbol{x})=\sum_{\mathbf{z}}q_{t|s}(\boldsymbol{z}|\boldsymbol{x})r_t(\boldsymbol{y}|\boldsymbol{z})\quad\text{or}\quad\frac{d}{dt}\overline{Q}_{t|s}=\overline{Q}_{t|s}R_t\qquad\text{Kolmogorov Forward} \tag{56}$$

$$\frac{d}{ds}q_{t|s}(\boldsymbol{y}|\boldsymbol{x})=-\sum_{\boldsymbol{z}}r_s(\boldsymbol{z}|\boldsymbol{x})q_{t|s}(\boldsymbol{y}|\boldsymbol{z})\quad\text{or}\quad\frac{d}{ds}\overline{Q}_{t|s}=-R_s\overline{Q}_{t|s}\qquad\text{柯尔莫哥洛夫向后} \tag{57}$$

上述方程是常微分方程 (ODE)，并且具有唯一解。林多斯等人。 [1995]提到，当R_{t_1}和R_{t_2}对任何t_1,t_2交换（即$R_{t_1}R_{t_2}=R_{t_2}R_{t_1}$）时，转移概率矩阵可以写为

$$\overline{Q}_{t|s}=\exp\Big(\int_s^t R_a da\Big) \tag{58}$$

where $\exp(M) := \sum_{k=0}^{\infty} \frac{M^k}{k!}$ is the matrix exponential operation. The commutative property of R_t can be achieved by choosing $R_t = \beta(t)R_b$ where $R_b \in \mathbb{R}^{K\times K}$ is a time-independent base transition rate matrix satisfying properties in Eq. (54) and $\beta(t)$ is a positive scalar dependent on time. Now assume R_b is diagonalizable and $R_b = U\Sigma U^{-1}$ by eigen-decomposition, then we can simplify Eq. (58) as

$$\overline{Q}_{t|s} = \exp\Big(\int_s^t \beta(t)R_b da\Big) = \exp(\overline{\beta}_{t|s}R_b) = \exp(\overline{\beta}_{t|s}U\Sigma U^{-1}) = U\exp(\overline{\beta}_{t|s}\Sigma)U^{-1} \tag{59}$$

where $\overline{\beta}_{t|s} := \int_s^t \beta(a)da$.

A.11 Derivation of Eq. (18)

$$\begin{aligned}\overline{Q}_{t|s} = \exp(\overline{\beta}_{t|s}R_b) &= I + \sum_{k=1}^{\infty}\frac{(-\overline{\beta}_{t|s})^k(-R_b)^k}{k!} = I - (\sum_{k=1}^{\infty}\frac{(-\overline{\beta}_{t|s})^k}{k!})R_b \\ &= I - (e^{-\overline{\beta}_{t|s}} - 1)R_b = e^{-\overline{\beta}_{t|s}}I + (1 - e^{-\overline{\beta}_{t|s}})\mathbf{1}\boldsymbol{m}^\top \end{aligned}\tag{60}$$

A.12 Derivation of Eq. (25)

With the help of Eq. (19), we first show that g_t^θ can be simplified as follows.

$$\begin{aligned} g_t^\theta(\boldsymbol{z}|\boldsymbol{x}) &= \frac{1}{\langle \boldsymbol{x},\boldsymbol{m}\rangle}\Big[\langle \boldsymbol{z},\boldsymbol{m}\rangle + \frac{\overline{\alpha}_{t|0}}{1-\overline{\alpha}_{t|0}}p^\theta_{0|t}(\boldsymbol{z}|\boldsymbol{x}) - \frac{\overline{\alpha}_{t|0}\langle \boldsymbol{z},\boldsymbol{m}\rangle}{\overline{\alpha}_{t|0} + (1-\overline{\alpha}_{t|0})\langle \boldsymbol{x},\boldsymbol{m}\rangle}p^\theta_{0|t}(\boldsymbol{x}|\boldsymbol{x})\Big] \\ &= \frac{1}{\langle \boldsymbol{x},\boldsymbol{m}\rangle}\Big[(1 - \frac{\overline{\alpha}_{t|0}p^\theta_{0|t}(\boldsymbol{x}|\boldsymbol{x})}{\overline{\alpha}_{t|0} + (1-\overline{\alpha}_{t|0})\langle \boldsymbol{x},\boldsymbol{m}\rangle})\langle \boldsymbol{z},\boldsymbol{m}\rangle + \frac{\overline{\alpha}_{t|0}}{1-\overline{\alpha}_{t|0}}p^\theta_{0|t}(\boldsymbol{z}|\boldsymbol{x})\Big] \quad \forall \boldsymbol{z}\neq\boldsymbol{x} \end{aligned}\tag{61}$$

Proof.

$$\frac{q_{t|0}(\boldsymbol{z}|\boldsymbol{x}_0)}{q_{t|0}(\boldsymbol{x}|\boldsymbol{x}_0)} = \frac{\boldsymbol{z}^\top\overline{Q}_{t|0}^\top\boldsymbol{x}_0}{\boldsymbol{x}^\top\overline{Q}_{t|0}^\top\boldsymbol{x}_0} = \frac{\overline{\alpha}_{t|0}\boldsymbol{z}^\top\boldsymbol{x}_0 + (1-\overline{\alpha}_{t|0})\langle \boldsymbol{z},\boldsymbol{m}\rangle}{\overline{\alpha}_{t|0}\boldsymbol{x}^\top\boldsymbol{x}_0 + (1-\overline{\alpha}_{t|0})\langle \boldsymbol{x},\boldsymbol{m}\rangle} \tag{62}$$

Then when $\boldsymbol{z}\neq\boldsymbol{x}$, we can write it as

$$\frac{q_{t|0}(\boldsymbol{z}|\boldsymbol{x}_0)}{q_{t|0}(\boldsymbol{x}|\boldsymbol{x}_0)} = \begin{cases} \frac{\langle \boldsymbol{z},\boldsymbol{m}\rangle}{\langle \boldsymbol{x},\boldsymbol{m}\rangle} & \text{if } \boldsymbol{x}_0\neq\boldsymbol{x} \text{ and } \boldsymbol{x}_0\neq\boldsymbol{z} \\ \frac{\frac{\overline{\alpha}_{t|0}}{1-\overline{\alpha}_{t|0}}+\langle \boldsymbol{z},\boldsymbol{m}\rangle}{\langle \boldsymbol{x},\boldsymbol{m}\rangle} & \text{if } \boldsymbol{x}_0 = \boldsymbol{z} \\ \frac{\langle \boldsymbol{z},\boldsymbol{m}\rangle}{\frac{\overline{\alpha}_{t|0}}{1-\overline{\alpha}_{t|0}}+\langle \boldsymbol{x},\boldsymbol{m}\rangle} & \text{if } \boldsymbol{x}_0 = \boldsymbol{x} \end{cases} \tag{63}$$

Hence,

$$\begin{aligned} g_t^\theta(\boldsymbol{z}|\boldsymbol{x}) &= \sum_{\boldsymbol{x}_0}\frac{q_{t|0}(\boldsymbol{z}|\boldsymbol{x}_0)}{q_{t|0}(\boldsymbol{x}|\boldsymbol{x}_0)}p^\theta_{0|t}(\boldsymbol{x}_0|\boldsymbol{x}) \\ &= \frac{q_{t|0}(\boldsymbol{z}|\boldsymbol{x})}{q_{t|0}(\boldsymbol{x}|\boldsymbol{x})}p^\theta_{0|t}(\boldsymbol{x}|\boldsymbol{x}) + \frac{q_{t|0}(\boldsymbol{z}|\boldsymbol{z})}{q_{t|0}(\boldsymbol{x}|\boldsymbol{z})}p^\theta_{0|t}(\boldsymbol{z}|\boldsymbol{x}) + \sum_{\boldsymbol{x}_0\neq\boldsymbol{z},\boldsymbol{x}}\frac{q_{t|0}(\boldsymbol{z}|\boldsymbol{x}_0)}{q_{t|0}(\boldsymbol{x}|\boldsymbol{x}_0)}p^\theta_{0|t}(\boldsymbol{x}_0|\boldsymbol{x}) \\ &= \frac{\langle \boldsymbol{z},\boldsymbol{m}\rangle}{\frac{\overline{\alpha}_{t|0}}{1-\overline{\alpha}_{t|0}}+\langle \boldsymbol{x},\boldsymbol{m}\rangle}p^\theta_{0|t}(\boldsymbol{x}|\boldsymbol{x}) + \frac{\frac{\overline{\alpha}_{t|0}}{1-\overline{\alpha}_{t|0}}+\langle \boldsymbol{z},\boldsymbol{m}\rangle}{\langle \boldsymbol{x},\boldsymbol{m}\rangle}p^\theta_{0|t}(\boldsymbol{z}|\boldsymbol{x}) + \frac{\langle \boldsymbol{z},\boldsymbol{m}\rangle}{\langle \boldsymbol{x},\boldsymbol{m}\rangle}(1 - p^\theta_{0|t}(\boldsymbol{z}|\boldsymbol{x}) - p^\theta_{0|t}(\boldsymbol{x}|\boldsymbol{x})) \\ &= \frac{1}{\langle \boldsymbol{x},\boldsymbol{m}\rangle}\Big[(1 - \frac{\overline{\alpha}_{t|0}p^\theta_{0|t}(\boldsymbol{x}|\boldsymbol{x})}{\overline{\alpha}_{t|0} + (1-\overline{\alpha}_{t|0})\langle \boldsymbol{x},\boldsymbol{m}\rangle})\langle \boldsymbol{z},\boldsymbol{m}\rangle + \frac{\overline{\alpha}_{t|0}}{1-\overline{\alpha}_{t|0}}p^\theta_{0|t}(\boldsymbol{z}|\boldsymbol{x})\Big] \quad \forall \boldsymbol{z}\neq\boldsymbol{x} \end{aligned}\tag{64}$$

Additionally, when $\boldsymbol{z}=\boldsymbol{x}$, $g_t^\theta(\boldsymbol{z}|\boldsymbol{x}) = \sum_{\boldsymbol{x}_0}p^\theta_{0|t}(\boldsymbol{x}_0|\boldsymbol{x}) = 1$. ☐

We can also compute the vectorization $g_t^\theta(\cdot|\boldsymbol{x})$ directly as

$$g_t^\theta(\cdot|\boldsymbol{x}) = \frac{1}{\langle \boldsymbol{x},\boldsymbol{m}\rangle}\Big[(1 - \frac{\overline{\alpha}_{t|0}\langle f_t^\theta(\boldsymbol{x}),\boldsymbol{x}\rangle}{\overline{\alpha}_{t|0} + (1-\overline{\alpha}_{t|0})\langle \boldsymbol{x},\boldsymbol{m}\rangle})\boldsymbol{m} + \frac{\overline{\alpha}_{t|0}}{1-\overline{\alpha}_{t|0}}f_t^\theta(\boldsymbol{x})\Big]\odot(\mathbf{1}-\boldsymbol{x}) + \boldsymbol{x} \tag{65}$$

其中 $\exp(M) := \sum_{k=0}^{\infty} \frac{M^k}{k!}$ 是矩阵指数运算。 R_t 的交换律可以通过选择 $R_t = \beta(t) R_b$ 来实现，其中 $R_b \in \mathbb{R}^{K\times K}$ 是满足等式（1）中性质的与时间无关的基本转移率矩阵。(54) 和 $\beta(t)$ 是依赖于时间的正标量。现在假设 R_b 可对角化，而 $R_b = U\Sigma U^{-1}$ 可通过特征分解进行对角化，那么我们可以简化方程：（58）作为

$$\overline{Q}_{t|s} = \exp\Big(\int_s^t \beta(t) R_b da\Big) = \exp(\overline{\beta}_{t|s} R_b) = \exp(\overline{\beta}_{t|s} U\Sigma U^{-1}) = U \exp(\overline{\beta}_{t|s}\Sigma) U^{-1} \tag{59}$$

其中 $\overline{\beta_{t|s}} := \int_s^t \beta(a) da$。

A.11 方程的推导(18)

$$\begin{aligned}\overline{Q}_{t|s} = \exp(\overline{\beta}_{t|s} R_b) &= I + \sum_{k=1}^{\infty} \frac{(-\overline{\beta}_{t|s})^k (-R_b)^k}{k!} = I - (\sum_{k=1}^{\infty} \frac{(-\overline{\beta}_{t|s})^k}{k!}) R_b \\ &= I - (e^{-\overline{\beta}_{t|s}} - 1) R_b = e^{-\overline{\beta}_{t|s}} I + (1 - e^{-\overline{\beta}_{t|s}}) \mathbf{1}\boldsymbol{m}^\top \end{aligned}\tag{60}$$

A.12 方程的推导(25)

在方程式的帮助下。(19)，我们首先证明g_t^θ可以简化如下。

$$\begin{aligned} g_t^\theta(\boldsymbol{z}|\boldsymbol{x}) &= \frac{1}{\langle \boldsymbol{x}, \boldsymbol{m}\rangle}\Big[\langle \boldsymbol{z}, \boldsymbol{m}\rangle + \frac{\overline{\alpha}_{t|0}}{1-\overline{\alpha}_{t|0}} p_{0|t}^\theta(\boldsymbol{z}|\boldsymbol{x}) - \frac{\overline{\alpha}_{t|0}\langle \boldsymbol{z}, \boldsymbol{m}\rangle}{\overline{\alpha}_{t|0} + (1-\overline{\alpha}_{t|0})\langle \boldsymbol{x}, \boldsymbol{m}\rangle} p_{0|t}^\theta(\boldsymbol{x}|\boldsymbol{x})\Big] \\ &= \frac{1}{\langle \boldsymbol{x}, \boldsymbol{m}\rangle}\Big[\big(1 - \frac{\overline{\alpha}_{t|0} p_{0|t}^\theta(\boldsymbol{x}|\boldsymbol{x})}{\overline{\alpha}_{t|0} + (1-\overline{\alpha}_{t|0})\langle \boldsymbol{x}, \boldsymbol{m}\rangle}\big)\langle \boldsymbol{z}, \boldsymbol{m}\rangle + \frac{\overline{\alpha}_{t|0}}{1-\overline{\alpha}_{t|0}} p_{0|t}^\theta(\boldsymbol{z}|\boldsymbol{x})\Big] \quad \forall \boldsymbol{z} \neq \boldsymbol{x} \end{aligned}\tag{61}$$

Proof.

$$\frac{q_{t|0}(\boldsymbol{z}|\boldsymbol{x}_0)}{q_{t|0}(\boldsymbol{x}|\boldsymbol{x}_0)} = \frac{\boldsymbol{z}^\top \overline{Q}_{t|0}^\top \boldsymbol{x}_0}{\boldsymbol{x}^\top \overline{Q}_{t|0}^\top \boldsymbol{x}_0} = \frac{\overline{\alpha}_{t|0}\boldsymbol{z}^\top \boldsymbol{x}_0 + (1-\overline{\alpha}_{t|0})\langle \boldsymbol{z}, \boldsymbol{m}\rangle}{\overline{\alpha}_{t|0}\boldsymbol{x}^\top \boldsymbol{x}_0 + (1-\overline{\alpha}_{t|0})\langle \boldsymbol{x}, \boldsymbol{m}\rangle} \tag{62}$$

那么当$\boldsymbol{z} \neq \boldsymbol{x}$时，我们可以写成

$$\frac{q_{t|0}(\boldsymbol{z}|\boldsymbol{x}_0)}{q_{t|0}(\boldsymbol{x}|\boldsymbol{x}_0)} = \begin{cases} \frac{\langle \boldsymbol{z}, \boldsymbol{m}\rangle}{\langle \boldsymbol{x}, \boldsymbol{m}\rangle} & \text{if } \boldsymbol{x}_0 \neq \boldsymbol{x} \text{ and } \boldsymbol{x}_0 \neq \boldsymbol{z} \\ \frac{\frac{\overline{\alpha}_{t|0}}{1-\overline{\alpha}_{t|0}} + \langle \boldsymbol{z}, \boldsymbol{m}\rangle}{\langle \boldsymbol{x}, \boldsymbol{m}\rangle} & \text{if } \boldsymbol{x}_0 = \boldsymbol{z} \\ \frac{\langle \boldsymbol{z}, \boldsymbol{m}\rangle}{\frac{\overline{\alpha}_{t|0}}{1-\overline{\alpha}_{t|0}} + \langle \boldsymbol{x}, \boldsymbol{m}\rangle} & \text{if } \boldsymbol{x}_0 = \boldsymbol{x} \end{cases}\tag{63}$$

因此，

$$\begin{aligned} g_t^\theta(\boldsymbol{z}|\boldsymbol{x}) &= \sum_{\boldsymbol{x}_0} \frac{q_{t|0}(\boldsymbol{z}|\boldsymbol{x}_0)}{q_{t|0}(\boldsymbol{x}|\boldsymbol{x}_0)} p_{0|t}^\theta(\boldsymbol{x}_0|\boldsymbol{x}) \\ &= \frac{q_{t|0}(\boldsymbol{z}|\boldsymbol{x})}{q_{t|0}(\boldsymbol{x}|\boldsymbol{x})} p_{0|t}^\theta(\boldsymbol{x}|\boldsymbol{x}) + \frac{q_{t|0}(\boldsymbol{z}|\boldsymbol{z})}{q_{t|0}(\boldsymbol{x}|\boldsymbol{z})} p_{0|t}^\theta(\boldsymbol{z}|\boldsymbol{x}) + \sum_{\boldsymbol{x}_0 \neq \boldsymbol{z}, \boldsymbol{x}} \frac{q_{t|0}(\boldsymbol{z}|\boldsymbol{x}_0)}{q_{t|0}(\boldsymbol{x}|\boldsymbol{x}_0)} p_{0|t}^\theta(\boldsymbol{x}_0|\boldsymbol{x}) \\ &= \frac{\langle \boldsymbol{z}, \boldsymbol{m}\rangle}{\frac{\overline{\alpha}_{t|0}}{1-\overline{\alpha}_{t|0}} + \langle \boldsymbol{x}, \boldsymbol{m}\rangle} p_{0|t}^\theta(\boldsymbol{x}|\boldsymbol{x}) + \frac{\frac{\overline{\alpha}_{t|0}}{1-\overline{\alpha}_{t|0}} + \langle \boldsymbol{z}, \boldsymbol{m}\rangle}{\langle \boldsymbol{x}, \boldsymbol{m}\rangle} p_{0|t}^\theta(\boldsymbol{z}|\boldsymbol{x}) + \frac{\langle \boldsymbol{z}, \boldsymbol{m}\rangle}{\langle \boldsymbol{x}, \boldsymbol{m}\rangle}(1 - p_{0|t}^\theta(\boldsymbol{z}|\boldsymbol{x}) - p_{0|t}^\theta(\boldsymbol{x}|\boldsymbol{x})) \\ &= \frac{1}{\langle \boldsymbol{x}, \boldsymbol{m}\rangle}\Big[\big(1 - \frac{\overline{\alpha}_{t|0} p_{0|t}^\theta(\boldsymbol{x}|\boldsymbol{x})}{\overline{\alpha}_{t|0} + (1-\overline{\alpha}_{t|0})\langle \boldsymbol{x}, \boldsymbol{m}\rangle}\big)\langle \boldsymbol{z}, \boldsymbol{m}\rangle + \frac{\overline{\alpha}_{t|0}}{1-\overline{\alpha}_{t|0}} p_{0|t}^\theta(\boldsymbol{z}|\boldsymbol{x})\Big] \quad \forall \boldsymbol{z} \neq \boldsymbol{x} \end{aligned}\tag{64}$$

另外，当 $\boldsymbol{z} = \boldsymbol{x}$ 时，$g_t^\theta(\boldsymbol{z}|\boldsymbol{x}) = \sum_{\boldsymbol{x}_0} p_{0|t}^\theta(\boldsymbol{x}_0|\boldsymbol{x}) = 1$。 □

我们还可以直接计算向量化 $g_t^\theta(\cdot|\boldsymbol{x})$

$$g_t^\theta(\cdot|\boldsymbol{x}) = \frac{1}{\langle \boldsymbol{x}, \boldsymbol{m}\rangle}\Big[\big(1 - \frac{\overline{\alpha}_{t|0}\langle f_t^\theta(\boldsymbol{x}), \boldsymbol{x}\rangle}{\overline{\alpha}_{t|0} + (1-\overline{\alpha}_{t|0})\langle \boldsymbol{x}, \boldsymbol{m}\rangle}\big)\boldsymbol{m} + \frac{\overline{\alpha}_{t|0}}{1-\overline{\alpha}_{t|0}} f_t^\theta(\boldsymbol{x})\Big] \odot (\mathbf{1} - \boldsymbol{x}) + \boldsymbol{x} \tag{65}$$

where $f_t^\theta(\boldsymbol{x})$ is the parameterized neural network for $p^\theta_{0|t}(\cdot|\boldsymbol{x})$ that outputs the distribution of $\mathbf{x}_{0|t}$. Eq. (25) combines Eq. (61) and $g_t^\theta(\boldsymbol{x}|\boldsymbol{x}) = 1, \forall \boldsymbol{x}$.

A.13 Derivation of Forward and Backward Transition Rate in Multi-element Case

In this section, we show how to extend transition rates, and and the ratio g_t^θ, into multi-element case. We let $\boldsymbol{x}^{\backslash d}$ represent $\boldsymbol{x}^{1:D\backslash d}$, i.e. the object without d-th element, for simplicity.

Forward Transition Rates: First, the transition rates for forward sampling has a specific decomposition formulation in multi-element case as proven by Campbell et al. [2022], thus, we summarize the result as follows. The key assumption for CTMC is that at a single time, only one dimension can change.

$$r_t^{1:D}(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) = \sum_{d=1}^{D} r_t^d(\boldsymbol{y}^d|\boldsymbol{x}^d)\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}} \tag{66}$$

where $\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}}$ is the Kronecker delta and it is 1 if and only if $\boldsymbol{x}^{\backslash d} = \boldsymbol{y}^{\backslash d}$. As we also assume that all dimension processes are indepedent, r_t^d denotes the transition rate of the CTMC process at d-th element/dimension.

Backward Transition Rates: Now let us work on $\hat{r}_t^{1:D}(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D})$. Notice that as the backward process is also a CTMC, it also satisfies that only one dimension can change at a time. We summarize two equivalent formulations as follows.

$$\hat{r}_t^{1:D}(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) = \sum_{d=1}^{D} r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d)\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}} \sum_{\boldsymbol{x}_0^d} \frac{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)} q_{0|t}(\boldsymbol{x}_0^d|\boldsymbol{x}^{1:D}) \tag{67}$$

$$\hat{r}_t^{1:D}(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) = \sum_{d=1}^{D} \frac{r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d)\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}}}{\sum_{\boldsymbol{x}_0^d} \frac{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)} q_{0|t}(\boldsymbol{x}_0^d|\boldsymbol{y}^{1:D})} \tag{68}$$

Notice that these two formulations should be equivalent. In practice, we use the first formulation to parameterize the reverse transition rate in learning.

Proof.

$$\frac{q_t(\boldsymbol{y}^{1:D})}{q_t(\boldsymbol{x}^{1:D})} = \sum_{\boldsymbol{x}_0^{1:D}} \frac{q_{t|0}(\boldsymbol{y}^{1:D}|\boldsymbol{x}_0^{1:D})}{q_{t|0}(\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D})} q_{0|t}(\boldsymbol{x}_0^{1:D}|\boldsymbol{x}^{1:D}) = \sum_{\boldsymbol{x}_0^{1:D}} q_{0|t}(\boldsymbol{x}_0^{1:D}|\boldsymbol{x}^{1:D}) \prod_{d=1}^{D} \frac{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)} \tag{69}$$

$$\sum_{\boldsymbol{x}_0^{\backslash d}} q_{0|t}(\boldsymbol{x}_0^{1:D}|\boldsymbol{x}^{1:D}) = \sum_{\boldsymbol{x}_0^{\backslash d}} q_{0|t}(\boldsymbol{x}_0^d|\boldsymbol{x}^{1:D}) q_{0|t}(\boldsymbol{x}_0^{\backslash d}|\boldsymbol{x}^{1:D}, \boldsymbol{x}_0^d) = q_{0|t}(\boldsymbol{x}_0^d|\boldsymbol{x}^{1:D}) \tag{70}$$

其中 $f_t^\theta(\boldsymbol{x})$ 是 $p_{0|t}^\theta(\cdot|\boldsymbol{x})$ 的参数化神经网络，输出 $\mathbf{x}_{0|t}$ 的分布。等式。(25) 结合等式。(61) 和 $g_t^\theta(\boldsymbol{x}|\boldsymbol{x}) = 1, \forall \boldsymbol{x}$。

A.13 多元素情况下前向和后向转移率的推导

在本节中，我们将展示如何将转换率和比率 g_t^θ 扩展到多元素情况。为了简单起见，我们让 $\boldsymbol{x}^{\backslash d}$ 代表 $\boldsymbol{x}^{1:D\backslash d}$，即没有第 d 个元素的对象。

正向转换率：首先，正向采样的转换率在多元素情况下具有特定的分解公式，如 Campbell 等人所证明的。[2022]，因此，我们将结果总结如下。CTMC 的关键假设是，在同一时间，只有一个维度可以改变。

$$r_t^{1:D}(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) = \sum_{d=1}^{D} r_t^d(\boldsymbol{y}^d|\boldsymbol{x}^d)\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}} \tag{66}$$

其中 $\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}}$ 是克罗内克增量，当且仅当 $\boldsymbol{x}^{\backslash d} = \boldsymbol{y}^{\backslash d}$ 时它为 1。由于我们还假设所有维度过程都是独立的，因此 r_t^d 表示 CTMC 过程在第 d 个元素/维度的转换率。

向后转换率：现在让我们研究 $\hat{r}_t^{1:D}(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D})$。请注意，由于后向过程也是 CTMC，因此它也满足一次只能改变一个维度。我们总结了两个等效的公式如下。

$$\hat{r}_t^{1:D}(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) = \sum_{d=1}^{D} r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d)\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}} \sum_{\boldsymbol{x}_0^d} \frac{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)} q_{0|t}(\boldsymbol{x}_0^d|\boldsymbol{x}^{1:D}) \tag{67}$$

$$\hat{r}_t^{1:D}(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) = \sum_{d=1}^{D} \frac{r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d)\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}}}{\sum_{\boldsymbol{x}_0^d} \frac{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)} q_{0|t}(\boldsymbol{x}_0^d|\boldsymbol{y}^{1:D})} \tag{68}$$

请注意，这两个公式应该是等效的。在实践中，我们使用第一个公式来参数化学习中的反向转移率。

Proof.

$$\frac{q_t(\boldsymbol{y}^{1:D})}{q_t(\boldsymbol{x}^{1:D})} = \sum_{\boldsymbol{x}_0^{1:D}} \frac{q_{t|0}(\boldsymbol{y}^{1:D}|\boldsymbol{x}_0^{1:D})}{q_{t|0}(\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D})} q_{0|t}(\boldsymbol{x}_0^{1:D}|\boldsymbol{x}^{1:D}) = \sum_{\boldsymbol{x}_0^{1:D}} q_{0|t}(\boldsymbol{x}_0^{1:D}|\boldsymbol{x}^{1:D}) \prod_{d=1}^{D} \frac{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)} \tag{69}$$

$$\sum_{\boldsymbol{x}_0^{\backslash d}} q_{0|t}(\boldsymbol{x}_0^{1:D}|\boldsymbol{x}^{1:D}) = \sum_{\boldsymbol{x}_0^{\backslash d}} q_{0|t}(\boldsymbol{x}_0^d|\boldsymbol{x}^{1:D}) q_{0|t}(\boldsymbol{x}_0^{\backslash d}|\boldsymbol{x}^{1:D}, \boldsymbol{x}_0^d) = q_{0|t}(\boldsymbol{x}_0^d|\boldsymbol{x}^{1:D}) \tag{70}$$

(Case 1)

$$
\begin{aligned}
\hat{r}_t^{1:D}(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) &= r_t^{1:D}(\boldsymbol{x}^{1:D}|\boldsymbol{y}^{1:D})\frac{q_t(\boldsymbol{y}^{1:D})}{q_t(\boldsymbol{x}^{1:D})} \\
&= \Big(\sum_{d=1}^{D} r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d)\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}}\Big)\Big(\sum_{\boldsymbol{x}_0^{1:D}} q_{0|t}(\boldsymbol{x}_0^{1:D}|\boldsymbol{x}^{1:D})\prod_{d=1}^{D}\frac{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}\Big) \\
&= \sum_{d=1}^{D}\sum_{\boldsymbol{x}_0^{1:D}} r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d)\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}} q_{0|t}(\boldsymbol{x}_0^{1:D}|\boldsymbol{x}^{1:D})\frac{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)} \\
&= \sum_{d=1}^{D} r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d)\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}}\sum_{\boldsymbol{x}_0^d}\frac{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}\sum_{\boldsymbol{x}_0^{\backslash d}} q_{0|t}(\boldsymbol{x}_0^{1:D}|\boldsymbol{x}^{1:D}) \\
&= \sum_{d=1}^{D} r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d)\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}}\sum_{\boldsymbol{x}_0^d}\frac{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)} q_{0|t}(\boldsymbol{x}_0^d|\boldsymbol{x}^{1:D}) \qquad (71)
\end{aligned}
$$

(Case 2)

$$
\begin{aligned}
\hat{r}_t^{1:D}(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) &= r_t^{1:D}(\boldsymbol{x}^{1:D}|\boldsymbol{y}^{1:D})/\frac{q_t(\boldsymbol{x}^{1:D})}{q_t(\boldsymbol{y}^{1:D})} \\
&= \sum_{d=1}^{D}\frac{r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d)\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}}}{\sum_{\boldsymbol{x}_0^{1:D}} q_{0|t}(\boldsymbol{x}_0^{1:D}|\boldsymbol{y}^{1:D})\prod_{i=1}^{D}\frac{q_{t|0}(\boldsymbol{x}^i|\boldsymbol{x}_0^i)}{q_{t|0}(\boldsymbol{y}^i|\boldsymbol{x}_0^i)}} \\
&= \sum_{d=1}^{D}\frac{r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d)\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}}}{\sum_{\boldsymbol{x}_0^{1:D}} q_{0|t}(\boldsymbol{x}_0^{1:D}|\boldsymbol{y}^{1:D})\frac{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)}} \\
&= \sum_{d=1}^{D}\frac{r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d)\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}}}{\sum_{\boldsymbol{x}_0^d}\frac{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)} q_{0|t}(\boldsymbol{x}_0^d|\boldsymbol{y}^{1:D})} \qquad (72)
\end{aligned}
$$

$\square$

Ratio: We now define an estimator $g_t^{\theta,d}(\boldsymbol{x}^d|\boldsymbol{y}^{1:D})$ as follows.

$$
g_t^{\theta,d}(\boldsymbol{x}^d|\boldsymbol{y}^{1:D}) := \sum_{\boldsymbol{x}_0^d}\frac{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)} p_{0|t}^{\theta}(\boldsymbol{x}_0^d|\boldsymbol{y}^{1:D}) \approx \sum_{\boldsymbol{x}_0^d}\frac{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)} q_{0|t}(\boldsymbol{x}_0^d|\boldsymbol{y}^{1:D}) = \frac{q_t(\boldsymbol{x}^d|\boldsymbol{y}^{\backslash d})}{q_t(\boldsymbol{y}^d|\boldsymbol{y}^{\backslash d})} \qquad (73)
$$

$$
g_t^{\theta}(\boldsymbol{x}^{1:D}|\boldsymbol{y}^{1:D}) := \prod_{d=1}^{D} g_t^{\theta,d}(\boldsymbol{x}^d|\boldsymbol{y}^{1:D}) = \sum_{\boldsymbol{x}_0^{1:D}}\frac{q_{t|0}(\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D})}{q_{t|0}(\boldsymbol{y}^{1:D}|\boldsymbol{x}_0^{1:D})} p_{0|t}^{\theta}(\boldsymbol{x}_0^{1:D}|\boldsymbol{y}^{1:D}) \approx \frac{q_t(\boldsymbol{x}^{1:D})}{q_t(\boldsymbol{y}^{1:D})} \qquad (74)
$$

We can extend the vector formulation Eq. (25) in Proposition 4 to $g_t^{\theta,d}(\boldsymbol{x}^d|\boldsymbol{y}^{1:D})$:

$$
g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D}) = \frac{1}{\langle \boldsymbol{x}^d, \boldsymbol{m}^d\rangle}\Big[\big(1 - \frac{\overline{\alpha}_{t|0}\langle f_t^{\theta,d}(\boldsymbol{x}^{1:D}), \boldsymbol{x}^d\rangle}{\overline{\alpha}_{t|0} + (1-\overline{\alpha}_{t|0})\langle \boldsymbol{x}^d, \boldsymbol{m}^d\rangle}\big)\boldsymbol{m}^d + \frac{\overline{\alpha}_{t|0}}{1-\overline{\alpha}_{t|0}} f_t^{\theta,d}(\boldsymbol{x}^{1:D})\Big] \odot (\boldsymbol{1} - \boldsymbol{x}^d) + \boldsymbol{x}^d \qquad (75)
$$

Then, we can derive two approximators for transition rate $\hat{r}_t^{1:D}(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D})$ as follows.

$$
[\hat{r}_t^{\theta,1:D}]^1(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) := \sum_{d=1}^{D} r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d) g_t^{\theta,d}(\boldsymbol{y}^d|\boldsymbol{x}^{1:D})\cdot\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}} \approx \text{Eq. (67)} \qquad (76)
$$

$$
[\hat{r}_t^{\theta,1:D}]^2(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) := \sum_{d=1}^{D}\frac{r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d)}{g_t^{\theta,d}(\boldsymbol{x}^d|\boldsymbol{y}^{1:D})}\cdot\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}} \approx \text{Eq. (68)} \qquad (77)
$$

（案例一）

$$
\begin{aligned}
\hat{r}_t^{1:D}(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) &= r_t^{1:D}(\boldsymbol{x}^{1:D}|\boldsymbol{y}^{1:D})\frac{q_t(\boldsymbol{y}^{1:D})}{q_t(\boldsymbol{x}^{1:D})} \\
&= \Big(\sum_{d=1}^{D} r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d)\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}}\Big)\Big(\sum_{\boldsymbol{x}_0^{1:D}} q_{0|t}(\boldsymbol{x}_0^{1:D}|\boldsymbol{x}^{1:D})\prod_{d=1}^{D}\frac{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}\Big) \\
&= \sum_{d=1}^{D}\sum_{\boldsymbol{x}_0^{1:D}} r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d)\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}} q_{0|t}(\boldsymbol{x}_0^{1:D}|\boldsymbol{x}^{1:D})\frac{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)} \\
&= \sum_{d=1}^{D} r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d)\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}}\sum_{\boldsymbol{x}_0^d}\frac{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}\sum_{\boldsymbol{x}_0^{\backslash d}} q_{0|t}(\boldsymbol{x}_0^{1:D}|\boldsymbol{x}^{1:D}) \\
&= \sum_{d=1}^{D} r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d)\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}}\sum_{\boldsymbol{x}_0^d}\frac{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)} q_{0|t}(\boldsymbol{x}_0^d|\boldsymbol{x}^{1:D}) \qquad (71)
\end{aligned}
$$

(案例2)

$$
\begin{aligned}
\hat{r}_t^{1:D}(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) &= r_t^{1:D}(\boldsymbol{x}^{1:D}|\boldsymbol{y}^{1:D})/\frac{q_t(\boldsymbol{x}^{1:D})}{q_t(\boldsymbol{y}^{1:D})} \\
&= \sum_{d=1}^{D}\frac{r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d)\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}}}{\sum_{\boldsymbol{x}_0^{1:D}} q_{0|t}(\boldsymbol{x}_0^{1:D}|\boldsymbol{y}^{1:D})\prod_{i=1}^{D}\frac{q_{t|0}(\boldsymbol{x}^i|\boldsymbol{x}_0^i)}{q_{t|0}(\boldsymbol{y}^i|\boldsymbol{x}_0^i)}} \\
&= \sum_{d=1}^{D}\frac{r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d)\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}}}{\sum_{\boldsymbol{x}_0^{1:D}} q_{0|t}(\boldsymbol{x}_0^{1:D}|\boldsymbol{y}^{1:D})\frac{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)}} \\
&= \sum_{d=1}^{D}\frac{r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d)\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}}}{\sum_{\boldsymbol{x}_0^d}\frac{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)} q_{0|t}(\boldsymbol{x}_0^d|\boldsymbol{y}^{1:D})} \qquad (72)
\end{aligned}
$$

□

比率：我们现在定义一个估计器 $g_t^{\theta,d}(\boldsymbol{x}^d|\boldsymbol{y}^{1:D})$ 如下。

$$
g_t^{\theta,d}(\boldsymbol{x}^d|\boldsymbol{y}^{1:D}) := \sum_{\boldsymbol{x}_0^d}\frac{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)} p_{0|t}^{\theta}(\boldsymbol{x}_0^d|\boldsymbol{y}^{1:D}) \approx \sum_{\boldsymbol{x}_0^d}\frac{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)} q_{0|t}(\boldsymbol{x}_0^d|\boldsymbol{y}^{1:D}) = \frac{q_t(\boldsymbol{x}^d|\boldsymbol{y}^{\backslash d})}{q_t(\boldsymbol{y}^d|\boldsymbol{y}^{\backslash d})} \tag{73}
$$

$$
g_t^{\theta}(\boldsymbol{x}^{1:D}|\boldsymbol{y}^{1:D}) := \prod_{d=1}^{D} g_t^{\theta,d}(\boldsymbol{x}^d|\boldsymbol{y}^{1:D}) = \sum_{\boldsymbol{x}_0^{1:D}}\frac{q_{t|0}(\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D})}{q_{t|0}(\boldsymbol{y}^{1:D}|\boldsymbol{x}_0^{1:D})} p_{0|t}^{\theta}(\boldsymbol{x}_0^{1:D}|\boldsymbol{y}^{1:D}) \approx \frac{q_t(\boldsymbol{x}^{1:D})}{q_t(\boldsymbol{y}^{1:D})} \tag{74}
$$

我们可以扩展向量公式：(25) 命题 4 至 $g_t^{\theta,d}(\boldsymbol{x}^d|\boldsymbol{y}^{1:D})$：

$$
g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D}) = \frac{1}{\langle \boldsymbol{x}^d, \boldsymbol{m}^d\rangle}\Big[\big(1 - \frac{\overline{\alpha}_{t|0}\langle f_t^{\theta,d}(\boldsymbol{x}^{1:D}), \boldsymbol{x}^d\rangle}{\overline{\alpha}_{t|0} + (1-\overline{\alpha}_{t|0})\langle \boldsymbol{x}^d, \boldsymbol{m}^d\rangle}\big)\boldsymbol{m}^d + \frac{\overline{\alpha}_{t|0}}{1-\overline{\alpha}_{t|0}} f_t^{\theta,d}(\boldsymbol{x}^{1:D})\Big] \odot (\mathbf{1} - \boldsymbol{x}^d) + \boldsymbol{x}^d \tag{75}
$$

然后，我们可以导出转移率 $\hat{r}_t^{1:D}(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D})$ 的两个近似器，如下所示。

$$
[\hat{r}_t^{\theta,1:D}]^1(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) := \sum_{d=1}^{D} r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d) g_t^{\theta,d}(\boldsymbol{y}^d|\boldsymbol{x}^{1:D}) \cdot \delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}} \approx \text{Eq. (67)} \tag{76}
$$

$$
[\hat{r}_t^{\theta,1:D}]^2(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) := \sum_{d=1}^{D}\frac{r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d)}{g_t^{\theta,d}(\boldsymbol{x}^d|\boldsymbol{y}^{1:D})} \cdot \delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{y}^{\backslash d}} \approx \text{Eq. (68)} \tag{77}
$$

A.14 Derivation of Negative VLB Loss in Multi-element Case

As forward process is defined in Eq. (19), in multi-element case we can easily get

$$r_t^d(\boldsymbol{x}^d|\cdot) = R_t^d \boldsymbol{x}^d = \beta(t)\big(\langle \boldsymbol{x}^d, \boldsymbol{m}^d\rangle \mathbf{1} - \boldsymbol{x}^d\big) \tag{78}$$

$$r_t^d(\cdot|\boldsymbol{x}^d) = (R_t^d)^\top \boldsymbol{x}^d = \beta(t)\big(\boldsymbol{m}^d - \boldsymbol{x}^d\big) \tag{79}$$

The $r_t^d(\boldsymbol{x}|\cdot)$ and $r_t^d(\cdot|\boldsymbol{x})$ are essentially the $\boldsymbol{x}$-th column and row of the transition rate matrix R_t^d.

In Mult-element case, the negative VLB loss in Eq. (23) can be written as

$$T\, \mathbb{E}_{\substack{t\sim \text{Uni}(0,T) \\ \boldsymbol{x}^{1:D}\sim q_{t|0}}}\Big[\sum_{\boldsymbol{z}^{1:D}\neq \boldsymbol{x}^{1:D}} \hat{r}_t^{\theta,1:D}(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D}) - \sum_{\boldsymbol{z}^{1:D}\neq \boldsymbol{x}^{1:D}} r_t^{1:D}(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D}) \log \hat{r}_t^{\theta,1:D}(\boldsymbol{x}^{1:D}|\boldsymbol{z}^{1:D})\Big] \tag{80}$$

As there are two terms, let's work on each term separately.

A.14.1 Term 1

Based on the formulation of $r_t^{1:D}$ and $\hat{r}_t^{\theta,1:D}$, we can rewrite the first term as

$$\begin{aligned}
\text{Term1} &= T\, \mathbb{E}_{t,\boldsymbol{x}^{1:D}} \sum_{\boldsymbol{z}^{1:D}\neq \boldsymbol{x}^{1:D}} \hat{r}_t^{\theta,1:D}(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D}) \\
&= T\, \mathbb{E}_{t,\boldsymbol{x}^{1:D}} \sum_{\boldsymbol{z}^{1:D}\neq \boldsymbol{x}^{1:D}} \sum_{d=1}^{D} r_t^d(\boldsymbol{x}^d|\boldsymbol{z}^d) g_t^{\theta,d}(\boldsymbol{z}^d|\boldsymbol{x}^{1:D}) \cdot \delta_{\boldsymbol{x}^{\setminus d},\boldsymbol{z}^{\setminus d}} \\
&= T\, \mathbb{E}_{t,\boldsymbol{x}^{1:D}} \Big[\sum_{d=1}^{D} \sum_{\boldsymbol{z}^d\neq \boldsymbol{x}^d} r_t^d(\boldsymbol{x}^d|\boldsymbol{z}^d) g_t^{\theta,d}(\boldsymbol{z}^d|\boldsymbol{x}^{1:D})\Big] \\
&= T\, \mathbb{E}_{t,\boldsymbol{x}^{1:D}} \Big[\sum_{d=1}^{D} r_t^d(\boldsymbol{x}^d|\cdot)^\top g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D})\Big] + \text{const.} \\
&= T\, \mathbb{E}_{\substack{t\sim \text{Uni.}(0,T) \\ \boldsymbol{x}^{1:D}\sim q_{t|0}}} \beta(t) \sum_{d=1}^{D} \langle \boldsymbol{x}^d, \boldsymbol{m}^d\rangle \Big[\mathbf{1}^\top g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D})\Big] + \text{const.}
\end{aligned} \tag{81}$$

A.14.2 Term 2

As the evaluation of $\hat{r}_t^\theta(\cdot|\boldsymbol{x})$ for any $\boldsymbol{x}$ requires a single forward pass of the parameterized network $p_{0|t}^\theta(\cdot|\boldsymbol{x})$, the second term within the expectation of Eq. (23) requires multiple passes of the model. This complexity is even greatly amplified in cases with multi-element objects. Campbell et al. [2022] avoids the multiple passes by changing the expectation variable through importance sampling. We take a similar approach to simplify the second term. Differently, Campbell et al. [2022] uses a specific sampling distribution (same as the forward transition rate) to introduce the auxiliary variable for changing the expectation variable, we generalize it to use a general sampling process S_t defined below.

Let $\boldsymbol{y}^{1:D}$ be the new variable upon which the exchanged expectation is based, and assume that $\boldsymbol{y}^{1:D}$ is sampled from an unnormalized joint distribution $S_t(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D})$. We restrict $S_t(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D})$ to be a unnormalized probability that is nonzero if and only if $\boldsymbol{y}^{1:D}$ and $\boldsymbol{x}^{1:D}$ are different at a single element. Formally, we can write the unnormalized distribution as

$$S_t(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) = (1-\delta_{\boldsymbol{y}^{1:D},\boldsymbol{x}^{1:D}}) \sum_{d=1}^{D} S_t^d(\boldsymbol{y}^d|\boldsymbol{x}^d)\delta_{\boldsymbol{y}^{\setminus d},\boldsymbol{x}^{\setminus d}}$$

$$\text{with normalizer } \mathcal{S}_t(\boldsymbol{x}^{1:D}) = \sum_{\boldsymbol{y}^{1:D}} S_t(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) = \sum_{d=1}^{D}\sum_{\boldsymbol{y}^d\neq \boldsymbol{x}^d} S_t^d(\boldsymbol{y}^d|\boldsymbol{x}^d) \tag{82}$$

where $S_t^d(\boldsymbol{y}^d|\boldsymbol{x}^d)$ is any unnormalized probability at dimension d.

A.14 多元素情况下负VLB损失的推导

正如方程中定义的前向过程。(19) 在多元素情况下我们很容易得到

$$r_t^d(\boldsymbol{x}^d|\cdot) = R_t^d \boldsymbol{x}^d = \beta(t)\big(\langle \boldsymbol{x}^d, \boldsymbol{m}^d\rangle \mathbf{1} - \boldsymbol{x}^d\big) \tag{78}$$

$$r_t^d(\cdot|\boldsymbol{x}^d) = (R_t^d)^\top \boldsymbol{x}^d = \beta(t)\big(\boldsymbol{m}^d - \boldsymbol{x}^d\big) \tag{79}$$

$r_t^d(\boldsymbol{x}|\cdot)$ 和 $r_t^d(\cdot|\boldsymbol{x})$ 本质上是转移率矩阵 R_t^d 的第 $\boldsymbol{x}$ 列和行。

在多元素情况下，方程中的负 VLB 损失。(23) 可以写成

$$T\,\mathbb{E}_{\substack{t\sim\text{Uni}(0,T)\\ \boldsymbol{x}^{1:D}\sim q_{t|0}}}\Big[\sum_{\boldsymbol{z}^{1:D}\neq\boldsymbol{x}^{1:D}} \hat{r}_t^{\theta,1:D}(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D}) - \sum_{\boldsymbol{z}^{1:D}\neq\boldsymbol{x}^{1:D}} r_t^{1:D}(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D})\log \hat{r}_t^{\theta,1:D}(\boldsymbol{x}^{1:D}|\boldsymbol{z}^{1:D})\Big] \tag{80}$$

由于有两个术语，让我们分别研究每个术语。

A.14.1 第 1 条

根据 $r_t^{1:D}$ 和 $\hat{r}_t^{\theta,1:D}$ 的公式，我们可以将第一项重写为

$$\begin{aligned}
\text{Term1} &= T\,\mathbb{E}_{t,\boldsymbol{x}^{1:D}} \sum_{\boldsymbol{z}^{1:D}\neq\boldsymbol{x}^{1:D}} \hat{r}_t^{\theta,1:D}(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D}) \\
&= T\,\mathbb{E}_{t,\boldsymbol{x}^{1:D}} \sum_{\boldsymbol{z}^{1:D}\neq\boldsymbol{x}^{1:D}} \sum_{d=1}^{D} r_t^d(\boldsymbol{x}^d|\boldsymbol{z}^d) g_t^{\theta,d}(\boldsymbol{z}^d|\boldsymbol{x}^{1:D}) \cdot \delta_{\boldsymbol{x}^{\setminus d},\boldsymbol{z}^{\setminus d}} \\
&= T\,\mathbb{E}_{t,\boldsymbol{x}^{1:D}} \Big[\sum_{d=1}^{D} \sum_{\boldsymbol{z}^d\neq\boldsymbol{x}^d} r_t^d(\boldsymbol{x}^d|\boldsymbol{z}^d) g_t^{\theta,d}(\boldsymbol{z}^d|\boldsymbol{x}^{1:D})\Big] \\
&= T\,\mathbb{E}_{t,\boldsymbol{x}^{1:D}} \Big[\sum_{d=1}^{D} r_t^d(\boldsymbol{x}^d|\cdot)^\top g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D})\Big] + \text{const.} \\
&= T\,\mathbb{E}_{\substack{t\sim\text{Uni.}(0,T)\\ \boldsymbol{x}^{1:D}\sim q_{t|0}}} \beta(t) \sum_{d=1}^{D} \langle \boldsymbol{x}^d, \boldsymbol{m}^d\rangle \Big[\mathbf{1}^\top g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D})\Big] + \text{const.}
\end{aligned} \tag{81}$$

A.14.2 第 2 条

由于对任何 $\boldsymbol{x}$ 的 $\hat{r}_t^\theta(\cdot|\boldsymbol{x})$ 的评估需要参数化网络 $p_{0|t}^\theta(\cdot|\boldsymbol{x})$ 的单次前向传递，因此方程的期望中的第二项。(23) 需要模型的多次传递。在多元素对象的情况下，这种复杂性甚至会大大增加。坎贝尔等人。[2022]通过重要性采样改变期望变量来避免多次传递。我们采用类似的方法来简化第二项。不同的是，坎贝尔等人。[2022]使用特定的采样分布（与前向转移率相同）引入用于改变期望变量的辅助变量，我们将其推广为使用下面定义的通用采样过程S_t。

令 $\boldsymbol{y}^{1:D}$ 为交换期望所基于的新变量，并假设 $\boldsymbol{y}^{1:D}$ 是从非标准化联合分布 $S_t(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D})$ 中采样的。当且仅当 $\boldsymbol{y}^{1:D}$ 和 $\boldsymbol{x}^{1:D}$ 在单个元素上不同时，我们将 $S_t(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D})$ 限制为非零非标准化概率。形式上，我们可以将非标准化分布写为

$$\begin{aligned}
S_t(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) &= (1-\delta_{\boldsymbol{y}^{1:D},\boldsymbol{x}^{1:D}}) \sum_{d=1}^{D} S_t^d(\boldsymbol{y}^d|\boldsymbol{x}^d)\delta_{\boldsymbol{y}^{\setminus d},\boldsymbol{x}^{\setminus d}} \\
\text{with normalizer } \mathcal{S}_t(\boldsymbol{x}^{1:D}) &= \sum_{\boldsymbol{y}^{1:D}} S_t(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) = \sum_{d=1}^{D}\sum_{\boldsymbol{y}^d\neq\boldsymbol{x}^d} S_t^d(\boldsymbol{y}^d|\boldsymbol{x}^d)
\end{aligned} \tag{82}$$

其中 $S_t^d(\boldsymbol{y}^d|\boldsymbol{x}^d)$ 是维度 d 上的任何非标准化概率。

Now for the second term, we have

$$
\begin{aligned}
\text{Term2} &= T\,\mathbb{E}_{t,\boldsymbol{x}^{1:D}} \sum_{\boldsymbol{z}^{1:D}\neq \boldsymbol{x}^{1:D}} r_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D}) \log \hat{r}_t^{\theta}(\boldsymbol{x}^{1:D}|\boldsymbol{z}^{1:D}) \\
&= T\,\mathbb{E}_{t,\boldsymbol{x}^{1:D}} \sum_{\boldsymbol{z}^{1:D}\neq \boldsymbol{x}^{1:D}} \frac{S_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D})}{\mathcal{S}(\boldsymbol{x}^{1:D})} \cdot \frac{\mathcal{S}_t(\boldsymbol{x}^{1:D})}{S_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D})} \cdot r_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D}) \log \hat{r}_t^{\theta}(\boldsymbol{x}^{1:D}|\boldsymbol{z}^{1:D}) \\
&= T\,\mathbb{E}_{\substack{t,\boldsymbol{x}^{1:D} \\ \boldsymbol{z}^{1:D}\sim S_t}} \Big[\frac{\mathcal{S}_t(\boldsymbol{x}^{1:D})}{S_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D})} \cdot r_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D}) \log \hat{r}_t^{\theta}(\boldsymbol{x}^{1:D}|\boldsymbol{z}^{1:D})\Big] \qquad (\text{As } S_t \text{ samples } \boldsymbol{z}^{1:D}\neq \boldsymbol{x}^{1:D}\ ) \\
&= T\,\mathbb{E}_{t,\boldsymbol{z}^{1:D}\sim S_t} \sum_{\boldsymbol{x}^{1:D}} \Big[q_{S_t}(\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D},\boldsymbol{z}^{1:D}) \cdot \frac{\mathcal{S}_t(\boldsymbol{x}^{1:D})}{S_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D})} \cdot r_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D}) \log \hat{r}_t^{\theta}(\boldsymbol{x}^{1:D}|\boldsymbol{z}^{1:D})\Big]
\end{aligned} \tag{83}
$$

where $q_{S_t}(\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D},\boldsymbol{z}^{1:D})$ is the conditional posterior distribution such that (for clearity we replace $\boldsymbol{x}^{1:D},\boldsymbol{z}^{1:D}$ with $\boldsymbol{x}_t^{1:D},\boldsymbol{z}_t^{1:D}$ respectively, as they are variables at time t)

$$
\begin{aligned}
& q_{S_t}(\boldsymbol{x}_t^{1:D}|\boldsymbol{x}_0^{1:D},\boldsymbol{z}_t^{1:D}) \\
&= \frac{q_{S_t}(\boldsymbol{x}_t^{1:D},\boldsymbol{z}_t^{1:D}|\boldsymbol{x}_0^{1:D})}{\sum_{\boldsymbol{y}_t^{1:D}} q_{S_t}(\boldsymbol{y}_t^{1:D},\boldsymbol{z}_t^{1:D}|\boldsymbol{x}_0^{1:D})} = \frac{q_{t|0}(\boldsymbol{x}_t^{1:D}|\boldsymbol{x}_0^{1:D}) \cdot S_t(\boldsymbol{z}_t^{1:D}|\boldsymbol{x}_t^{1:D})/\mathcal{S}_t(\boldsymbol{x}_t^{1:D})}{\sum_{\boldsymbol{y}_y^{1:D}} q_{t|0}(\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D}) \cdot S_t(\boldsymbol{z}_t^{1:D}|\boldsymbol{y}_t^{1:D})/\mathcal{S}_t(\boldsymbol{y}_t^{1:D})} \\
&= \frac{(1-\delta_{\boldsymbol{z}_t,\boldsymbol{x}_t})\sum_{d=1}^{D}\delta_{\boldsymbol{z}_t^{\backslash d},\boldsymbol{x}_t^{\backslash d}} S_t^d(\boldsymbol{z}_t^d|\boldsymbol{x}_t^d)/\mathcal{S}_t(\boldsymbol{x}_t^{1:D}) \cdot q_{t|0}(\boldsymbol{x}_t^d \circ \boldsymbol{z}_t^{\backslash d}|\boldsymbol{x}_0^{1:D})}{\sum_{\boldsymbol{y}_t^{1:D}}\big[(1-\delta_{\boldsymbol{z}_t,\boldsymbol{y}_t})\sum_{d=1}^{D}\delta_{\boldsymbol{z}_t^{\backslash d},\boldsymbol{y}_t^{\backslash d}} S_t^d(\boldsymbol{z}_t^d|\boldsymbol{y}_t^d)/\mathcal{S}_t(\boldsymbol{y}_t^{1:D}) \cdot q_{t|0}(\boldsymbol{y}_t^d \circ \boldsymbol{z}_t^{\backslash d}|\boldsymbol{x}_0^{1:D})\big]} \\
&= \frac{(1-\delta_{\boldsymbol{z}_t,\boldsymbol{x}_t})\sum_{d=1}^{D}\delta_{\boldsymbol{z}_t^{\backslash d},\boldsymbol{x}_t^{\backslash d}} \frac{S_t^d(\boldsymbol{z}_t^d|\boldsymbol{x}_t^d)}{\mathcal{S}_t(\boldsymbol{x}_t^{1:D})} \cdot \frac{q_{t|0}(\boldsymbol{x}_t^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}_t^d|\boldsymbol{x}_0^d)}}{\sum_{\boldsymbol{y}_t^{1:D}}\big[(1-\delta_{\boldsymbol{z}_t,\boldsymbol{y}_t})\sum_{d=1}^{D}\delta_{\boldsymbol{z}_t^{\backslash d},\boldsymbol{y}_t^{\backslash d}} \frac{S_t^d(\boldsymbol{z}_t^d|\boldsymbol{y}_t^d)}{\mathcal{S}_t(\boldsymbol{y}_t^{1:D})} \cdot \frac{q_{t|0}(\boldsymbol{y}_t^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}_t^d|\boldsymbol{x}_0^d)}\big]} \\
&= \frac{(1-\delta_{\boldsymbol{z}_t,\boldsymbol{x}_t})\sum_{d=1}^{D}\delta_{\boldsymbol{z}_t^{\backslash d},\boldsymbol{x}_t^{\backslash d}} \frac{S_t^d(\boldsymbol{z}_t^d|\boldsymbol{x}_t^d)}{\mathcal{S}_t(\boldsymbol{x}_t^{1:D})} \cdot \frac{q_{t|0}(\boldsymbol{x}_t^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}_t^d|\boldsymbol{x}_0^d)}}{\sum_{d=1}^{D}\sum_{\boldsymbol{y}_t^d\neq \boldsymbol{z}^d} \frac{S_t^d(\boldsymbol{z}_t^d|\boldsymbol{y}_t^d)}{\mathcal{S}_t(\boldsymbol{y}_t^d\circ \boldsymbol{z}_t^{\backslash d})} \cdot \frac{q_{t|0}(\boldsymbol{y}_t^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}_t^d|\boldsymbol{x}_0^d)}}
\end{aligned} \tag{84}
$$

Now taking the formulation of q_{S_t} back into Term 2, we further simplify Term 2 as

$$
\begin{aligned}
&\text{Term2} \\
&= T\,\mathbb{E}_{t,\boldsymbol{z}^{1:D}\sim S_t} \frac{\sum_{\boldsymbol{x}^{1:D}}\Big[(1-\delta_{\boldsymbol{z},\boldsymbol{x}})\sum_{d=1}^{D}\delta_{\boldsymbol{z}^{\backslash d},\boldsymbol{x}^{\backslash d}} \frac{S_t^d(\boldsymbol{z}^d|\boldsymbol{x}^d)}{S_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D})} \cdot \frac{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)} \cdot r_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D}) \log \hat{r}_t^{\theta}(\boldsymbol{x}^{1:D}|\boldsymbol{z}^{1:D})\Big]}{\sum_{d=1}^{D}\sum_{\boldsymbol{y}^d\neq \boldsymbol{z}^d} \frac{S_t^d(\boldsymbol{z}^d|\boldsymbol{y}^d)}{\mathcal{S}_t(\boldsymbol{y}^d\circ \boldsymbol{z}^{\backslash d})} \cdot \frac{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}} \\
&= \mathbb{E}_{t,\boldsymbol{z}^{1:D}\sim S_t} \frac{\sum_{d=1}^{D}\sum_{\boldsymbol{x}^d\neq \boldsymbol{z}^d} \frac{S_t^d(\boldsymbol{z}^d|\boldsymbol{x}^d)}{S_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^d\circ \boldsymbol{z}^{\backslash d})} \cdot \frac{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)} \cdot r_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^d\circ \boldsymbol{z}^{\backslash d}) \log \hat{r}_t^{\theta}(\boldsymbol{x}^d\circ \boldsymbol{z}^{\backslash d}|\boldsymbol{z}^{1:D})}{\sum_{d=1}^{D}\sum_{\boldsymbol{y}^d\neq \boldsymbol{z}^d} \frac{S_t^d(\boldsymbol{z}^d|\boldsymbol{y}^d)}{\mathcal{S}_t(\boldsymbol{y}^d\circ \boldsymbol{z}^{\backslash d})} \cdot \frac{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}} \\
&= \mathbb{E}_{t,\boldsymbol{z}^{1:D}\sim S_t} \frac{\sum_{d=1}^{D}\sum_{\boldsymbol{x}^d\neq \boldsymbol{z}^d} \frac{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)} \cdot r_t^d(\boldsymbol{z}^d|\boldsymbol{x}^d) \cdot \log r_t^d(\boldsymbol{z}^d|\boldsymbol{x}^d) g_t^{\theta,d}(\boldsymbol{x}^d|\boldsymbol{z}^{1:D})}{\sum_{d=1}^{D}\sum_{\boldsymbol{y}^d\neq \boldsymbol{z}^d} \frac{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)} \cdot \frac{S_t^d(\boldsymbol{z}^d|\boldsymbol{y}^d)}{\mathcal{S}_t(\boldsymbol{y}^d\circ \boldsymbol{z}^{\backslash d})}} \\
&= \mathbb{E}_{t,\boldsymbol{z}^{1:D}\sim S_t} \frac{\sum_{d=1}^{D}\sum_{\boldsymbol{x}^d\neq \boldsymbol{z}^d} \frac{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)} \cdot r_t^d(\boldsymbol{z}^d|\boldsymbol{x}^d) \cdot \log g_t^{\theta,d}(\boldsymbol{x}^d|\boldsymbol{z}^{1:D})}{\sum_{d=1}^{D}\sum_{\boldsymbol{y}^d\neq \boldsymbol{z}^d} \frac{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)} \cdot \frac{S_t^d(\boldsymbol{z}^d|\boldsymbol{y}^d)}{\mathcal{S}_t(\boldsymbol{y}^d\circ \boldsymbol{z}^{\backslash d})}} + \text{const.}
\end{aligned} \tag{85}
$$

The above equation further shows that the sampling distribution S_t for adding exchanging variable $\boldsymbol{z}^{1:D}$ only affect a weighting term of the loss computation. Let us define the scalar weighting term as

$$
\mathcal{M}_{S_t}(\boldsymbol{z}^{1:D}|\boldsymbol{x}_0^{1:D}) := \sum_{d=1}^{D}\sum_{\boldsymbol{y}^d\neq \boldsymbol{z}^d} \frac{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)} \cdot \frac{S_t^d(\boldsymbol{z}^d|\boldsymbol{y}^d)}{\mathcal{S}_t(\boldsymbol{y}^d\circ \boldsymbol{z}^{\backslash d})} \tag{86}
$$

现在，对于第二个学期，我们有

$$\text{Term2} = T\,\mathbb{E}_{t,\boldsymbol{x}^{1:D}} \sum_{\boldsymbol{z}^{1:D}\neq \boldsymbol{x}^{1:D}} r_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D}) \log \hat{r}_t^{\theta}(\boldsymbol{x}^{1:D}|\boldsymbol{z}^{1:D})$$

$$= T\,\mathbb{E}_{t,\boldsymbol{x}^{1:D}} \sum_{\boldsymbol{z}^{1:D}\neq \boldsymbol{x}^{1:D}} \frac{S_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D})}{\mathcal{S}(\boldsymbol{x}^{1:D})} \cdot \frac{\mathcal{S}_t(\boldsymbol{x}^{1:D})}{S_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D})} \cdot r_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D}) \log \hat{r}_t^{\theta}(\boldsymbol{x}^{1:D}|\boldsymbol{z}^{1:D})$$

$$= T\,\mathbb{E}_{\substack{t,\boldsymbol{x}^{1:D}\\ \boldsymbol{z}^{1:D}\sim S_t}} \left[\frac{\mathcal{S}_t(\boldsymbol{x}^{1:D})}{S_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D})} \cdot r_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D}) \log \hat{r}_t^{\theta}(\boldsymbol{x}^{1:D}|\boldsymbol{z}^{1:D})\right] \qquad (\text{As } S_t \text{ samples } \boldsymbol{z}^{1:D}\neq \boldsymbol{x}^{1:D})$$

$$= T\,\mathbb{E}_{t,\boldsymbol{z}^{1:D}\sim S_t} \sum_{\boldsymbol{x}^{1:D}} \left[q_{S_t}(\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D},\boldsymbol{z}^{1:D}) \cdot \frac{\mathcal{S}_t(\boldsymbol{x}^{1:D})}{S_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D})} \cdot r_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D}) \log \hat{r}_t^{\theta}(\boldsymbol{x}^{1:D}|\boldsymbol{z}^{1:D})\right] \tag{83}$$

其中 $q_{S_t}(\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D},\boldsymbol{z}^{1:D})$ 是条件后验分布，使得（为了清楚起见，我们分别用 $\boldsymbol{x}_t^{1:D},\boldsymbol{z}_t^{1:D}$ 替换 $\boldsymbol{x}^{1:D},\boldsymbol{z}^{1:D}$，因为它们是时间 t 的变量）

$$\begin{aligned}
&q_{S_t}(\boldsymbol{x}_t^{1:D}|\boldsymbol{x}_0^{1:D},\boldsymbol{z}_t^{1:D})\\
&= \frac{q_{S_t}(\boldsymbol{x}_t^{1:D},\boldsymbol{z}_t^{1:D}|\boldsymbol{x}_0^{1:D})}{\sum_{\boldsymbol{y}_t^{1:D}} q_{S_t}(\boldsymbol{y}_t^{1:D},\boldsymbol{z}_t^{1:D}|\boldsymbol{x}_0^{1:D})} = \frac{q_{t|0}(\boldsymbol{x}_t^{1:D}|\boldsymbol{x}_0^{1:D})\cdot S_t(\boldsymbol{z}_t^{1:D}|\boldsymbol{x}_t^{1:D})/\mathcal{S}_t(\boldsymbol{x}_t^{1:D})}{\sum_{\boldsymbol{y}_y^{1:D}} q_{t|0}(\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D})\cdot S_t(\boldsymbol{z}_t^{1:D}|\boldsymbol{y}_t^{1:D})/\mathcal{S}_t(\boldsymbol{y}_t^{1:D})}\\
&= \frac{(1-\delta_{\boldsymbol{z}_t,\boldsymbol{x}_t})\sum_{d=1}^{D}\delta_{\boldsymbol{z}_t^{\setminus d},\boldsymbol{x}_t^{\setminus d}} S_t^d(\boldsymbol{z}_t^d|\boldsymbol{x}_t^d)/\mathcal{S}_t(\boldsymbol{x}_t^{1:D})\cdot q_{t|0}(\boldsymbol{x}_t^d\circ \boldsymbol{z}_t^{\setminus d}|\boldsymbol{x}_0^{1:D})}{\sum_{\boldsymbol{y}_t^{1:D}}\left[(1-\delta_{\boldsymbol{z}_t,\boldsymbol{y}_t})\sum_{d=1}^{D}\delta_{\boldsymbol{z}_t^{\setminus d},\boldsymbol{y}_t^{\setminus d}} S_t^d(\boldsymbol{z}_t^d|\boldsymbol{y}_t^d)/\mathcal{S}_t(\boldsymbol{y}_t^{1:D})\cdot q_{t|0}(\boldsymbol{y}_t^d\circ \boldsymbol{z}_t^{\setminus d}|\boldsymbol{x}_0^{1:D})\right]}\\
&= \frac{(1-\delta_{\boldsymbol{z}_t,\boldsymbol{x}_t})\sum_{d=1}^{D}\delta_{\boldsymbol{z}_t^{\setminus d},\boldsymbol{x}_t^{\setminus d}} \frac{S_t^d(\boldsymbol{z}_t^d|\boldsymbol{x}_t^d)}{\mathcal{S}_t(\boldsymbol{x}_t^{1:D})}\cdot \frac{q_{t|0}(\boldsymbol{x}_t^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}_t^d|\boldsymbol{x}_0^d)}}{\sum_{\boldsymbol{y}_t^{1:D}}\left[(1-\delta_{\boldsymbol{z}_t,\boldsymbol{y}_t})\sum_{d=1}^{D}\delta_{\boldsymbol{z}_t^{\setminus d},\boldsymbol{y}_t^{\setminus d}} \frac{S_t^d(\boldsymbol{z}_t^d|\boldsymbol{y}_t^d)}{\mathcal{S}_t(\boldsymbol{y}_t^{1:D})}\cdot \frac{q_{t|0}(\boldsymbol{y}_t^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}_t^d|\boldsymbol{x}_0^d)}\right]}\\
&= \frac{(1-\delta_{\boldsymbol{z}_t,\boldsymbol{x}_t})\sum_{d=1}^{D}\delta_{\boldsymbol{z}_t^{\setminus d},\boldsymbol{x}_t^{\setminus d}} \frac{S_t^d(\boldsymbol{z}_t^d|\boldsymbol{x}_t^d)}{\mathcal{S}_t(\boldsymbol{x}_t^{1:D})}\cdot \frac{q_{t|0}(\boldsymbol{x}_t^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}_t^d|\boldsymbol{x}_0^d)}}{\sum_{d=1}^{D}\sum_{\boldsymbol{y}_t^d\neq \boldsymbol{z}^d} \frac{S_t^d(\boldsymbol{z}_t^d|\boldsymbol{y}_t^d)}{\mathcal{S}_t(\boldsymbol{y}_t^d\circ \boldsymbol{z}_t^{\setminus d})}\cdot \frac{q_{t|0}(\boldsymbol{y}_t^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}_t^d|\boldsymbol{x}_0^d)}}
\end{aligned} \tag{84}$$

现在将 q_{S_t} 的公式带回第 2 项，我们进一步将第 2 项简化为 Term2

$$\begin{aligned}
&= T\,\mathbb{E}_{t,\boldsymbol{z}^{1:D}\sim S_t} \frac{\sum_{\boldsymbol{x}^{1:D}}\left[(1-\delta_{\boldsymbol{z},\boldsymbol{x}})\sum_{d=1}^{D}\delta_{\boldsymbol{z}^{\setminus d},\boldsymbol{x}^{\setminus d}} \frac{S_t^d(\boldsymbol{z}^d|\boldsymbol{x}^d)}{S_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D})}\cdot \frac{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}\cdot r_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D})\log \hat{r}_t^{\theta}(\boldsymbol{x}^{1:D}|\boldsymbol{z}^{1:D})\right]}{\sum_{d=1}^{D}\sum_{\boldsymbol{y}^d\neq \boldsymbol{z}^d} \frac{S_t^d(\boldsymbol{z}^d|\boldsymbol{y}^d)}{\mathcal{S}_t(\boldsymbol{y}^d\circ \boldsymbol{z}^{\setminus d})}\cdot \frac{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}}\\
&= \mathbb{E}_{t,\boldsymbol{z}^{1:D}\sim S_t} \frac{\sum_{d=1}^{D}\sum_{\boldsymbol{x}^d\neq \boldsymbol{z}^d} \frac{S_t^d(\boldsymbol{z}^d|\boldsymbol{x}^d)}{S_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^d\circ \boldsymbol{z}^{\setminus d})}\cdot \frac{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}\cdot r_t(\boldsymbol{z}^{1:D}|\boldsymbol{x}^d\circ \boldsymbol{z}^{\setminus d})\log \hat{r}_t^{\theta}(\boldsymbol{x}^d\circ \boldsymbol{z}^{\setminus d}|\boldsymbol{z}^{1:D})}{\sum_{d=1}^{D}\sum_{\boldsymbol{y}^d\neq \boldsymbol{z}^d} \frac{S_t^d(\boldsymbol{z}^d|\boldsymbol{y}^d)}{\mathcal{S}_t(\boldsymbol{y}^d\circ \boldsymbol{z}^{\setminus d})}\cdot \frac{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}}\\
&= \mathbb{E}_{t,\boldsymbol{z}^{1:D}\sim S_t} \frac{\sum_{d=1}^{D}\sum_{\boldsymbol{x}^d\neq \boldsymbol{z}^d} \frac{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}\cdot r_t^d(\boldsymbol{z}^d|\boldsymbol{x}^d)\cdot \log r_t^d(\boldsymbol{z}^d|\boldsymbol{x}^d) g_t^{\theta,d}(\boldsymbol{x}^d|\boldsymbol{z}^{1:D})}{\sum_{d=1}^{D}\sum_{\boldsymbol{y}^d\neq \boldsymbol{z}^d} \frac{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}\cdot \frac{S_t^d(\boldsymbol{z}^d|\boldsymbol{y}^d)}{\mathcal{S}_t(\boldsymbol{y}^d\circ \boldsymbol{z}^{\setminus d})}}\\
&= \mathbb{E}_{t,\boldsymbol{z}^{1:D}\sim S_t} \frac{\sum_{d=1}^{D}\sum_{\boldsymbol{x}^d\neq \boldsymbol{z}^d} \frac{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}\cdot r_t^d(\boldsymbol{z}^d|\boldsymbol{x}^d)\cdot \log g_t^{\theta,d}(\boldsymbol{x}^d|\boldsymbol{z}^{1:D})}{\sum_{d=1}^{D}\sum_{\boldsymbol{y}^d\neq \boldsymbol{z}^d} \frac{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}\cdot \frac{S_t^d(\boldsymbol{z}^d|\boldsymbol{y}^d)}{\mathcal{S}_t(\boldsymbol{y}^d\circ \boldsymbol{z}^{\setminus d})}} + \text{const.}
\end{aligned} \tag{85}$$

上式进一步表明，添加交换变量$\boldsymbol{z}^{1:D}$的采样分布S_t仅影响损失计算的权重项。让我们将标量加权项定义为

$$\mathcal{M}_{S_t}(\boldsymbol{z}^{1:D}|\boldsymbol{x}_0^{1:D}) := \sum_{d=1}^{D}\sum_{\boldsymbol{y}^d\neq \boldsymbol{z}^d} \frac{q_{t|0}(\boldsymbol{y}^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}\cdot \frac{S_t^d(\boldsymbol{z}^d|\boldsymbol{y}^d)}{\mathcal{S}_t(\boldsymbol{y}^d\circ \boldsymbol{z}^{\setminus d})} \tag{86}$$

With this definition, the Term 2 can be further rewrited as

$$\begin{aligned}\text{Term2} &= \mathbb{E}_{t,\boldsymbol{z}^{1:D}\sim S_t}\Big[\frac{1}{\mathcal{M}_{S_t}(\boldsymbol{z}^{1:D}|\boldsymbol{x}_0^{1:D})}\sum_{d=1}^{D}\frac{1}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}\sum_{\boldsymbol{x}^d\neq\boldsymbol{z}^d} q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)\cdot r_t^d(\boldsymbol{z}^d|\boldsymbol{x}^d)\cdot\log g_t^{\theta,d}(\boldsymbol{x}^d|\boldsymbol{z}^{1:D})\Big]\\ &= \mathbb{E}_{t,\boldsymbol{z}^{1:D}\sim S_t}\Big[\frac{1}{\mathcal{M}_{S_t}(\boldsymbol{z}^{1:D}|\boldsymbol{x}_0^{1:D})}\sum_{d=1}^{D}\frac{\mathbf{1}^\top[q_{t|0}(\cdot|\boldsymbol{x}_0^d)\odot r_t^d(\boldsymbol{z}^d|\cdot)\odot\log g_t^{\theta,d}(\cdot|\boldsymbol{z}^{1:D})]}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}\Big]+\text{const.}\\ &= \mathbb{E}_{t,\boldsymbol{z}^{1:D}\sim S_t}\Big[\frac{\beta(t)}{\mathcal{M}_{S_t}(\boldsymbol{z}^{1:D}|\boldsymbol{x}_0^{1:D})}\sum_{d=1}^{D}\frac{\langle\boldsymbol{z}^d,\boldsymbol{m}^d\rangle}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}\mathbf{1}^\top[q_{t|0}(\cdot|\boldsymbol{x}_0^d)\odot\log g_t^{\theta,d}(\cdot|\boldsymbol{z}^{1:D})]+\text{const.}\Big]\end{aligned} \tag{87}$$

A.14.3 All Terms

Combine term 1 and term 2 together, we can write the negative VLB loss as

$$\begin{aligned}&T\,\mathbb{E}_{\substack{t\sim\text{Uni}(0,T)\\ \boldsymbol{x}^{1:D}\sim q_{t|0}}}\Big[\sum_{\boldsymbol{z}^{1:D}\neq\boldsymbol{x}^{1:D}}\hat{r}_t^{\theta,1:D}(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D})-\sum_{\boldsymbol{z}^{1:D}\neq\boldsymbol{x}^{1:D}} r_t^{1:D}(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D})\log\hat{r}_t^{\theta,1:D}(\boldsymbol{x}^{1:D}|\boldsymbol{z}^{1:D})\Big]\\ &= T\,\mathbb{E}_{\substack{t\sim\text{Uni}(0,T)\\ \boldsymbol{x}^{1:D}\sim q_{t|0}(\boldsymbol{x}_0^{1:D})\\ \boldsymbol{z}^{1:D}\sim S_t(\boldsymbol{x}^{1:D})}}\Big[\sum_{d=1}^{D} r_t^d(\boldsymbol{x}^d|\cdot)^\top g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D})\\ &-\frac{1}{\mathcal{M}_{S_t}(\boldsymbol{z}^{1:D}|\boldsymbol{x}_0^{1:D})}\sum_{d=1}^{D}\frac{\mathbf{1}^\top[q_{t|0}(\cdot|\boldsymbol{x}_0^d)\odot r_t^d(\boldsymbol{z}^d|\cdot)\odot\log g_t^{\theta,d}(\cdot|\boldsymbol{z}^{1:D})]}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}\Big]+\text{const.}\end{aligned} \tag{88}$$

A.15 Further Simplification of Continuous VLB

Proposition A.1. *The loss in Eq. (26) can be further simplified as*

$$T\,\mathbb{E}_{\substack{t\sim Uni(0,T)\\ \boldsymbol{x}^{1:D}\sim q_{t|0}(\cdot|\boldsymbol{x}_0^{1:D})}}\Big[\sum_{d=1}^{D} r_t^d(\boldsymbol{x}^d|\cdot)^\top\Big(g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D})-\frac{q_{t|0}(\cdot|\boldsymbol{x}_0^d)\odot\log g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D})}{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}\Big)\Big] \tag{89}$$

where the loss only requires a single forward pass of the model and no auxiliary variable is needed.

Proof. We use $q_{t|0}(\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D})$ to denote $P(\mathbf{x}_t^{1:D}=\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D})$. Similarly, let $\tilde{q}_{t|0}(\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D})$ denotes $P(\mathbf{z}_t^{1:D}=\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D})$. Based on the definition of S_t in Eq. (82) and the procedure of sampling $\mathbf{z}_t^{1:D}$ conditional on $\mathbf{x}_t^{1:D}$ described in Proposition 3.2, we can compute the conditional probability $\tilde{q}_{t|0}(\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D})$ as follows:

$$\begin{aligned}\tilde{q}_{t|0}(\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D}) &= \sum_{\boldsymbol{x}_t^{1:D}} P(\mathbf{z}_t^{1:D}=\boldsymbol{x}^{1:D},\boldsymbol{x}_t^{1:D}|\boldsymbol{x}_0^{1:D}) = \sum_{\boldsymbol{x}_t^{1:D}}\frac{S_t(\boldsymbol{x}^{1:D}|\boldsymbol{x}_t^{1:D})}{S_t(\boldsymbol{x}_t^{1:D})}q_{t|0}(\boldsymbol{x}_t^{1:D}|\boldsymbol{x}_0^{1:D})\\ &= \sum_{\boldsymbol{x}_t^{1:D}}(1-\delta_{\boldsymbol{x}^{1:D},\boldsymbol{x}_t^{1:D}})\Big(\sum_{d=1}^{D}\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{x}_t^{\backslash d}}\cdot\frac{S_t^d(\boldsymbol{x}^d|\boldsymbol{x}_t^d)}{S_t(\boldsymbol{x}^{\backslash d}\circ\boldsymbol{x}_t^d)}q_{t|0}(\boldsymbol{x}^{\backslash d}\circ\boldsymbol{x}_t^d|\boldsymbol{x}_0^{1:D})\Big)\\ &= q_{t|0}(\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D})\sum_{\boldsymbol{x}_t^{1:D}}(1-\delta_{\boldsymbol{x}^{1:D},\boldsymbol{x}_t^{1:D}})\Big(\sum_{d=1}^{D}\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{x}_t^{\backslash d}}\cdot\frac{S_t^d(\boldsymbol{x}^d|\boldsymbol{x}_t^d)}{S_t(\boldsymbol{x}^{\backslash d}\circ\boldsymbol{x}_t^d)}\frac{q_{t|0}(\boldsymbol{x}_t^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}\Big)\\ &= q_{t|0}(\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D})\,\mathcal{M}_{S_t}(\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D})\ \text{(Based on } \mathcal{M}_{S_t} \text{ in Eq. (86)).}\end{aligned} \tag{90}$$

The original loss in Eq. (26) can be written as

$$\begin{aligned}Eq.\ (26) = T\,\mathbb{E}_{\substack{t\sim\text{Uni}(0,T)\\ \boldsymbol{x}^{1:D}\sim q_{t|0}(\cdot|\boldsymbol{x}_0^{1:D})}}\Big[\sum_{d=1}^{D} r_t^d(\boldsymbol{x}^d|\cdot)^\top g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D})\Big]-\\ T\,\mathbb{E}_{\substack{t\sim\text{Uni}(0,T)\\ \boldsymbol{x}^{1:D}\sim q_{t|0}(\cdot|\boldsymbol{x}_0^{1:D})\\ \boldsymbol{z}^{1:D}\sim S_t(\cdot|\boldsymbol{x}^{1:D})}}\Big[\frac{1}{\mathcal{M}_{S_t}(\boldsymbol{z}^{1:D}|\boldsymbol{x}_0^{1:D})}\sum_{d=1}^{D}\frac{\mathbf{1}^\top[q_{t|0}(\cdot|\boldsymbol{x}_0^d)\odot r_t^d(\boldsymbol{z}^d|\cdot)\odot\log g_t^{\theta,d}(\cdot|\boldsymbol{z}^{1:D})]}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}\Big]\end{aligned}$$

根据这个定义，第 2 项可以进一步重写为

$$\begin{aligned}\text{Term2} &= \mathbb{E}_{t,\boldsymbol{z}^{1:D}\sim S_t}\Big[\frac{1}{\mathcal{M}_{S_t}(\boldsymbol{z}^{1:D}|\boldsymbol{x}_0^{1:D})}\sum_{d=1}^{D}\frac{1}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}\sum_{\boldsymbol{x}^d\neq\boldsymbol{z}^d} q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)\cdot r_t^d(\boldsymbol{z}^d|\boldsymbol{x}^d)\cdot\log g_t^{\theta,d}(\boldsymbol{x}^d|\boldsymbol{z}^{1:D})\Big] \\ &= \mathbb{E}_{t,\boldsymbol{z}^{1:D}\sim S_t}\Big[\frac{1}{\mathcal{M}_{S_t}(\boldsymbol{z}^{1:D}|\boldsymbol{x}_0^{1:D})}\sum_{d=1}^{D}\frac{\mathbf{1}^\top[q_{t|0}(\cdot|\boldsymbol{x}_0^d)\odot r_t^d(\boldsymbol{z}^d|\cdot)\odot\log g_t^{\theta,d}(\cdot|\boldsymbol{z}^{1:D})]}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}\Big] + \text{const.} \\ &= \mathbb{E}_{t,\boldsymbol{z}^{1:D}\sim S_t}\Big[\frac{\beta(t)}{\mathcal{M}_{S_t}(\boldsymbol{z}^{1:D}|\boldsymbol{x}_0^{1:D})}\sum_{d=1}^{D}\frac{\langle\boldsymbol{z}^d,\boldsymbol{m}^d\rangle}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}\mathbf{1}^\top[q_{t|0}(\cdot|\boldsymbol{x}_0^d)\odot\log g_t^{\theta,d}(\cdot|\boldsymbol{z}^{1:D})] + \text{const.}\Big]\end{aligned} \tag{87}$$

A.14.3 所有条款

康姆宾　第 1 项和第 2 项一起，我们可以写出负 VLB los　　　　　作为

$$\begin{aligned}&T\,\mathbb{E}_{\substack{t\sim\text{Uni}(0,T)\\ \boldsymbol{x}^{1:D}\sim q_{t|0}}}\Big[\sum_{\boldsymbol{z}^{1:D}\neq\boldsymbol{x}^{1:D}}\hat{r}_t^{\theta,1:D}(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D}) - \sum_{\boldsymbol{z}^{1:D}\neq\boldsymbol{x}^{1:D}} r_t^{1:D}(\boldsymbol{z}^{1:D}|\boldsymbol{x}^{1:D})\log\hat{r}_t^{\theta,1:D}(\boldsymbol{x}^{1:D}|\boldsymbol{z}^{1:D})\Big] \\ &= T\,\mathbb{E}_{\substack{t\sim\text{Uni}(0,T)\\ \boldsymbol{x}^{1:D}\sim q_{t|0}(\boldsymbol{x}_0^{1:D})\\ \boldsymbol{z}^{1:D}\sim S_t(\boldsymbol{x}^{1:D})}}\Big[\sum_{d=1}^{D} r_t^d(\boldsymbol{x}^d|\cdot)^\top g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D}) \\ &- \frac{1}{\mathcal{M}_{S_t}(\boldsymbol{z}^{1:D}|\boldsymbol{x}_0^{1:D})}\sum_{d=1}^{D}\frac{\mathbf{1}^\top[q_{t|0}(\cdot|\boldsymbol{x}_0^d)\odot r_t^d(\boldsymbol{z}^d|\cdot)\odot\log g_t^{\theta,d}(\cdot|\boldsymbol{z}^{1:D})]}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}\Big] + \text{const.}\end{aligned} \tag{88}$$

A.15 连续VLB的进一步简化

提案 A.1。 *The loss in Eq. (26) can be further simplified as*

$$T\,\mathbb{E}_{\substack{t\sim Uni(0,T)\\ \boldsymbol{x}^{1:D}\sim q_{t|0}(\cdot|\boldsymbol{x}_0^{1:D})}}\Big[\sum_{d=1}^{D} r_t^d(\boldsymbol{x}^d|\cdot)^\top\Big(g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D}) - \frac{q_{t|0}(\cdot|\boldsymbol{x}_0^d)\odot\log g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D})}{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}\Big)\Big] \tag{89}$$

where the loss only requires a single forward pass of the model and no auxiliary variable is needed.

Proof. 我们用$q_{t|0}(\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D})$来表示$P(\mathbf{x}_t^{1:D} = \boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D})$。类似地，让$\tilde{q}_{t|0}(\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D})$表示$P(\mathbf{z}_t^{1:D} = \boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D})$。基于等式中$S_t$的定义。 (82) 和命题3.2中描述的以$\mathbf{x}_t^{1:D}$为条件的采样$\mathbf{z}_t^{1:D}$的过程，我们可以计算条件概率$\tilde{q}_{t|0}(\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D})$如下:

$$\begin{aligned}\tilde{q}_{t|0}(\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D}) &= \sum_{\boldsymbol{x}_t^{1:D}} P(\mathbf{z}_t^{1:D} = \boldsymbol{x}^{1:D}, \boldsymbol{x}_t^{1:D}|\boldsymbol{x}_0^{1:D}) = \sum_{\boldsymbol{x}_t^{1:D}}\frac{S_t(\boldsymbol{x}^{1:D}|\boldsymbol{x}_t^{1:D})}{S_t(\boldsymbol{x}_t^{1:D})}q_{t|0}(\boldsymbol{x}_t^{1:D}|\boldsymbol{x}_0^{1:D}) \\ &= \sum_{\boldsymbol{x}_t^{1:D}}(1-\delta_{\boldsymbol{x}^{1:D},\boldsymbol{x}_t^{1:D}})\Big(\sum_{d=1}^{D}\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{x}_t^{\backslash d}}\cdot\frac{S_t^d(\boldsymbol{x}^d|\boldsymbol{x}_t^d)}{S_t(\boldsymbol{x}^{\backslash d}\circ\boldsymbol{x}_t^d)}q_{t|0}(\boldsymbol{x}^{\backslash d}\circ\boldsymbol{x}_t^d|\boldsymbol{x}_0^{1:D})\Big) \\ &= q_{t|0}(\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D})\sum_{\boldsymbol{x}_t^{1:D}}(1-\delta_{\boldsymbol{x}^{1:D},\boldsymbol{x}_t^{1:D}})\Big(\sum_{d=1}^{D}\delta_{\boldsymbol{x}^{\backslash d},\boldsymbol{x}_t^{\backslash d}}\cdot\frac{S_t^d(\boldsymbol{x}^d|\boldsymbol{x}_t^d)}{S_t(\boldsymbol{x}^{\backslash d}\circ\boldsymbol{x}_t^d)}\frac{q_{t|0}(\boldsymbol{x}_t^d|\boldsymbol{x}_0^d)}{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}\Big) \\ &= q_{t|0}(\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D})\,\mathcal{M}_{S_t}(\boldsymbol{x}^{1:D}|\boldsymbol{x}_0^{1:D})\ \text{(Based on } \mathcal{M}_{S_t} \text{ in Eq. (86)).}\end{aligned} \tag{90}$$

方程中的原始损失。 (26) 可以写成

$$\begin{aligned}Eq.\ (26) = T\,\mathbb{E}_{\substack{t\sim\text{Uni}(0,T)\\ \boldsymbol{x}^{1:D}\sim q_{t|0}(\cdot|\boldsymbol{x}_0^{1:D})}}\Big[\sum_{d=1}^{D} r_t^d(\boldsymbol{x}^d|\cdot)^\top g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D})\Big] - \\ T\,\mathbb{E}_{\substack{t\sim\text{Uni}(0,T)\\ \boldsymbol{x}^{1:D}\sim q_{t|0}(\cdot|\boldsymbol{x}_0^{1:D})\\ \boldsymbol{z}^{1:D}\sim S_t(\cdot|\boldsymbol{x}^{1:D})}}\Big[\frac{1}{\mathcal{M}_{S_t}(\boldsymbol{z}^{1:D}|\boldsymbol{x}_0^{1:D})}\sum_{d=1}^{D}\frac{\mathbf{1}^\top[q_{t|0}(\cdot|\boldsymbol{x}_0^d)\odot r_t^d(\boldsymbol{z}^d|\cdot)\odot\log g_t^{\theta,d}(\cdot|\boldsymbol{z}^{1:D})]}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}\Big]\end{aligned}$$

Notice that the second part of the loss can rewritten to

$$
\begin{aligned}
&\mathbb{E}_{\substack{t\sim\text{Uni}(0,T)\\ \boldsymbol{z}^{1:D}\sim\tilde{q}_{t|0}(\cdot|\boldsymbol{x}_0^{1:D})}} \Big[\frac{1}{\mathcal{M}_{S_t}(\boldsymbol{z}^{1:D}|\boldsymbol{x}_0^{1:D})}\sum_{d=1}^{D} \frac{\mathbf{1}^\top[q_{t|0}(\cdot|\boldsymbol{x}_0^d)\odot r_t^d(\boldsymbol{z}^d|\cdot)\odot\log g_t^{\theta,d}(\cdot|\boldsymbol{z}^{1:D})]}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}\Big]\\
=&\mathbb{E}_t \sum_{\boldsymbol{z}^{1:D}} \Big[\frac{\tilde{q}_{t|0}(\boldsymbol{z}^{1:D}|\boldsymbol{x}_0^{1:D})}{\mathcal{M}_{S_t}(\boldsymbol{z}^{1:D}|\boldsymbol{x}_0^{1:D})}\sum_{d=1}^{D} \frac{\mathbf{1}^\top[q_{t|0}(\cdot|\boldsymbol{x}_0^d)\odot r_t^d(\boldsymbol{z}^d|\cdot)\odot\log g_t^{\theta,d}(\cdot|\boldsymbol{z}^{1:D})]}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}\Big] \text{(now take Eq. (90) into)}\\
=&\mathbb{E}_t \sum_{\boldsymbol{z}^{1:D}} \Big[q_{t|0}(\boldsymbol{z}^{1:D}|\boldsymbol{x}_0^{1:D})\sum_{d=1}^{D} \frac{\mathbf{1}^\top[q_{t|0}(\cdot|\boldsymbol{x}_0^d)\odot r_t^d(\boldsymbol{z}^d|\cdot)\odot\log g_t^{\theta,d}(\cdot|\boldsymbol{z}^{1:D})]}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}\Big]\\
=&\mathbb{E}_{\substack{t\sim\text{Uni}(0,T)\\ \boldsymbol{z}^{1:D}\sim q_{t|0}(\cdot|\boldsymbol{x}_0^{1:D})}} \Big[\sum_{d=1}^{D} \frac{\mathbf{1}^\top[q_{t|0}(\cdot|\boldsymbol{x}_0^d)\odot r_t^d(\boldsymbol{z}^d|\cdot)\odot\log g_t^{\theta,d}(\cdot|\boldsymbol{z}^{1:D})]}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}\Big]
\end{aligned}
\tag{91}
$$

Taking Eq. (91) back into Eq. (26) we can get the following further simplified loss that only have a single forward pass of the model:

$$
\begin{aligned}
&T\,\mathbb{E}_{\substack{t\sim\text{Uni}(0,T)\\ \boldsymbol{x}^{1:D}\sim q_{t|0}(\cdot|\boldsymbol{x}_0^{1:D})}} \Big[\sum_{d=1}^{D} r_t^d(\boldsymbol{x}^d|\cdot)^\top g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D}) - \sum_{d=1}^{D}\frac{\mathbf{1}^\top[q_{t|0}(\cdot|\boldsymbol{x}_0^d)\odot r_t^d(\boldsymbol{x}^d|\cdot)\odot\log g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D})]}{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}\Big]\\
=&\,T\,\mathbb{E}_{\substack{t\sim\text{Uni}(0,T)\\ \boldsymbol{x}^{1:D}\sim q_{t|0}(\cdot|\boldsymbol{x}_0^{1:D})}} \Big[\sum_{d=1}^{D} r_t^d(\boldsymbol{x}^d|\cdot)^\top \Big(g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D}) - \frac{q_{t|0}(\cdot|\boldsymbol{x}_0^d)\odot\log g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D})}{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}\Big)\Big]
\end{aligned}
\tag{92}
$$

□

A.16 Algorithm for Unified Training and Generation

Alg. 1 and Alg. 2 demonstrate our unified training and generation framework.

Algorithm 1 USD3 Unified Training: Red: discrete-time step, and blue: continuous-time step.

Input: A stationary distribution $\boldsymbol{m}$, samples $\sim p_{\text{data}}$, weight factor λ, max time T
repeat
    Draw $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x}_0)$
    Draw $t \sim$ **Uniform**$(0, ..., T)$ or $t \sim$ **Uniform**$(0, 1)$
    Compute $\overline{\alpha}_t$ from Table 3
    Draw $\mathbf{m}_0^{1:D} \sim \text{Cat}(\mathbf{m}_0^{1:D}; \boldsymbol{m})$
    $\mathbf{x}_{t|0}^{1:D} = \delta^{1:D}_{1,\mathbf{b}_t^{1:D}} \odot \mathbf{x}_0^{1:D} + (1-\delta^{1:D}_{1,\mathbf{b}_t^{1:D}}) \odot \mathbf{m}_0^{1:D}$ where $\mathbf{b}_t \sim \text{Bernoulli}(\overline{\alpha}_t)$, from Eq. (49)
    Compute $f_t^\theta(\boldsymbol{x}_t^{1:D})$ as parameterization of $p_\theta(\mathbf{x}_0^{1:D}|\mathbf{x}_t^{1:D})$
    Take gradient descent step on $\nabla_\theta\{\mathcal{L}_t(\theta) + \lambda\mathcal{L}_t^{CE}(\theta)\}$ (from Eq. (11) + Eq. (12))
    or $\nabla_\theta\{\mathcal{L}_t^{CTMC}(\theta) + \lambda\mathcal{L}_t^{CE}(\theta)\}$ (from Eq. (23) + Eq. (12))
until convergence

Algorithm 2 USD3 Unified Sampling

Input: A stationary distribution $\boldsymbol{m}$, $\{t_i\}_{i=0}^n$ s.t. $0 = t_0 < t_1 < ... < t_n = T$, learned $f_{t_i}^\theta$.
MCMC Input: use_MCMC, step size $\triangle n$, total steps N.
Set $\mathbf{x}_n^{1:D} \sim \text{Cat}(\mathbf{x}_n^{1:D}; \boldsymbol{m})$
for $i \in \{n, ..., 0\}$ **do**
    Compute $f_{t_i}^\theta(\boldsymbol{x}_i^{1:D})$ and $p_\theta(\mathbf{x}_{i-1}^{1:D}|\mathbf{x}_i^{1:D})$ from Eq. (10)
    Draw $\mathbf{x}_{i-1}^{1:D} \sim \text{Cat}(\mathbf{x}_{i-1}^{1:D}; p_\theta(\mathbf{x}_{i-1}^{1:D}|\mathbf{x}_i^{1:D}))$
    If use_MCMC:
        $\mathbf{x}_{i-1}^{1:D} \leftarrow \text{MCMC\_Corrector}(t_i, \mathbf{x}_{i-1}^{1:D}, \triangle n, f_{t_i}^\theta, N)$ (Algo. 3)
return $\mathbf{x}_0$

Notice that the second part of the loss can rewritten to

$$
\begin{aligned}
&\mathbb{E}_{\substack{t\sim\text{Uni}(0,T)\\ \boldsymbol{z}^{1:D}\sim\tilde{q}_{t|0}(\cdot|\boldsymbol{x}_0^{1:D})}}\Big[\frac{1}{\mathcal{M}_{S_t}(\boldsymbol{z}^{1:D}|\boldsymbol{x}_0^{1:D})}\sum_{d=1}^{D}\frac{\mathbf{1}^\top[q_{t|0}(\cdot|\boldsymbol{x}_0^d)\odot r_t^d(\boldsymbol{z}^d|\cdot)\odot\log g_t^{\theta,d}(\cdot|\boldsymbol{z}^{1:D})]}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}\Big]\\
=&\mathbb{E}_t\sum_{\boldsymbol{z}^{1:D}}\Big[\frac{\tilde{q}_{t|0}(\boldsymbol{z}^{1:D}|\boldsymbol{x}_0^{1:D})}{\mathcal{M}_{S_t}(\boldsymbol{z}^{1:D}|\boldsymbol{x}_0^{1:D})}\sum_{d=1}^{D}\frac{\mathbf{1}^\top[q_{t|0}(\cdot|\boldsymbol{x}_0^d)\odot r_t^d(\boldsymbol{z}^d|\cdot)\odot\log g_t^{\theta,d}(\cdot|\boldsymbol{z}^{1:D})]}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}\Big]\text{(now take Eq. (90) into)}\\
=&\mathbb{E}_t\sum_{\boldsymbol{z}^{1:D}}\Big[q_{t|0}(\boldsymbol{z}^{1:D}|\boldsymbol{x}_0^{1:D})\sum_{d=1}^{D}\frac{\mathbf{1}^\top[q_{t|0}(\cdot|\boldsymbol{x}_0^d)\odot r_t^d(\boldsymbol{z}^d|\cdot)\odot\log g_t^{\theta,d}(\cdot|\boldsymbol{z}^{1:D})]}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}\Big]\\
=&\mathbb{E}_{\substack{t\sim\text{Uni}(0,T)\\ \boldsymbol{z}^{1:D}\sim q_{t|0}(\cdot|\boldsymbol{x}_0^{1:D})}}\Big[\sum_{d=1}^{D}\frac{\mathbf{1}^\top[q_{t|0}(\cdot|\boldsymbol{x}_0^d)\odot r_t^d(\boldsymbol{z}^d|\cdot)\odot\log g_t^{\theta,d}(\cdot|\boldsymbol{z}^{1:D})]}{q_{t|0}(\boldsymbol{z}^d|\boldsymbol{x}_0^d)}\Big] \qquad (91)
\end{aligned}
$$

取方程。(91) 回到方程。(26) 我们可以得到以下进一步简化的损失，该损失仅具有模型的单个前向传播：

$$
\begin{aligned}
&T\,\mathbb{E}_{\substack{t\sim\text{Uni}(0,T)\\ \boldsymbol{x}^{1:D}\sim q_{t|0}(\cdot|\boldsymbol{x}_0^{1:D})}}\Big[\sum_{d=1}^{D}r_t^d(\boldsymbol{x}^d|\cdot)^\top g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D})-\sum_{d=1}^{D}\frac{\mathbf{1}^\top[q_{t|0}(\cdot|\boldsymbol{x}_0^d)\odot r_t^d(\boldsymbol{x}^d|\cdot)\odot\log g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D})]}{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}\Big]\\
=&T\,\mathbb{E}_{\substack{t\sim\text{Uni}(0,T)\\ \boldsymbol{x}^{1:D}\sim q_{t|0}(\cdot|\boldsymbol{x}_0^{1:D})}}\Big[\sum_{d=1}^{D}r_t^d(\boldsymbol{x}^d|\cdot)^\top\Big(g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D})-\frac{q_{t|0}(\cdot|\boldsymbol{x}_0^d)\odot\log g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D})}{q_{t|0}(\boldsymbol{x}^d|\boldsymbol{x}_0^d)}\Big)\Big] \qquad (92)
\end{aligned}
$$

□

A.16 统一训练和生成的算法

藻类。和藻类。2 展示我们的统一训练和生成框架 电子工作。

算法1 USD3统一训练：红色：离散时间步长，蓝色：连续时间步长。

1: 输入：平稳分布 $\boldsymbol{m}$、样本 $\sim p_{\text{data}}$、权重因子 λ、最大时间 T 2: 重复 3: 绘制 $\mathbf{x}_0\sim p_{\text{data}}(\mathbf{x}_0)$ 4: 绘制 $t\sim$ Uniform$(0,...,T)$ 或 $t\sim$ Uniform(0, 1) 5: 根据表 3 计算 α_t 6: 绘制 $\mathbf{m}_0^{1:D}\sim$ Cat($\mathbf{m}_0^{1:D}$; $\boldsymbol{m}$) 7: $\mathbf{x}_{t|0}^{1:D}=\delta_{1,\mathbf{b}_t^{1:D}}^{1:D}\odot\mathbf{x}_0^{1:D}+(1-\delta_{1,\mathbf{b}_t^{1:D}}^{1:D})\odot\mathbf{m}_0^{1:D}$ 其中 $\mathbf{b}_t\sim$ 伯努利 (α_t)，来自方程。(49) 8：计算 $f_t^\theta(\boldsymbol{x}_t^{1:D})$ 作为 $p_\theta(\mathbf{x}_0^{1:D}|\mathbf{x}_t^{1:D})$ 的参数化 9：从等式 1 中对 $\nabla_\theta\{\mathcal{L}_t(\theta)+\lambda\mathcal{L}_t^{CE}(\theta)\}$ (采取梯度下降步骤。(11) + 方程 (12)) 10: 或 $\nabla_\theta\{\mathcal{L}_t^{CTMC}(\theta)+\lambda\mathcal{L}_t^{CE}(\theta)\}$ (来自方程 (12)) (23) + 等式(12))

11：直到收敛

算法2 USD3统一抽样

1：输入：平稳分布 $\boldsymbol{m}$, $\{t_i\}_{i=0}^n$ s.t. $0=t_0<t_1<...<t_n=T$，学会了 $f_{t_i}^\theta$。 2：MC MC输入：use_MCMC，步长$\triangle n$，总步数N。 3: 设置 $\mathbf{x}_n^{1:D}\sim$ Cat($\mathbf{x}_n^{1:D}$; $\boldsymbol{m}$) 4: 对于 $i\in\{n,...,0\}$ 执行 5: 根据等式计算 $f_{t_i}^\theta(\boldsymbol{x}_i^{1:D})$ 和 $p_\theta(\mathbf{x}_{i-1}^{1:D}|\mathbf{x}_i^{1:D})$ (10) 6: 绘制 $\mathbf{x}_{i-1}^{1:D}\sim$ Cat ($\mathbf{x}_{i-1}^{1:D}$; $p_\theta(\mathbf{x}_{i-1}^{1:D}|\mathbf{x}_i^{1:D})$) 7: 如果 use_MCMC: 8: $\mathbf{x}_{i-1}^{1:D}\leftarrow$ MCMC_Corrector($t_i,\mathbf{x}_{i-1}^{1:D},\triangle n,f_{t_i}^\theta,N$) (算法 3) 9: 返回 $\mathbf{x}_0$

A.17 Continuous-Time MCMC Sampling Corrector & Noise Scheduling

Campbell et al. [2022] show that the MCMC for discrete data can be done by a predictor step to simulate $\widehat{r}_t^\theta$ and a corrector step using $r_t + \widehat{r}_t^\theta$. We extend this result with improved derivation and show that, thanks to the shared parameterized form, *both* discrete- and continuous-time discrete diffusion can leverage the same transition probability calculation (see Eq. (98)), leading to a shared MCMC scheme, with detailed derivation provided in Appx. §A.17.

A.17.1 The MCMC Sampling Corrector

Song et al. [2021b] introduced a predictor-corrector step to further improve the quality of generated data based on score-based Markov Chain Monte Carlo (MCMC) for continuous-time diffusion over continuous distribution. Campbell et al. [2022] showed that there is a similar MCMC based corrector that can be used for CTMC to improve reverse sampling at any time t. Although we use different reverse sampling than Campbell et al. [2022], the similar corrector step can also be developed to improve the quality of reverse sampling introduced in §3.4. In this section, we derive the corrector formally and simplify it based the multi-element formulations summarized in Eq. (98).

Formally, at any time t, Campbell et al. [2022] proved that a *time-homogeneous* CTMC with transition rate being $c_t := r_t + \hat{r}_t$ has its stationary distribution being $q_t(\mathbf{x}_t^{1:D})$. To avoid ambiguity, we use $n \in [0, +\infty)$ as the time variable for that CTMC with stationary distribution $q_t(\mathbf{x}_t^{1:D})$. Then for any sample $\boldsymbol{z}_t^{1:D}$ generated from reverse sampling process at time t, we can push it closer to the target marginal distribution $q_t(\mathbf{x}_t^{1:D})$ by sampling from the corrector CTMC with initial value being $\boldsymbol{z}_t^{1:D}$, named as $\boldsymbol{z}_{t,n=0}^{1:D}$. Let N be the maximum time allocated in the corrector CTMC, then after the corrector step $\boldsymbol{z}_{t,n=N}^{1:D}$ is used to replace the original $\boldsymbol{z}_t^{1:D}$.

We now introduce how to sampling from the CTMC. Let $\triangle n$ be the time incremental for each sampling step of the corrector CTMC. Solving the Kolmogorov forward equation of this time-homogeneous CTMC can derive the transition probability at any time n as

$$\forall n \text{ and } \triangle n,\ p_{n+\triangle n|n}(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) = \exp(\triangle n \cdot C_t^{1:D})[\boldsymbol{x}^{1:D}, \boldsymbol{y}^{1:D}] \tag{93}$$

Where C_t is the transition rate matrix of the corrector CTMC, and $\exp(\triangle n \cdot C_t^{1:D})$ is the transition probability matrix at time n. Notice that this matrix exponential does not have analytical formulation. Instead, we propose to control $\triangle n$ to be small enough such that

$$\exp(\triangle n \cdot C_t^{1:D}) \approx I + \triangle n \cdot C_t^{1:D} \tag{94}$$

Then taking it back we can obtain

$$\begin{aligned} p_{n+\triangle n|n}(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) &\approx \delta_{\boldsymbol{x}^{1:D},\boldsymbol{y}^{1:D}} + \triangle n \cdot c_t^{1:D}(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) \\ &= \delta_{\boldsymbol{x}^{1:D},\boldsymbol{y}^{1:D}} + \triangle n \sum_{d=1}^{D} \Big[r_t^d(\boldsymbol{y}^d|\boldsymbol{x}^d) + r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d) \cdot g_t^d(\boldsymbol{y}^d|\boldsymbol{x}^{1:D}) \Big] \delta_{\boldsymbol{x}^{\setminus d},\boldsymbol{y}^{\setminus d}} \end{aligned} \tag{95}$$

Instead of sampling all elements jointly, we propose to sample each element of the object independently from their individual marginal distribution, which can be analytically formulated as

$$\begin{aligned} p_{n+\triangle n|n}(\boldsymbol{y}^d|\boldsymbol{x}^{1:D}) &= \sum_{\boldsymbol{y}^{\setminus d}} p_{n+\triangle n|n}(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) \\ &= \triangle n \Big[r_t^d(\boldsymbol{y}^d|\boldsymbol{x}^d) + r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d) \cdot g_t^d(\boldsymbol{y}^d|\boldsymbol{x}^{1:D}) \Big] \text{ if } \boldsymbol{y}^d \neq \boldsymbol{x}^d \\ &= \triangle n(\boldsymbol{y}^d)^\top \Big[r_t^d(\cdot|\boldsymbol{x}^d) + r_t^d(\boldsymbol{x}^d|\cdot) \odot g_t^d(\cdot|\boldsymbol{x}^{1:D}) \Big] \text{ if } \boldsymbol{y}^d \neq \boldsymbol{x}^d \\ &= \triangle n \beta(t)(\boldsymbol{y}^d)^\top \Big[\boldsymbol{m}^d - \boldsymbol{x}^d + \big(\langle \boldsymbol{x}^d, \boldsymbol{m}^d \rangle \mathbf{1} - \boldsymbol{x}^d \big) \odot g_t^d(\cdot|\boldsymbol{x}^{1:D}) \Big] \text{ if } \boldsymbol{y}^d \neq \boldsymbol{x}^d \\ &= \triangle n \beta(t)(\boldsymbol{y}^d)^\top \Big[\boldsymbol{m}^d + \langle \boldsymbol{x}^d, \boldsymbol{m}^d \rangle g_t^d(\cdot|\boldsymbol{x}^{1:D}) \Big] \text{ if } \boldsymbol{y}^d \neq \boldsymbol{x}^d \\ &\approx \triangle n \beta(t)(\boldsymbol{y}^d)^\top \Big[\boldsymbol{m}^d + \langle \boldsymbol{x}^d, \boldsymbol{m}^d \rangle g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D}) \Big] \text{ if } \boldsymbol{y}^d \neq \boldsymbol{x}^d \end{aligned} \tag{96}$$

A.17 连续时间 MCMC 采样校正器和噪声调度

坎贝尔等人。 [2022]表明离散数据的 MCMC 可以通过模拟 $\hat{r}_t^\theta$ 的预测器步骤和使用 $r_t + \hat{r}_t^\theta$ 的校正器步骤来完成。我们通过改进的推导扩展了这个结果，并表明，由于共享参数化形式，*both*离散和连续时间离散扩散可以利用相同的转移概率计算（参见方程（98）），从而形成共享的 MCMC 方案，详细推导在 Appx.1 中提供。 §A.17。

A.17.1 MCMC 采样校正器

宋等人。 [2021b]引入了预测校正步骤，以进一步提高基于分数的马尔可夫链蒙特卡罗（MCMC）生成的数据的质量，以实现连续分布上的连续时间扩散。坎贝尔等人。 [2022]表明存在一种类似的基于 MCMC 的校正器，可用于 CTMC 来改进任意时间 t 的反向采样。尽管我们使用与 Campbell 等人不同的反向采样。 [2022]，还可以开发类似的校正器步骤来提高§ 3.4中引入的反向采样的质量。在本节中，我们正式推导校正器并根据方程（1）中总结的多元素公式对其进行简化。 （98）。

正式地，在任何时间t，坎贝尔等人。 [2022] 证明了转移率为 $c_t := r_t + \hat{r}_t$ 的 *time-homogeneous* CTMC 的平稳分布为 $q_t(\mathbf{x}_t^{1:D})$。为了避免歧义，我们使用 $n \in [0, +\infty)$ 作为具有平稳分布 $q_t(\mathbf{x}_t^{1:D})$ 的 CTMC 的时间变量。然后，对于在时间t通过反向采样过程生成的任何样本$\boldsymbol{z}_t^{1:D}$，我们可以通过从校正器CTMC采样来将其推近目标边际分布$q_t(\mathbf{x}_t^{1:D})$，初始值为$\boldsymbol{z}_t^{1:D}$，命名为$\boldsymbol{z}_{t,n=0}^{1:D}$。令$N$为校正器CTMC中分配的最大时间，然后在校正器步骤之后使用$\boldsymbol{z}_{t,n=N}^{1:D}$来替换原始的$\boldsymbol{z}_t^{1:D}$。

现在介绍如何从CTMC采样。令 $\triangle n$ 为校正器 CTMC 的每个采样步骤的时间增量。求解该时间齐次 CTMC 的 Kolmogorov 前向方程可以导出任意时刻 n 的转移概率为

$$\forall n \text{ and } \triangle n,\ p_{n+\triangle n|n}(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) = \exp(\triangle n \cdot C_t^{1:D})[\boldsymbol{x}^{1:D}, \boldsymbol{y}^{1:D}] \tag{93}$$

其中C_t是校正器CTMC的转移率矩阵，$\exp(\triangle n \cdot C_t^{1:D})$是在时间$n$的转移概率矩阵。请注意，该矩阵指数没有解析公式。相反，我们建议将 $\triangle n$ 控制得足够小，以便

$$\exp(\triangle n \cdot C_t^{1:D}) \approx I + \triangle n \cdot C_t^{1:D} \tag{94}$$

然后拿回来我们可以得到

$$\begin{aligned} p_{n+\triangle n|n}(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) &\approx \delta_{\boldsymbol{x}^{1:D},\boldsymbol{y}^{1:D}} + \triangle n \cdot c_t^{1:D}(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) \\ &= \delta_{\boldsymbol{x}^{1:D},\boldsymbol{y}^{1:D}} + \triangle n \sum_{d=1}^{D} \Big[r_t^d(\boldsymbol{y}^d|\boldsymbol{x}^d) + r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d) \cdot g_t^d(\boldsymbol{y}^d|\boldsymbol{x}^{1:D}) \Big] \delta_{\boldsymbol{x}^{\setminus d},\boldsymbol{y}^{\setminus d}} \end{aligned} \tag{95}$$

我们建议对对象的每个元素独立于其各自的边缘分布进行采样，而不是对所有元素进行联合采样，这可以分析地表述为

$$\begin{aligned} p_{n+\triangle n|n}(\boldsymbol{y}^d|\boldsymbol{x}^{1:D}) &= \sum_{\boldsymbol{y}^{\setminus d}} p_{n+\triangle n|n}(\boldsymbol{y}^{1:D}|\boldsymbol{x}^{1:D}) \\ &= \triangle n \Big[r_t^d(\boldsymbol{y}^d|\boldsymbol{x}^d) + r_t^d(\boldsymbol{x}^d|\boldsymbol{y}^d) \cdot g_t^d(\boldsymbol{y}^d|\boldsymbol{x}^{1:D}) \Big] \text{ if } \boldsymbol{y}^d \neq \boldsymbol{x}^d \\ &= \triangle n (\boldsymbol{y}^d)^\top \Big[r_t^d(\cdot|\boldsymbol{x}^d) + r_t^d(\boldsymbol{x}^d|\cdot) \odot g_t^d(\cdot|\boldsymbol{x}^{1:D}) \Big] \text{ if } \boldsymbol{y}^d \neq \boldsymbol{x}^d \\ &= \triangle n \beta(t) (\boldsymbol{y}^d)^\top \Big[\boldsymbol{m}^d - \boldsymbol{x}^d + \big(\langle \boldsymbol{x}^d, \boldsymbol{m}^d \rangle \mathbf{1} - \boldsymbol{x}^d \big) \odot g_t^d(\cdot|\boldsymbol{x}^{1:D}) \Big] \text{ if } \boldsymbol{y}^d \neq \boldsymbol{x}^d \\ &= \triangle n \beta(t) (\boldsymbol{y}^d)^\top \Big[\boldsymbol{m}^d + \langle \boldsymbol{x}^d, \boldsymbol{m}^d \rangle g_t^d(\cdot|\boldsymbol{x}^{1:D}) \Big] \text{ if } \boldsymbol{y}^d \neq \boldsymbol{x}^d \\ &\approx \triangle n \beta(t) (\boldsymbol{y}^d)^\top \Big[\boldsymbol{m}^d + \langle \boldsymbol{x}^d, \boldsymbol{m}^d \rangle g_t^{\theta,d}(\cdot|\boldsymbol{x}^{1:D}) \Big] \text{ if } \boldsymbol{y}^d \neq \boldsymbol{x}^d \end{aligned} \tag{96}$$

Now we define the notation $\overline{p_{\triangle n}(\cdot|\boldsymbol{x}^{1:D})}$ to derive the distributional form of $p_{n+\triangle n|n}(\boldsymbol{y}^d|\boldsymbol{x}^{1:D})$.

$$\overline{p^{\theta,d}_{\triangle n}(\cdot|\boldsymbol{x}^{1:D})} := \triangle n \beta(t)\Big[\big(2 - \frac{\overline{\alpha}_{t|0}\langle f_t^{\theta,d}(\boldsymbol{x}^{1:D}), \boldsymbol{x}^d\rangle}{\overline{\alpha}_{t|0} + (1-\overline{\alpha}_{t|0})\langle \boldsymbol{x}^d, \boldsymbol{m}^d\rangle}\big)\boldsymbol{m}^d + \frac{\overline{\alpha}_{t|0}}{1-\overline{\alpha}_{t|0}} f_t^{\theta,d}(\boldsymbol{x}^{1:D})\Big] \odot (\boldsymbol{1} - \boldsymbol{x}^d) \tag{97}$$

With the above notation, the sampling probability can be further simplified as

$$p_{n+\triangle n|n}(\boldsymbol{y}^d|\boldsymbol{x}^{1:D}) = \mathrm{Cat}\Big(\boldsymbol{y}^d; \overline{p^{\theta,d}_{\triangle n}(\cdot|\boldsymbol{x}^{1:D})} + \big(1 - \boldsymbol{1}^\top \overline{p^{\theta,d}_{\triangle n}(\cdot|\boldsymbol{x}^{1:D})}\big)\boldsymbol{x}^d\Big) \tag{98}$$

Notice that $\triangle n$ should be set small enough such that $\boldsymbol{1}^\top \overline{p^{\theta,d}_{\triangle n}(\cdot|\boldsymbol{x}^{1:D})} \leq 1$. This condition can be used to derive $\triangle n$ dynamically. In practice, we can also easily clip the scale of $\overline{p^{\theta,d}_{\triangle n}(\cdot|\boldsymbol{x}^{1:D})}$ to 1 when $\boldsymbol{1}^\top \overline{p^{\theta,d}_{\triangle n}(\cdot|\boldsymbol{x}^{1:D})} > 1$ to prevent illness condition. Intuitively, $1 - \boldsymbol{1}^\top \overline{p^{\theta,d}_{\triangle n}(\cdot|\boldsymbol{x}^{1:D})}$ defines the keeping rate of the d-th element during correction step, and it should be larger with increasing t and n during the reverse sampling and correction period.

Algorithm 3 The MCMC correcting algorithm at time t

Input: The sample at time t from reverse sampling, $\boldsymbol{z}_t^{1:D}$; step size $\triangle n$; learned f_t^θ; total steps N.

Initialize $\boldsymbol{x}_{t,0}^{1:D} \leftarrow \boldsymbol{z}_t^{1:D}$
for i from 1 to N **do**
  Compute $\overline{p^{\theta,d}_{\triangle n}(\cdot|\boldsymbol{x}^{1:D}_{t,i-1})}$ from Eq. (97);
  $\forall d$, Draw $\boldsymbol{x}^d_{t,i} \sim \mathrm{Cat}\Big(\cdot\,; \overline{p^{\theta,d}_{\triangle n}(\cdot|\boldsymbol{x}^{1:D}_{t,i-1})} + \big(1 - \boldsymbol{1}^\top \overline{p^{\theta,d}_{\triangle n}(\cdot|\boldsymbol{x}^{1:D}_{t,i-1})}\big)\boldsymbol{x}^d_{t,i-1}\Big)$
end for
Output: the improved (corrected) sample $\boldsymbol{x}_{t,N}^{1:D}$

A.17.2 Noise Scheduling in Training

Notice that $\beta(t) = -\frac{1}{\overline{\alpha}_t} \cdot \frac{d\overline{\alpha}_t}{dt}$ as $\overline{\alpha}_{t|s} = \exp(-\int_s^t \beta(a)da)$. We now first present a general way to design the scheduling of $\overline{\alpha}_t$ based on Chen [2023]. For any continuous function $h(t)$, we define $\overline{\alpha}_t$ as the follows that satisfies $\overline{\alpha}_0 = 1$ and $\overline{\alpha}_T = 0$:

$$\overline{\alpha}_t = \frac{h(T) - h(t)}{h(T) - h(0)} \tag{99}$$

We can easily derive that

$$\beta(t) = -\frac{1}{\overline{\alpha}_t} \cdot \frac{h'(t)}{h(0) - h(T)} = \frac{h'(t)}{h(T) - h(t)} \tag{100}$$

$$\overline{\alpha}_{t|s} = \exp(-\int_s^t \frac{dh(t)}{h(T) - h(t)}) = \exp(\int_s^t \frac{d(h(T) - h(t))}{h(T) - h(t)}) = \frac{h(T) - h(t)}{h(T) - h(s)} \tag{101}$$

Based on the above general formulation, we now present some widely used noise schedules

Table 3: Widely used noise scheduling with maximum time T

Type	Param.	Reference	$h(t)$	$\overline{\alpha}_t$	$\beta(t)$	$\overline{\alpha}_{t\|s}$
Cosine	$a = 0.008$	[Hoogeboom et al., 2021]	$\cos(\frac{t/T+a}{1+a}\frac{\pi}{2})$	$\frac{h(t)}{h(0)}$	$\frac{\pi \tan(\frac{t/T+a}{1+a}\frac{\pi}{2})}{2T(1+a)}$	$\frac{h(t)}{h(s)}$
Linear	-	[Ho et al., 2020a]	t	$1 - \frac{t}{T}$	$\frac{1}{T-t}$	$\frac{T-t}{T-s}$
Exp.	a, b	[Campbell et al., 2022]	-	$\exp(Ta(1 - b^{\frac{t}{T}}))$	$ab^{\frac{t}{T}} \log b$	$\exp(Ta(b^{\frac{s}{T}} - b^{\frac{t}{T}}))$

A.18 Experiment Details

A.18.1 Lakh Piano Dataset Details

The Lakh pianoroll dataset contains $6,000$ training and 973 evaluating piano sequences, with each music sequence spanning a length of 256 in total. Each music note in the sequence can take on a value

现在我们定义符号 $\overline{p_{\triangle n}(\cdot|\boldsymbol{x}^{1:D})}$ 来导出 $p_{n+\triangle n|n}(\boldsymbol{y}^d|\boldsymbol{x}^{1:D})$ 的分布形式。

$$\overline{p^{\theta,d}_{\triangle n}(\cdot|\boldsymbol{x}^{1:D})} := \triangle n\beta(t)\Big[\big(2-\frac{\overline{\alpha}_{t|0}\langle f_t^{\theta,d}(\boldsymbol{x}^{1:D}),\boldsymbol{x}^d\rangle}{\overline{\alpha}_{t|0}+(1-\overline{\alpha}_{t|0})\langle\boldsymbol{x}^d,\boldsymbol{m}^d\rangle}\big)\boldsymbol{m}^d+\frac{\overline{\alpha}_{t|0}}{1-\overline{\alpha}_{t|0}}f_t^{\theta,d}(\boldsymbol{x}^{1:D})\Big]\odot(\boldsymbol{1}-\boldsymbol{x}^d) \tag{97}$$

利用上述表示法，采样概率可以进一步简化为

$$p_{n+\triangle n|n}(\boldsymbol{y}^d|\boldsymbol{x}^{1:D}) = \text{Cat}\Big(\boldsymbol{y}^d;\overline{p^{\theta,d}_{\triangle n}(\cdot|\boldsymbol{x}^{1:D})}+\big(1-\boldsymbol{1}^\top\overline{p^{\theta,d}_{\triangle n}(\cdot|\boldsymbol{x}^{1:D})}\big)\boldsymbol{x}^d\Big) \tag{98}$$

请注意，$\triangle n$ 应设置得足够小，以便 $\boldsymbol{1}^\top\overline{p^{\theta,d}_{\triangle n}(\cdot|\boldsymbol{x}^{1:D})} \leq$ 为 1。此条件可用于动态导出 $\triangle n$。在实践中，当$\boldsymbol{1}^\top\overline{p^{\theta,d}_{\triangle n}(\cdot|\boldsymbol{x}^{1:D})} >$为1时，我们也可以轻松地将$\overline{p^{\theta,d}_{\triangle n}(\cdot|\boldsymbol{x}^{1:D})}$的范围修剪为1，以防止出现疾病情况。直观上，$1-\boldsymbol{1}^\top\overline{p^{\theta,d}_{\triangle n}(\cdot|\boldsymbol{x}^{1:D})}$定义了校正步骤中第$d$个元素的保留率，并且在反向采样和校正期间，它应该随着t和n的增加而变大。

算法3 t时刻的MCMC校正算法

输入：反向采样在时间t时的样本，$\boldsymbol{z}_t^{1:D}$；步长$\triangle n$；学到了 f_t^θ;总步数 N。

Initialize $\boldsymbol{x}_{t,0}^{1:D} \leftarrow \boldsymbol{z}_t^{1:D}$
for i from 1 to N **do**
  Compute $\overline{p^{\theta,d}_{\triangle n}(\cdot|\boldsymbol{x}^{1:D}_{t,i-1})}$ from Eq. (97);
  $\forall d$, Draw $\boldsymbol{x}^d_{t,i} \sim \text{Cat}\Big(\cdot;\overline{p^{\theta,d}_{\triangle n}(\cdot|\boldsymbol{x}^{1:D}_{t,i-1})}+\big(1-\boldsymbol{1}^\top\overline{p^{\theta,d}_{\triangle n}(\cdot|\boldsymbol{x}^{1:D}_{t,i-1})}\big)\boldsymbol{x}^d_{t,i-1}\Big)$
结束于
输出：改进（修正）的样本$\boldsymbol{x}^{1:D}_{t,N}$

A.17.2 训练中的噪声调度

请注意，$\beta(t) = -\frac{1}{\overline{\alpha}_t}\cdot\frac{d\overline{\alpha}_t}{dt}$ 为 $\overline{\alpha_{t|s}} = \exp(-\int_s^t\beta(a)da)$。我们现在首先基于 Chen [2023] 提出一种设计 $\overline{\alpha}_t$ 调度的通用方法。对于任何连续函数 $h(t)$，我们将 $\overline{\alpha}_t$ 定义如下，满足 $\overline{\alpha}_0 = 1$ 和 $\overline{\alpha}_T = 0$：

$$\overline{\alpha}_t = \frac{h(T)-h(t)}{h(T)-h(0)} \tag{99}$$

我们可以很容易地得出

$$\beta(t) = -\frac{1}{\overline{\alpha}_t}\cdot\frac{h'(t)}{h(0)-h(T)} = \frac{h'(t)}{h(T)-h(t)} \tag{100}$$

$$\overline{\alpha}_{t|s} = \exp(-\int_s^t\frac{dh(t)}{h(T)-h(t)}) = \exp(\int_s^t\frac{d(h(T)-h(t))}{h(T)-h(t)}) = \frac{h(T)-h(t)}{h(T)-h(s)} \tag{101}$$

基于上述一般公式，我们现在提出一些广泛使用的噪声表

表3：广泛使用的最大时间T的噪声调度

Type	Param.	Reference	$h(t)$	$\overline{\alpha}_t$	$\beta(t)$	$\overline{\alpha}_{t\|s}$
Cosine	$a=0.008$	[Hoogeboom et al., 2021]	$\cos(\frac{t/T+a}{1+a}\frac{\pi}{2})$	$\frac{h(t)}{h(0)}$	$\frac{\pi\tan(\frac{t/T+a}{1+a}\frac{\pi}{2})}{2T(1+a)}$	$\frac{h(t)}{h(s)}$
Linear	-	[Ho et al., 2020a]	t	$1-\frac{t}{T}$	$\frac{1}{T-t}$	$\frac{T-t}{T-s}$
Exp.	a,b	[Campbell et al., 2022]	-	$\exp(Ta(1-b^{\frac{t}{T}}))$	$ab^{\frac{t}{T}}\log b$	$\exp(Ta(b^{\frac{s}{T}}-b^{\frac{t}{T}}))$

A.18 实验细节

A.18.1 十万钢琴数据集详细信息

Lakh Pianoroll 数据集包含 6, 000 个训练钢琴序列和 973 个评估钢琴序列，每个音乐序列总共跨越 256 个长度。序列中的每个音符都可以具有一个值

of the 128 music notes plus 1 additional class meaning an empty note. The music note orderings are scrambled into the same random order as described in Campbell et al. [2022] such that the ordinal structure of the music notes are destroyed.

For evaluation, the first 32 notes of the 967 evaluation sequences are given to the model, while the model is asked to generate the resting 224 notes. Upon an analysis of the training and evaluation music sequences, we find that a total of 124 evaluation samples can be found with at least one matching training samples that have the same 32 dimensions. We separate out these samples and call the set PIANO-P. Among PIANO-P, 20 samples can be found to contain the same first 32 notes with 2 samples from the training sequences, three of them contain the same first 32 notes with 4 training samples each, one of them shares the same with 6 training samples, and one shares the same with 8 training samples. If the same 32 notes appear both in the training samples and as quests for the model to provide the inferences, the model is likely to directly memorizing the remaining 224 notes from the training set, and "parroting" music sequences according to the training samples. The rest 843 evaluation samples that do not have matching training samples are constituent of PIANO.

A.18.2 Pre-training VQGAN

To generate images in a categorical discrete latent space, we follow the implementation of VQGAN Esser et al. [2020] in MaskGIT Chang et al. [2022]. Specifically, we use the same VQGAN setting as mentioned in Sun et al. [2023]. VQGAN is a variant of Vector Quantized Variational Autoencoder (VQ-VAE) by van den Oord et al. [2018]. In our setup for CIFAR10, a VQGAN encodes an image of shape $H \times W \times 3$ to $(\frac{H}{4} \times \frac{W}{4})$ tokens with vocabulary size of 512. For the encoder, we use three convolutional blocks, with filter sizes of 64,128 and 256, and an average pooling between each blocks. For each block, it consists of two residual blocks. After an image is encoded, the output is mapped to a token index with a codebook of 512×256. For VQGAN loss objective, the additional GAN loss and perceptual loss are added with weight 0.1. To train a general VQGAN model that allows us to embed CIFAR10 images without overfitting, we apply data augmentation (random flipping and cropping) to the 64×64 version of ImageNet dataset [Deng et al. [2009]], and train for 90 epochs. The VQGAN is trained with Adam optimizer ($\beta_1 = 0$, $\beta_2 = 0.99$), and the learning rate is linearly warmup to the peak of $1e^{-4}$ and then drops with cosine decay.

After VQGAN is trained, we freeze the VQGAN and apply encoder to the CIFAR10 images to create 8×8 latent codes. The latent codes then flattened to a vector of size 64 as the input of the diffusion model. Before we evaluate our diffusion methods, we will feed the generated latent codes back to the VQGAN decoder to reconstruct a sample in the image spaces. We test the effectivenss of VQGAN by trying to reconstruct the CIFAR10 dataset. The reconstruction gives a FID of 7.68 and IS of 10.42, using the Inception V3 Model [4].

A.18.3 Music Generation Eval Metrics

In evaluation of the conditional music generation task, we apply the following metrics to measure generation quality:

- 1-gram Hellinger Distance (↓) and 1-gram Proportion of Outliers (↓): for these two metrics, we following the same evaluation as described in Campbell et al. [2022] and Chen [2023].
- $\{2, 3\}$-gram Hellinger Distance (↓) and $\{2, 3\}$-gram Proportion of Outliers (↓): similar to n-gram models, we first convert the music sequences into tuples of neighboring nodes. Then for Hellinger Distance, we compute the distance of the empirical probabilities of conditional generated samples to the ground truth samples. The empirical probabilities are constructed based on the histograms of the neighboring tuples, with bins being all possible $\{2, 3\}$-gram nodes. Similarly, for the Proportion of Outliers, we count the fractions of newly appeared tuples that are not seen in the training samples. With these metrics, we are able to capture the sequence information instead of just measuring the single node distributions.
- Diverse Edit Distance (↑): which accounts for the creativity/novelty of generated samples across multiple generation runs. For the conditionally generated music samples given the same first 32 notes, we calculate the edit distance, which is the minimum number of single-character edits (insertions, deletions, or substitutions) required to change one music sequence into the other, between each two of the generation samples. The mean and standard deviation are obtained for all edit distances between pairs. The higher the diverse edit distance, the further apart the music

[4] https://github.com/openai/consistency_models/tree/main/evaluations

128 个音符加上 1 个额外的类（表示空音符）。音符顺序被打乱为与 Campbell 等人所述相同的随机顺序。 [2022]使得音符的顺序结构被破坏。

为了进行评估，将 967 个评估序列中的前 32 个音符提供给模型，同时要求模型生成其余 224 个音符。通过对训练和评估音乐序列的分析，我们发现总共可以找到 124 个评估样本，其中至少有一个匹配的训练样本具有相同的 32 个维度。我们分离出这些样本并将其称为 PIANO-P。在PIANO-P中，可以发现20个样本与训练序列中的2个样本包含相同的前32音符，其中3个样本包含相同的前32音符，每个样本有4个训练样本，其中1个样本与6个训练样本相同，1个样本与8个训练样本相同。如果相同的 32 个音符同时出现在训练样本中，并且作为模型提供推论的任务，模型很可能会直接记住训练集中剩余的 224 个音符，并根据训练样本“鹦鹉学舌”地模仿音乐序列。其余843个没有匹配训练样本的评估样本是PIANO的组成部分。

A.18.2 预训练 VQGAN

为了在分类离散潜在空间中生成图像，我们遵循 VQGAN Esser 等人的实现。 [2020]MaskGIT Chang 等人。 [2022]。具体来说，我们使用 Sun 等人中提到的相同的 VQGAN 设置。 [2023]。 VQGAN 是 van den Oord 等人提出的矢量量化变分自编码器 (VQ-VAE) 的变体。 [2018]。在我们的 CIFAR10 设置中，VQGAN 将形状为 $H \times W \times 3$ 的图像编码为词汇大小为 512 的 ($\frac{H}{4} \times \frac{W}{4}$) 标记。对于编码器，我们使用三个卷积块，滤波器大小分别为 64,128 和 256，以及每个块之间的平均池化。对于每个块，它由两个残差块组成。图像编码后，输出映射到码本为 512 $\times$ 256 的令牌索引。对于 VQGAN 损失目标，附加 GAN 损失和感知损失添加权重 0.1。为了训练一个通用的 VQGAN 模型，使我们能够嵌入 CIFAR10 图像而不会过度拟合，我们将数据增强（随机翻转和裁剪）应用于 ImageNet 数据集的 64 $\times$ 64 版本 [Deng 等人，2017]。 [2009]]，并训练 90 个 epoch。 VQGAN 使用 Adam 优化器 ($\beta_1 = 0, \beta_2 = 0.99$) 进行训练，学习率线性预热到 $1e^{-4}$ 的峰值，然后随着余弦衰减而下降。

训练完 VQGAN 后，我们冻结 VQGAN 并将编码器应用于 CIFAR10 图像以创建 8 $\times$ 8 潜在代码。然后，潜在代码被展平为大小为 64 的向量，作为扩散模型的输入。在评估扩散方法之前，我们将生成的潜在代码反馈给 VQGAN 解码器以重建图像空间中的样本。我们通过尝试重建 CIFAR10 数据集来测试 VQGAN 的有效性。使用 Inception V3 模型 [4]，重建给出的 FID 为 7.68，IS 为 10.42。

A.18.3 音乐生成评估指标

在评估条件音乐生成任务时，我们应用以下指标来衡量生成质量:

- 1-gram Hellinger 距离 (↓) 和 1-gram 异常值比例 (↓)：对于这两个指标，我们遵循 Campbell 等人中所述的相同评估。 [2022]和陈[2023]。
- $\{2, 3\}$-gram Hellinger 距离 (↓) 和 $\{2, 3\}$-gram 异常值比例 (↓)：与 n-gram 模型类似，我们首先将音乐序列转换为相邻节点的元组。然后对于海灵格距离，我们计算条件生成样本的经验概率与地面真实样本的距离。经验概率是基于相邻元组的直方图构建的，箱是所有可能的 $\{2, 3\}$-gram 节点。类似地，对于异常值的比例，我们计算训练样本中未出现的新出现元组的分数。通过这些指标，我们能够捕获序列信息，而不仅仅是测量单节点分布。
- 多样化编辑距离 (↑)：它解释了多代运行中生成的样本的创造力/新颖性。对于给定相同前 32 个音符的条件生成的音乐样本，我们计算编辑距离，即每两个生成样本之间将一个音乐序列更改为另一个音乐序列所需的单字符编辑（插入、删除或替换）的最小数量。获得对之间所有编辑距离的平均值和标准差。多样化编辑距离越高，音乐相距越远

[4] `https://github.com/openai/consistency_models/tree/main/evaluations`

sequences are and more creativity are enforced in the generation process. However, there is also a trade-off between diverse edit distance and other accuracy measurements when the model processes large uncertainty about the underlying distribution.

- Train-to-Test Ratios (↑) for $\{1, 2, 3\}$-gram Hellinger as well as Proportion of Outliers, which compare the weighted distance of a generated sample to its evaluation (test) vs. training sequence that share the same first 32 notes. We evaluate the ratios only for PIANO-P. Denote the evaluation ground truth set in PIANO-P as *tr* , the corresponding set of training samples as *ts*, and conditionally generated samples as *gs*. For the selected distance metrics *dist()* (from n-gram Hellinger or n-gram Proportion of Outliers), we calculate the ratio as: $\frac{1}{dist(tr,gs)+dist(ts,gs)} * \frac{dist(tr,gs)}{dist(ts,gs)}$. Such ratio measures quantify the extent of "parroting" in PIANO-P. The larger the ratio, the more equally distant the generated examples is from its training and evaluating set, and the less "parroting" occurs by simply memorizing all training sequences. We apply an additional coefficient to the ratio such that it will also penalize the models that provide unrealistic examples that do not conform both training and evaluation distributions.

A.18.4 Training Details

USD3 Lakh Pianoroll Training Details.

For the backbone sequence transformer structure, we adopt a similar transformer model as utilized by in τ-LDR-0 Campbell et al. [2022]. The sequence transformer is composed of several encoder blocks, where for each internal block, time is fed into the block through a FiLM layer and added to the input. The result is then fed into a multi-headed self-attention layer and fully connected layer. We use RELU for activation. At the output of self-attention layer and fully-connected layer, a dropout of 0.1 is applied. After obtaining the final embedding of each token, we feed that into a 1-block ResNet to obtain the predicted logits. In comparison with other baseline metrics, the transformer contains 6 encoder blocks, each containing 16 attention heads, input dimension of 1024 and MLP dimension of 4096. In ablation study, we also test our methods on a smaller architecture that contains 6 encoder blocks, each containing 8 attention heads, with input dimension of 128 and MLP dimension of 1024.

For training the pianoroll dataset, we use a batch size of 64, a learning rate of $5e^{-4}$, with a warmup of first 25 epochs. We adopt a constant learning rate decay scheduler, decaying the learning rate by half after every 500 epochs. The final result is given over 3000 epochs. We run our results with 2 A6000 GPUs. In discrete-time diffusion, we sample 1000 number of timesteps, fixing a cosine scheduler with $\alpha = 0.008$, In continuous-time diffusion, we sample time t between $[0, 1]$ and apply a constant scheduler with rate equals 0.007. We maintain an exponential moving average of parameters with decay factor 0.9999. We clip the gradient norm at a norm value of 1.0.

Baseline Training Details.

For **D3PM** and **τ-LDR-0**, for fair comparison, we use the same architecture, diffusion scheme and training scheme. For calculating the loss of both methods, we follow the previous literature and give 0.001 to CE loss, and 1 to the VLB loss. For **RDM**, we use the official implementation of loss function in reparam multinomial diffusion[5].We keep the same neural network architecture, and apply reweighting = 1.0, with no label smoothing, and let sampling strategy be cmlm-dep. Similar to D3PM and USD3, we apply cosine scheduler with uniform noise. We remove the padding functions and sampling temperatures when adapting RDM from language tasks to music and image generation.

For **SEDD**, we follow the official code implementation with uniform noise and loss functions[6]. While keeping the same neural architectures as D3PM and USD3, we add an additional masking to the output (the same masking as in SEDD's diffusion transformer), since SEDD models the ratio of two probabilities instead of direct log probabilities. We use log-linear noise schedules with $1e^{-3}$.

For **SDDM**, since it adopts a different architecture, directly utilizing SDDM with our experiment configurations is not feasible. Instead, we use the experiment configurations as given in the paper: the backbone structure is a hollow transformer Sun et al. [2023]. Each transformer block has 6 layers with embedding size of 256, 8 attention heads and hidden dimension of 2048. The batch size is 64 and number of training steps is 2 million. The weight decay is set to $1e^{-6}$. The learning rate is at constant $1e^{-3}$. For diffusion, SDDM adopts the constant noise scheduler with a uniform rate constant of 0.04.

[5] https://github.com/HKUNLP/reparam-discrete-diffusion

[6] https://github.com/louaaron/Score-Entropy-Discrete-Diffusion

序列和更多的创造力在生成过程中得到加强。然而，当模型处理潜在分布的巨大不确定性时，不同的编辑距离和其他精度测量之间也存在权衡。

- $\{1, 2, 3\}$-gram Hellinger 的训练与测试比率 (↑) 以及离群值比例，将生成样本的加权距离与其评估（测试）与共享相同前 32 个音符的训练序列进行比较。我们仅评估 PIANO-P 的比率。将 PIANO-P 中的评估地面真值集表示为 tr，相应的训练样本集表示为 ts，条件生成的样本表示为 gs。对于从 n-gram Hellinger 或 n-gram 异常值比例) 中选择的距离度量 $dist()$ (，我们将比率计算为：$\frac{1}{dist(tr,gs)+dist(ts,gs)} * \frac{dist(tr,gs)}{dist(ts,gs)}$。这种比率测量量化了 PIANO-P 中“鹦鹉学舌”的程度。该比率越大，生成的示例与其训练和评估集的距离越均匀，并且通过简单地记住所有训练序列而发生的“鹦鹉学舌”就越少。我们对比率应用了一个附加系数，这样它也会惩罚那些提供不符合训练和评估分布的不切实际示例的模型。

A.18.4 培训细节

价值 30 万美元的钢琴卷轴培训详情。

对于主干序列变压器结构，我们采用与 τ-LDR-0 Campbell 等人使用的类似变压器模型。 [2022]。序列变换器由多个编码器块组成，其中对于每个内部块，时间通过 FiLM 层馈送到块中并添加到输入中。然后将结果输入多头自注意力层和全连接层。我们使用 RELU 进行激活。在自注意力层和全连接层的输出处，应用 0.1 的 dropout。获得每个 token 的最终嵌入后，我们将其输入 1 块 ResNet 以获得预测的 logits。与其他基线指标相比，变压器包含 6 个编码器块，每个编码器块包含 16 个注意力头，输入维度为 1024，MLP 维度为 4096。在消融研究中，我们还在包含 6 个编码器块的较小架构上测试我们的方法，每个编码器块包含 8 个注意力头，输入维度为 128，MLP 维度为 1024。

为了训练钢琴卷轴数据集，我们使用 64 的批量大小、$5e^{-4}$ 的学习率、前 25 个时期的预热。我们采用恒定的学习率衰减调度器，每 500 个时期后将学习率衰减一半。最终结果给出了 3000 多个 epoch。我们使用 2 个 A6000 GPU 运行结果。在离散时间扩散中，我们对 1000 个时间步进行采样，使用 $\alpha = 0.008$ 固定余弦调度程序。在连续时间扩散中，我们对 [0, 1] 之间的时间 t 进行采样，并应用速率等于 0.007 的常量调度程序。我们维持参数的指数移动平均值，衰减因子为 0.9999。我们将梯度范数裁剪为范数 1.0。

基线培训详细信息。

对于D3PM和τ-LDR-0，为了公平比较，我们使用相同的架构、扩散方案和训练方案。为了计算这两种方法的损失，我们遵循之前的文献，将CE损失赋予0.001，将VLB损失赋予1。对于RDM，我们在重新参数多项式扩散[5]中使用损失函数的官方实现。我们保持相同的神经网络架构，并应用重新加权= 1.0，没有标签平滑，并让采样策略为cmlm-dep。与 D3PM 和 USD3 类似，我们应用具有均匀噪声的余弦调度器。当将 RDM 从语言任务调整为音乐和图像生成时，我们删除了填充函数和采样温度。

对于SEDD，我们遵循官方代码实现，具有统一的噪声和损失函数[6]。在保持与 D3PM 和 USD3 相同的神经架构的同时，我们在输出中添加了额外的掩蔽（与 SEDD 的扩散变压器中的掩蔽相同），因为 SEDD 建模的是两个概率的比率而不是直接对数概率。我们使用 $1e^{-3}$ 的对数线性噪声表。

对于SDDM，由于它采用不同的架构，因此在我们的实验配置中直接使用SDDM是不可行的。相反，我们使用论文中给出的实验配置：主干结构是空心变压器 Sun 等人。 [2023]。每个 Transformer 块有 6 层，嵌入大小为 256，8 个注意力头，隐藏维度为 2048。批量大小为 64，训练步骤数为 200 万。权重衰减设置为 $1e^{-6}$。学习率恒定为 $1e^{-3}$。对于扩散，SDDM 采用恒定噪声调度器，统一速率常数为 0.04。

[5] https://github.com/HKUNLP/reparam-discrete-diffusion
[6] https://github.com/louaaron/Score-Entropy-Discrete-Diffusion

USD3 VQCIFAR10 Training Details.

For image generation task, we parameterize $f_t^\theta(\mathbf{x}_t)$ with the same sequence transformer as mentioned before. The model is a 12 layer transformer, where each layer has 16 attention heads, input embedding of 768 and a hidden layer size of 3072 for the MLPs. We use ReLU for activation. At the output of each internal block, a dropout of 0.1 is applied. The time is input into the network through FiLM layers before the self attention block. We use a learning rate of $5e^{-4}$, with warmup of 50000 steps, and a cosine learning rate decay scheduler, to train 2 million steps. In discrete-time diffusion, all the USD3 and its variants use the same cosine noise scheduler with $\alpha = 0.008$. For continuous-time diffusion, USD3-CE and USD3* apply the cosine noise scheduler with $\alpha = 0.008$. We find that USD3 is extremely hard to optimize due to the scale differences of coefficient $\beta(t)$, so we provide USD3* which clips the $\beta(t) = \max(1, \beta(t))$. USD3 utilizes a constant noise scheduler with 0.007, to match the scheduler in τ-LDR-0 and SDDM. We maintain an exponential moving average of parameters with decay factor 0.9999. We clip the gradient norm at a norm value of 1.0.

Baseline Training Details.

For **τ-LDR-0** and **D3PM**, we again use the same transformer architecture and same training scheme. We apply a hybrid loss with cross entropy as a directly supervision, added with 0.001 CE loss. The τ-LDR-0 uses a constant rate noise scheduler of 0.007, while D3PM uses cosine scheduler with $\alpha = 0.008$. We train all models in parallel on 2 A6000 GPUs. For **D3PM** and **τ-LDR-0**, for fair comparison, we use the same architecture, diffusion scheme and training scheme. For calculating the loss of both methods, we follow the previous literature and give 0.001 to CE loss, and 1 to the VLB loss. For **RDM**, we apply reweighting = 1.0, with no label smoothing, and let sampling strategy be cmlm-dep. We apply cosine scheduler with uniform noise. For **SEDD**, we use log-linear noise schedules with $1e^{-3}$.

For **SDDM**, the model uses a masked modeling, where backbone neural network is BERT-based, with 12 layers of transformers. Each layer has 12 attention heads, embedding size of 768 and hidden layer size of 3072 for MLPs. After obtaining the final embedding of each token, the output is fed that into a 2-block ResNet to acquire the predicted logits. For diffusion, SDDM uses a constant uniform rate of 0.007 as the noise scheduler in the forward process. In Sun et al. [2023], the number of training step is set to $700,000$, where the learning rate is warmed up to $1e^{-4}$ during the first 3% steps, and then decays to 0 in a linear schedule. We extended the training to $2m$ steps, but did not observe improved performance.

A.19 Additional Experiment Results

A.19.1 Full Experiment Results on Music Generation

We present the full evaluation metrics and results, including mean and standard deviation, for Lakh Pianoroll music generation tasks, as shown in Table 4.

Table 4: Metrics comparing generated conditional samples and evaluating ground truths for Lakh Pianoroll. For each of n-gram Hellinger Distance (ng.-Hellinger) and Proportion of Outliers (ng.-Prop. Outlier) and Diverse Edit Distance metrics, we show mean $\pm$ std with respect to 3 generated samples. The top two are highlighted by **First**, **Second**.

	Method	1g.-Hellinger(↓)	2g.-Hellinger(↓)	3g.-Hellinger(↓)	1g.-Prop.Out.(↓)	2g.-Prop.Out.(↓)	3g.-Prop.Out.(↓)	Edit Distance(↑)
Discrete-time	D3PM	0.3982±0.0004	0.5303±0.0038	0.5918±0.0046	0.1209±0.0008	0.2532± 0.0053	0.3790±0.0059	**0.2950±0.0775**
	RDM	0.4123±0.0012	0.5964±0.0021	0.6198±0.0039	0.1401±0.0011	0.2459± 0.0023	0.3864±0.0030	0.2891±0.0690
	USD3-CE	0.3754±0.0007	0.4835±0.0009	0.5741±0.0008	0.1079±0.0002	0.2099±0.0005	0.3034±0.0007	0.0472±0.0465
	USD3-VLB	0.3790±0.0009	0.4640±0.0010	**0.5427±0.0009**	0.1174±0.0007	0.1845±0.0010	0.2734±0.0011	**0.0828±0.0567**
	USD3	0.3770±0.0011	0.4693±0.0015	0.5525±0.0015	**0.1077±0.0006**	0.1861±0.0011	0.2861±0.0013	0.0648±0.0459
	USD3*	0.3753±0.0013	0.4704±0.0010	0.5550±0.0012	0.1107±0.0005	0.1911±0.0005	0.2832±0.0011	0.0664±0.0471
Continuous-time	SDDM	0.3759±0.0020	0.4856±0.0015	0.5773±0.0009	0.1101±0.0015	0.2059±0.0005	0.3409±0.0017	0.0606±0.0249
	τ-LDR-0	0.3796±0.0009	0.4811±0.0008	0.5710±0.0007	0.1149±0.0008	0.2078±0.0014	0.3202±0.0014	0.0553±0.0637
	SEDD	0.3814±0.0012	0.4712 ±0.0020	0.5470 ±0.0013	**0.1052±0.0010**	**0.1823±0.0015**	0.2709±0.0013	0.0632±0.0532
	USD3-CE	**0.3734±0.0002**	0.4837±0.0003	0.5776±0.0013	0.1158±0.0004	0.2218±0.0003	0.3461±0.0005	0.0434±0.0357
	USD3-VLB	0.3764±0.0006	**0.4620±0.0014**	0.5487±0.0018	0.1198±0.0009	0.1866±0.0011	0.2952±0.0011	0.0661±0.0478
	USD3	**0.3735±0.0005**	**0.4617±0.0012**	**0.5432±0.0018**	0.1123±0.0009	**0.1840±0.0011**	**0.2704±0.0009**	0.0661±0.0401
	USD3*	0.3737±0.0004	0.4623±0.0009	0.5490±0.0017	0.1143±0.0008	0.1844±0.0008	**0.2707±0.0011**	0.0516±0.0365

A.19.2 Study of "Parroting" with Ratio Metrics

As shown in Table 5, we find that models with combined losses, USD3 and USD3*, achieve higher Train-to-Test ratio than pure USD3-CE or USD3-VLB, suggesting that the combination of two losses

3 美元 VQCIFAR10 培训详情。

对于图像生成任务，我们使用与前面提到的相同的序列转换器来参数化 $f_t^\theta(\mathbf{x}_t)$。该模型是一个 12 层 Transformer，其中每层有 16 个注意力头，输入嵌入为 768，MLP 的隐藏层大小为 3072。我们使用 ReLU 进行激活。在每个内部块的输出处，应用 0.1 的 dropout。时间通过自注意力块之前的 FiLM 层输入到网络中。我们使用 $5e^{-4}$ 的学习率、50000 步的预热和余弦学习率衰减调度程序来训练 200 万步。在离散时间扩散中，所有 USD3 及其变体都使用相同的余弦噪声调度器 $\alpha = 0.008$。对于连续时间扩散，USD3-CE 和 USD3* 应用带有 $\alpha = 0.008$ 的余弦噪声调度器。我们发现，由于系数$\beta(t)$的尺度差异，USD3极其难以优化，因此我们提供了USD3*，它对$\beta(t) = \max(1, \beta(t))$进行了剪辑。 USD3 使用 0.007 的恒定噪声调度程序，以匹配 τ-LDR-0 和 SDDM 中的调度程序。我们维持参数的指数移动平均值，衰减因子为 0.9999。我们将梯度范数裁剪为范数 1.0。

基线培训详细信息。

对于 τ-LDR-0 和 D3PM，我们再次使用相同的变压器架构和相同的训练方案。我们应用交叉熵的混合损失作为直接监督，并添加 0.001 CE 损失。 τ-LDR-0 使用恒定速率噪声调度器 0.007，而 D3PM 使用余弦调度器 $\alpha = 0.008$。我们在 2 个 A6000 GPU 上并行训练所有模型。对于D3PM和τ-LDR-0，为了公平比较，我们使用相同的架构、扩散方案和训练方案。为了计算这两种方法的损失，我们遵循之前的文献，将CE损失赋予0.001，将VLB损失赋予1。对于 RDM，我们应用重新加权 = 1.0，没有标签平滑，并让采样策略为 cmlm-dep。我们应用具有均匀噪声的余弦调度器。对于 SEDD，我们使用 $1e^{-3}$ 的对数线性噪声计划。

对于SDDM，该模型使用掩模建模，其中主干神经网络是基于BERT的，具有12层变压器。每层有 12 个注意力头，MLP 的嵌入大小为 768，隐藏层大小为 3072。获得每个 token 的最终嵌入后，将输出输入到 2 块 ResNet 中以获取预测的 logits。对于扩散，SDDM 使用恒定的均匀速率 0.007 作为前向过程中的噪声调度器。在孙等人。 [2023]，训练步骤数设置为 700, 000，其中学习率在前 3% 的步骤中预热到 $1e^{-4}$，然后以线性时间表衰减到 0。我们将训练扩展到 $2m$ 步骤，但没有观察到性能的提高。

A.19 附加实验结果

A.19.1 音乐生成的完整实验结果

我们展示了 Lakh Pianoroll 音乐生成任务的完整评估指标和结果，包括平均值和标准差，如表 4 所示。

表 4：比较生成的条件样本和评估 Lakh Pianoroll 的基本事实的指标。对于每个 n-gram Hellinger 距离 (ng.-Hellinger) 和离群值比例 (ng.-Prop. Outlier) 以及多样化编辑距离指标，我们显示了 3 个生成样本的平均值 ± std。前两个突出显示为“第一”、“第二”。

	Method	1g.-Hellinger(↓)	2g.-Hellinger(↓)	3g.-Hellinger(↓)	1g.-Prop.Out.(↓)	2g.-Prop.Out.(↓)	3g.-Prop.Out.(↓)	Edit Distance(↑)
Discrete-Time	D3PM	0.3982±0.0004	0.5303±0.0038	0.5918±0.0046	0.1209±0.0008	0.2532± 0.0053	0.3790±0.0059	**0.2950±0.0775**
	RDM	0.4123±0.0012	0.5964±0.0021	0.6198±0.0039	0.1401±0.0011	0.2459± 0.0023	0.3864±0.0030	0.2891±0.0690
	USD3-CE	0.3754±0.0007	0.4835±0.0009	0.5741±0.0008	0.1079±0.0002	0.2099±0.0005	0.3034±0.0007	0.0472±0.0465
	USD3-VLB	0.3790±0.0009	0.4640±0.0010	**0.5427±0.0009**	0.1174±0.0007	0.1845±0.0010	0.2734±0.0011	**0.0828±0.0567**
	USD3	0.3770±0.0011	0.4693±0.0015	0.5525±0.0015	**0.1077±0.0006**	0.1861±0.0011	0.2861±0.0013	0.0648±0.0459
	USD3*	0.3753±0.0013	0.4704±0.0010	0.5550±0.0012	0.1107±0.0005	0.1911±0.0005	0.2832±0.0011	0.0664±0.0471
Continuous-Time	SDDM	0.3759±0.0020	0.4856±0.0015	0.5773±0.0009	0.1101±0.0015	0.2059±0.0005	0.3409±0.0017	0.0606±0.0249
	τ-LDR-0	0.3796±0.0009	0.4811±0.0008	0.5710±0.0007	0.1149±0.0008	0.2078±0.0014	0.3202±0.0014	0.0553±0.0637
	SEDD	0.3814±0.0012	0.4712 ±0.0020	0.5470 ±0.0013	**0.1052±0.0010**	**0.1823±0.0015**	0.2709±0.0013	0.0632±0.0532
	USD3-CE	**0.3734±0.0002**	0.4837±0.0003	0.5776±0.0013	0.1158±0.0004	0.2218±0.0003	0.3461±0.0005	0.0434±0.0357
	USD3-VLB	0.3764±0.0006	**0.4620±0.0014**	0.5487±0.0018	0.1198±0.0009	0.1866±0.0011	0.2952±0.0011	0.0661±0.0478
	USD3	**0.3735±0.0005**	**0.4617±0.0012**	**0.5432±0.0018**	0.1123±0.0009	**0.1840±0.0011**	**0.2704±0.0009**	0.0661±0.0401
	USD3*	0.3737±0.0004	0.4623±0.0009	0.5490±0.0017	0.1143±0.0008	0.1844±0.0008	**0.2707±0.0011**	0.0516±0.0365

A.19.2 用比率度量进行“鹦鹉学舌”研究

如表 5 所示，我们发现具有组合损失的模型 USD3 和 USD3* 比纯 USD3-CE 或 USD3-VLB 实现了更高的训练测试比，这表明两种损失的组合

can alleviate overfitting, i.e. "parroting" the training data. Overall, τ-LDR-0 is a strong competitor in n-gram Hellinger Train-to-Test ratios, suggesting that its VLB focused loss optimization can also alleviate parroting and generate diverse samples.

Table 5: Train-to-test Ratio Metrics (↑) for Lakh Pianoroll, using evaluation sequences fron PIANO-P and the training sequences. The higher ratios indicate less "parroting" phenomenon and better generalization ability. Mean and std are calculated across 3 generated samples.The top two are highlighted by **First**, **Second**.

	Method	1g.-Hellinger Ratio	2g.-Hellinger Ratio	3g.-Hellinger Ratio	1g.-Prop.Out. Ratio	2g.-Prop.Out. Ratio	3g.-Prop.Out.r Ratio
Discrete-time	D3PM	1.7182±0.0500	1.3624±0.0411	1.1307±0.0304	5.6237±0.1463	3.7051±0.1162	2.2216±0.0616
	RDM	1.7534±0.0928	1.4023±0.00701	1.2023±0.0392	5.9981±0.0497	3.6053±0.1532	2.5211±0.0256
	USD3-CE	1.7922±0.0186	1.4028±0.0074	1.1906±0.0090	6.3964±0.4669	3.9206±0.1082	2.8887±0.0399
	USD3-VLB	1.7758±0.0097	1.4123±0.0118	1.2011±0.0101	6.2024±0.2584	3.9093± 0.0663	2.8635±0.0479
	USD3	1.8073±0.0880	1.4749±0.0708	1.2078±0.0518	**6.8742±0.7366**	**4.2448± 0.4161**	**3.0830±0.2520**
	USD3*	1.7705±0.0235	**1.4959±0.0176**	1.1928±0.0125	**6.6452±0.2702**	3.9867±0.1188	2.9591±0.0653
Continuous-time	SDDM	1.7872±0.0131	1.4516±0.0097	1.1812±0.0103	6.3019±0.2522	4.0652±0.0591	2.9014±0.0782
	τ-LDR-0	**1.8648±0.0306**	1.4726±0.0233	**1.2497±0.0185**	6.4586±0.3798	4.0863±0.2001	2.9651±0.1089
	SEDD	1.8230±0.0298	**1.4902±0.0090**	1.2152±0.0150	6.4361±0.2001	4.0598±0.0621	3.0421±0.0922
	USD3-CE	1.7951±0.0290	1.4023±0.0264	1.1474±0.0206	6.3712±0.4502	3.6232±0.1734	2.6379±0.0823
	USD3-VLB	1.7379±0.0220	1.4117±0.0250	1.1810±0.0232	6.2752±0.1232	3.9321±0.0949	2.9042±0.0853
	USD3	**1.8623±0.0340**	1.4516±0.0077	1.2017±0.0096	6.4218±0.1739	**4.0877±0.0553**	**3.0828±0.0502**
	USD3*	1.7970±0.0106	1.4898±0.0026	**1.2377±0.0015**	6.4361±0.2023	3.9340±0.0284	2.9234±0.0110

A.19.3 Model Sizes and Loss Analysis on Piano Dataset

Table 6 shows the evaluation metrics of different model sizes.This ablation study shows that CE loss is also preferred in combination with VLB, especially for smaller network structures, since VLB is harder to optimize alone.

Table 6: Metrics comparing different loss combinations and different model sizes for Lakh Pianoroll. For each of n-gram Hellinger Distance (ng.-Hellinger) and Proportion of Outliers (ng.-Prop. Outlier) metrics, we show mean ± std with respect to 3 generated samples. We use USD3-VLB to denote an additional variant of our model that only uses the exact VLB loss in training. "Small" refers to the backbone transformer model that has 6 Layers, 8 Attention Heads, Input Dimension of 128 and MLP dimension of 1024. The top two are highlighted by **First**, **Second**.

	Method	1g.-Hellinger(↓)	2g.-Hellinger(↓)	3g.-Hellinger(↓)	1g.-Prop.Out.(↓)	2g.-Prop.Out.(↓)	3g.-Prop.Out.(↓)	Edit Distance(↑)
Discrete-time	USD3-CE-S	0.3984±0.0006	0.4902 ±0.0004	0.5785±0.0004	0.1158±0.0002	0.1899±0.0006	0.3142±0.0005	0.1301±0.0613
	USD3-S	0.4011±0.0014	0.4902±0.0009	0.5707±0.0008	0.1215± 0.0007	0.1866±0.0006	0.3006±0.0013	0.1292±0.0623
	USD3-VLB-S	0.4115±0.0001	0.4954± 0.0001	0.5738±0.0005	0.1203±0.0006	0.1958±0.0009	0.3036±0.0010	0.2137±0.0843
	USD3-CE	0.3754±0.0007	0.4835±0.0009	0.5741±0.0008	**0.1079±0.0002**	0.2099±0.0005	0.3034±0.0007	0.0472±0.0465
	USD3	0.3770±0.0011	0.4693±0.0015	0.5525±0.0015	**0.1077±0.0006**	**0.1861±0.0011**	0.2861±0.0013	0.0648±0.0459
	USD3-VLB	0.3790±0.0009	0.4640±0.0010	**0.5427±0.0009**	0.1174±0.0007	0.1845±0.0010	**0.2734±0.0011**	0.0828±0.0567
Continuous-time	USD3-CE-S	0.4208±0.0170	0.5196±0.0078	0.6072±0.0005	0.1355±0.0147	0.2276±0.0007	0.3431±0.0181	0.2358±0.0619
	USD3-S	0.4239±0.0012	0.5083±0.0011	0.5852±0.0011	0.1403±0.0007	0.2070±0.0008	0.2943±0.0016	**0.2468±0.0907**
	USD3-VLB-S	0.4435±0.0012	0.5295±0.0011	0.6070±0.0009	0.1562±0.0014	0.2269±0.0013	0.3168±0.0013	**0.2809±0.0866**
	USD3-CE	**0.3734±0.0002**	0.4837±0.0003	0.5776±0.0013	0.1158±0.0004	0.2218±0.0003	0.3461±0.0005	0.0434±0.0357
	USD3	**0.3735±0.0005**	**0.4617±0.0012**	**0.5432±0.0018**	0.1123±0.0009	**0.1840±0.0011**	**0.2704±0.0009**	0.0661±0.0401
	USD3-VLB	0.3764±0.0006	**0.4620±0.0014**	0.5487±0.0018	0.1198±0.0009	0.1866±0.0011	0.2952±0.0011	0.0661±0.0478

A.19.4 Memory and Running-time Comparison

Table 7 provides the running time, number of parameters (in the transformer architecture) and GPU-memory. While D3PM, and τ-LDR-0 use the same backbone transformer structure as USD3 and contain the same number of parameters in the transformer, they require a passing of transition matrices, which incur additional memory and slow down the training process. RDM is similar to USD3-CE and USD3* in memory calculation and runtime. SDDM requires a special architecture and considerably larger memory during training. SEDD requires significantly less memory than USD3 (continuous version), and training time is almost equivalent to the discrete-version USD3-CE and USD3*. SEDD's loss calculation removes the redundant additional forward sampling, yet the loss calculation is less exact tham USD3 in continuous-time.

A.19.5 Qualitative Examples for VQCIFAR10 Image Generation

We provide 80 reconstructed images VQGAN decoder in Fig. 2. Most examples are easy to recognize from one of the 10 classes in the original CIFAR10: airplanes, cars, birds, cats, deer, dogs, frogs, horses, ships, and trucks.

可以缓解过度拟合，即“鹦鹉学舌”训练数据。总体而言，τ-LDR-0 在 n-gram Hellinger 训练与测试比率方面是强有力的竞争对手，这表明其以 VLB 为重点的损失优化也可以减轻鹦鹉学舌并生成多样化的样本。

表 5：Lakh Pianoroll 的训练与测试比率指标 (↑)，使用 PIANO-P 的评估序列和训练序列。比率越高，表明“鹦鹉学舌”现象越少，泛化能力越好。平均值和标准差是在 3 个生成的样本中计算的。前两个通过 First、Second 突出显示。

	Method	1g.-Hellinger Ratio	2g.-Hellinger Ratio	3g.-Hellinger Ratio	1g.-Prop.Out. Ratio	2g.-Prop.Out. Ratio	3g.-Prop.Out.r Ratio
Discrete-time	D3PM	1.7182±0.0500	1.3624±0.0411	1.1307±0.0304	5.6237±0.1463	3.7051±0.1162	2.2216±0.0616
	RDM	1.7534±0.0928	1.4023±0.00701	1.2023±0.0392	5.9981±0.0497	3.6053±0.1532	2.5211±0.0256
	USD3-CE	1.7922±0.0186	1.4028±0.0074	1.1906±0.0090	6.3964±0.4669	3.9206±0.1082	2.8887±0.0399
	USD3-VLB	1.7758±0.0097	1.4123±0.0118	1.2011±0.0101	6.2024±0.2584	3.9093± 0.0663	2.8635±0.0479
	USD3	1.8073±0.0880	1.4749±0.0708	1.2078±0.0518	**6.8742±0.7366**	**4.2448± 0.4161**	**3.0830±0.2520**
	USD3*	1.7705±0.0235	**1.4959±0.0176**	1.1928±0.0125	**6.6452±0.2702**	3.9867±0.1188	2.9591±0.0653
Continuous-time	SDDM	1.7872±0.0131	1.4516±0.0097	1.1812±0.0103	6.3019±0.2522	4.0652±0.0591	2.9014±0.0782
	τ-LDR-0	**1.8648±0.0306**	1.4726±0.0233	**1.2497±0.0185**	6.4586±0.3798	4.0863±0.2001	2.9651±0.1089
	SEDD	1.8230±0.0298	**1.4902±0.0090**	1.2152±0.0150	6.4361±0.2001	4.0598±0.0621	3.0421±0.0922
	USD3-CE	1.7951±0.0290	1.4023±0.0264	1.1474±0.0206	6.3712±0.4502	3.6232±0.1734	2.6379±0.0823
	USD3-VLB	1.7379±0.0220	1.4117±0.0250	1.1810±0.0232	6.2752±0.1232	3.9321±0.0949	2.9042±0.0853
	USD3	**1.8623±0.0340**	1.4516±0.0077	1.2017±0.0096	6.4218±0.1739	**4.0877±0.0553**	**3.0828±0.0502**
	USD3*	1.7970±0.0106	1.4898±0.0026	**1.2377±0.0015**	6.4361±0.2023	3.9340±0.0284	2.9234±0.0110

A.19.3 Piano 数据集的模型大小和损失分析

表6显示了不同模型大小的评估指标。这项消融研究表明，CE损失也优先与VLB结合使用，特别是对于较小的网络结构，因为VLB更难以单独优化。

表 6：比较 Lakh Pianoroll 的不同损耗组合和不同模型大小的指标。对于每个 n-gram Hellinger 距离 (ng.-Hellinger) 和离群值比例 (ng.-Prop. Outlier) 指标，我们显示了 3 个生成样本的平均值 ± std。我们使用 USD3-VLB 来表示模型的另一个变体，该变体仅在训练中使用精确的 VLB 损失。“小”是指骨干变压器模型有 6 层、8 个注意力头、输入维度为 128、MLP 维度为 1024。前两个用 First、Second 突出显示。

	Method	1g.-Hellinger(↓)	2g.-Hellinger(↓)	3g.-Hellinger(↓)	1g.-Prop.Out.(↓)	2g.-Prop.Out.(↓)	3g.-Prop.Out.(↓)	Edit Distance(↑)
Discrete-time	USD3-CE-S	0.3984±0.0006	0.4902 ±0.0004	0.5785±0.0004	0.1158±0.0002	0.1899±0.0006	0.3142±0.0005	0.1301±0.0613
	USD3-S	0.4011±0.0014	0.4902±0.0009	0.5707±0.0008	0.1215± 0.0007	0.1866±0.0006	0.3006±0.0013	0.1292±0.0623
	USD3-VLB-S	0.4115±0.0001	0.4954± 0.0001	0.5738±0.0005	0.1203±0.0006	0.1958±0.0009	0.3036±0.0010	0.2137±0.0843
	USD3-CE	0.3754±0.0007	0.4835±0.0009	0.5741±0.0008	**0.1079±0.0002**	0.2099±0.0005	0.3034±0.0007	0.0472±0.0465
	USD3	0.3770±0.0011	0.4693±0.0015	0.5525±0.0015	**0.1077±0.0006**	**0.1861±0.0011**	0.2861±0.0013	0.0648±0.0459
	USD3-VLB	0.3790±0.0009	0.4640±0.0010	**0.5427±0.0009**	0.1174±0.0007	0.1845±0.0010	**0.2734±0.0011**	0.0828±0.0567
Continuous-time	USD3-CE-S	0.4208±0.0170	0.5196±0.0078	0.6072±0.0005	0.1355±0.0147	0.2276±0.0007	0.3431±0.0181	0.2358±0.0619
	USD3-S	0.4239±0.0012	0.5083±0.0011	0.5852±0.0011	0.1403±0.0007	0.2070±0.0008	0.2943±0.0016	**0.2468±0.0907**
	USD3-VLB-S	0.4435±0.0012	0.5295±0.0011	0.6070±0.0009	0.1562±0.0014	0.2269±0.0013	0.3168±0.0013	**0.2809±0.0866**
	USD3-CE	**0.3734±0.0002**	0.4837±0.0003	0.5776±0.0013	0.1158±0.0004	0.2218±0.0003	0.3461±0.0005	0.0434±0.0357
	USD3	**0.3735±0.0005**	**0.4617±0.0012**	**0.5432±0.0018**	0.1123±0.0009	**0.1840±0.0011**	**0.2704±0.0009**	0.0661±0.0401
	USD3-VLB	0.3764±0.0006	**0.4620±0.0014**	0.5487±0.0018	0.1198±0.0009	0.1866±0.0011	0.2952±0.0011	0.0661±0.0478

A.19.4 内存和运行时间比较

表 7 提供了运行时间、参数数量（在 Transformer 架构中）和 GPU 内存。虽然 D3PM 和 τ-LDR-0 使用与 USD3 相同的主干变压器结构，并且变压器中包含相同数量的参数，但它们需要传递转换矩阵，这会产生额外的内存并减慢训练过程。 RDM在内存计算和运行时与USD3-CE和USD3*类似。 SDDM 在训练期间需要特殊的架构和相当大的内存。 SEDD 所需的内存明显少于 USD3（连续版本），并且训练时间几乎相当于离散版本 USD3-CE 和 USD3*。 SEDD的损失计算消除了多余的额外前向采样，但损失计算在连续时间上不如USD3精确。

A.19.5 VQCIFAR10 图像生成的定性示例

我们在图 2 中提供了 80 个重建图像 VQGAN 解码器。大多数示例很容易从原始 CIFAR10 中的 10 个类别之一中识别：飞机、汽车、鸟、猫、鹿、狗、青蛙、马、船和卡车。

Table 7: The GPU-memory, running time and number of network parameters in all methods.USD3 is easier to train and incurs the least GPU memory in both discrete- and continuous- time diffusions.

	Method	Num. Parameters	Memory	Runtime		Method	Num. Parameters	Memory	Runtime
Discrete-time	D3PM	$\sim$ 102,700,000	15669MiB	93 hrs	Continuous-time	SDDM	$\sim$ 12,350,000	85528MiB	96 hrs
	RDM	$\sim$ 102,700,000	9570MiB	86 hrs		SEDD	$\sim$ 102,700,000	4620MiB	85 hrs
	USD3-CE	$\sim$ 102,700,000	9735MiB	83 hrs		τ-LDR-0	$\sim$ 102,700,000	17669 MiB	129 hrs
	USD3*	$\sim$ 102,700,000	9735MiB	82 hrs		USD3-CE	$\sim$ 102,700,000	15703MiB	93 hrs
	USD3	$\sim$ 102,700,000	9735MiB	90 hrs		USD3*	$\sim$ 102,700,000	17620MiB	91 hrs
						USD3	$\sim$ 102,700,000	17620MiB	101 hrs

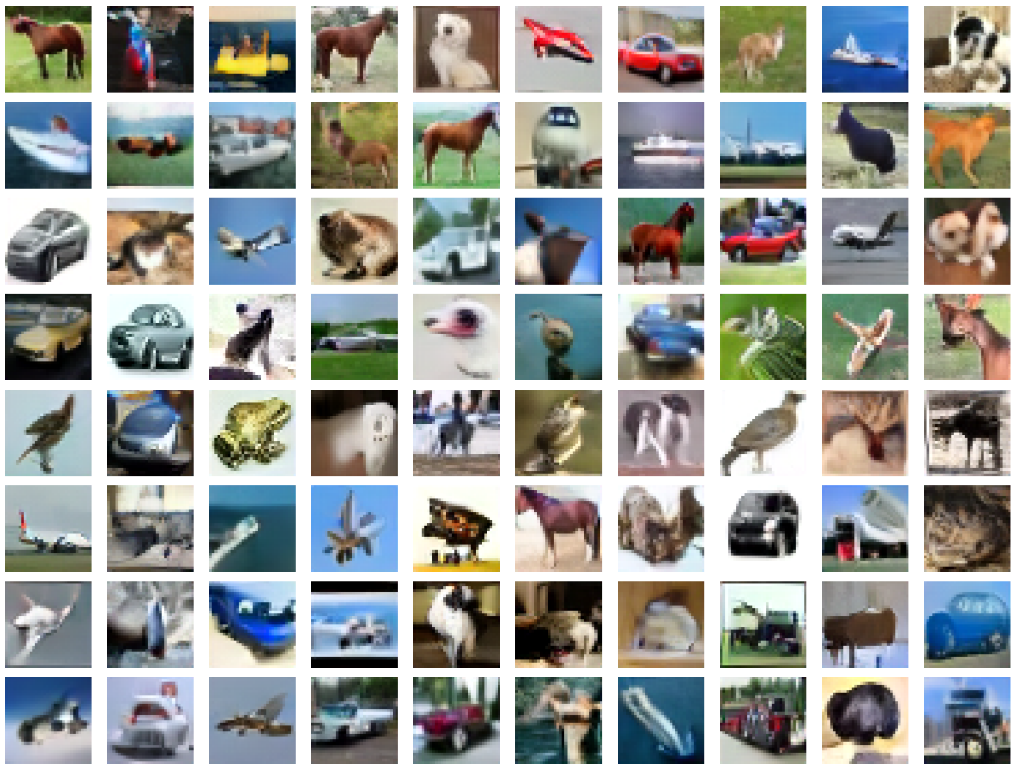


Figure 2: Example image samples generated by USD3* as trained on VQCIFAR10. Most images are easy to recognize as being from one of the 10 classes in CIFAR10.

A.19.6 MCMC for Discrete-time and Continuous-time Discrete Diffusion

To demonstrate the effectiveness of MCMC corrector steps, we take the top performing methods in discrete and continuous time diffusion models (for discrete time, USD3*, and for continuous time USD3-CE) and show improved quality metrics over generated images after MCMC corrector is applied. Due to extensive time required for MCMC corrector step, we could only conduct evaluation over $5,000$ images, and thus the results are not comparable to the main VQCIFAR10 result in Table 2. We set the number of generation steps to be 100 and use MCMC corrector on the last 10, 20 generation timesteps, respectively.

From the results shown in Table 8 and Table 9, we can see that MCMC corrector can significantly improve the quality of the generated samples. Specifically, for discrete-time generation, IS is showing a significant improvement than the sampling process without the MCMC corrector. For continuous-time case, both IS and FID scores are improving from the baseline (without MCMC).

表 7：所有方法中的 GPU 内存、运行时间和网络参数数量。USD3 更容易训练，并且在离散时间和连续时间扩散中占用的 GPU 内存最少。

	Method	Num. Parameters	Memory	Runtime
Discrete-time	D3PM	$\sim$ 102,700,000	15669MiB	93 hrs
	RDM	$\sim$ 102,700,000	9570MiB	86 hrs
	USD3-CE	$\sim$ 102,700,000	9735MiB	83 hrs
	USD3*	$\sim$ 102,700,000	9735MiB	82 hrs
	USD3	$\sim$ 102,700,000	9735MiB	90 hrs
Continuous-time	SDDM	$\sim$ 12,350,000	85528MiB	96 hrs
	SEDD	$\sim$ 102,700,000	4620MiB	85 hrs
	τ-LDR-0	$\sim$ 102,700,000	17669 MiB	129 hrs
	USD3-CE	$\sim$ 102,700,000	15703MiB	93 hrs
	USD3*	$\sim$ 102,700,000	17620MiB	91 hrs
	USD3	$\sim$ 102,700,000	17620MiB	101 hrs

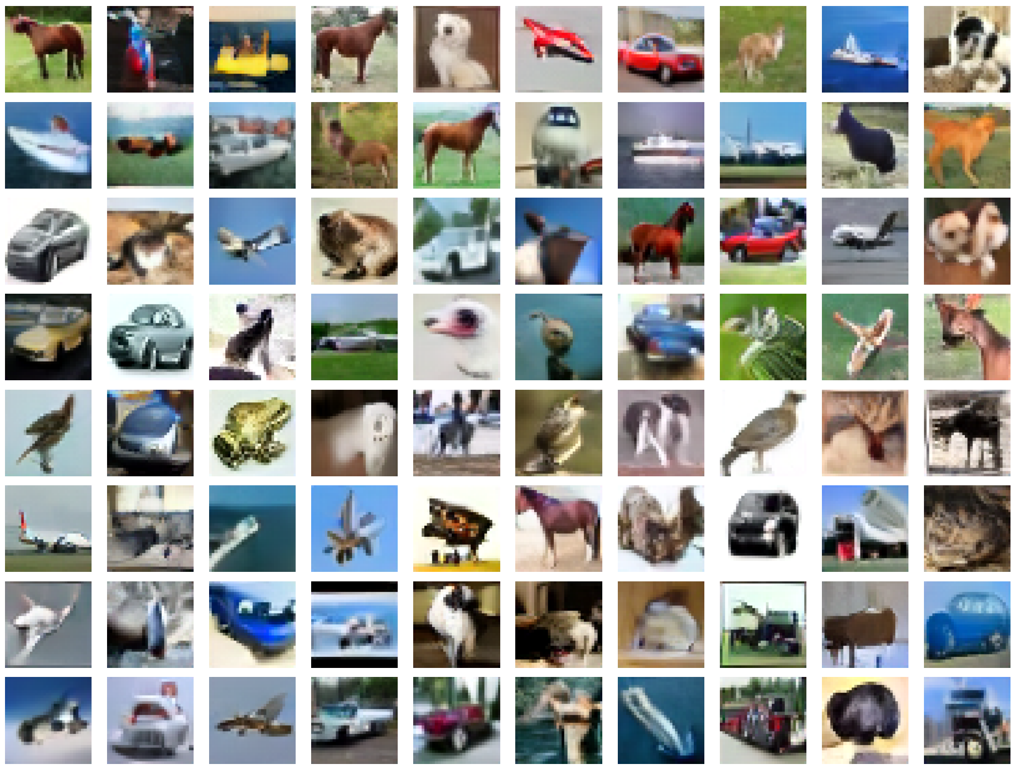


图 2：USD3* 在 VQCIFAR10 上训练生成的示例图像样本。大多数图像很容易被识别为来自 CIFAR10 中的 10 个类别之一。

A.19.6 离散时间和连续时间离散扩散的 MCMC

为了证明 MCMC 校正器步骤的有效性，我们采用了离散和连续时间扩散模型（对于离散时间 USD3* 和对于连续时间 USD3-CE）中表现最佳的方法，并显示了应用 MCMC 校正器后生成的图像质量指标的改进。由于MCMC校正器步骤需要大量时间，我们只能对5, 000张图像进行评估，因此结果与表2中的主要VQCIFAR10结果无法比较。我们将生成步骤数设置为100，并分别在最后10、20个生成时间步上使用MCMC校正器。

从表8和表9所示的结果可以看出，MCMC校正器可以显着提高生成样本的质量。具体来说，对于离散时间生成，IS 比没有 MCMC 校正器的采样过程显示出显着的改进。对于连续时间情况，IS 和 FID 分数均较基线（无 MCMC）有所提高。

Table 8: Image gen. quality w.r.t. Inception Score (IS) and the Frechet Inception Dist. (FID) over 5,000 samples unconditionally generated by USD3* in discrete-time case. MCMC corrector is conducted for the last 10,20 timesteps over 100 sampling steps.

MCMC Configuration	IS (↑)	FID (↓)
Without MCMC	9.01	19.79
$\Delta n = 0.001, N = 2$, Start Steps:10	9.28	18.46
$\Delta n = 0.001, N = 2$, Start Steps:20	**9.59**	18.36
$\Delta n = 0.005, N = 2$, Start Steps:10	9.43	18.26
$\Delta n = 0.005, N = 2$, Start Steps:20	9.43	20.37
$\Delta n = 0.001, N = 5$, Start Steps:10	9.29	**18.02**
$\Delta n = 0.001, N = 5$, Start Steps:20	9.47	18.56
$\Delta n = 0.002, N = 5$, Start Steps:10	9.35	18.18
$\Delta n = 0.002, N = 5$, Start Steps:20	9.48	20.37

Table 9: Image gen. quality w.r.t. Inception Score (IS) and the Frechet Inception Dist. (FID) over 5,000 samples generated by USD3-CE in continuous-time case. MCMC corrector is conducted for the last 10,20 timesteps over 100 sampling steps.

MCMC Configuration	IS (↑)	FID (↓)
Without MCMC	8.98	19.19
$\Delta n = 0.001, N = 2$, Start Steps:10	9.12	17.62
$\Delta n = 0.001, N = 2$, Start Steps:20	**9.23**	17.35
$\Delta n = 0.005, N = 2$, Start Steps:10	9.01	17.71
$\Delta n = 0.005, N = 2$, Start Steps:20	**9.23**	17.58
$\Delta n = 0.001, N = 5$, Start Steps:10	9.03	**17.26**
$\Delta n = 0.001, N = 5$, Start Steps:20	9.00	17.42
$\Delta n = 0.002, N = 5$, Start Steps:10	9.16	17.98
$\Delta n = 0.002, N = 5$, Start Steps:20	8.97	17.83

A.19.7 Sampling Time Discrete-time and Continuous-time Discrete Diffusion

In Table 10, we have calculated the time required for baselines and our models to sample 50,000 VQ-encoded images with 1000 timesteps on a single A6000 GPU. In addition, we calculate the time required when USD3 and τ-LDR need MCMC corrector steps (=5) in sampling. We compare our results in the following tables. In terms of sampling time, RDM yields the most sampling time due to its re-routing sampling scheme. SEDD and τ-LDR are similar in sampling time, due to the tau-leaping scheme. USD3 falls short of SDDM, which requires a special NN architecture and different code base(jaxlib) than other methods. But when considering MCMC sampling, USD3 brings faster sampling time than tau-leaping in τ-LDR, while allowing sampling for both continuous-time and discrete-time models.

Table 10: Sample times and MCMC sampling steps for different models.

Model	**Sample Time**	**MCMC Sampling Step=5**
RDM	~23 hours	-
D3PM	~8 hours	-
τ-LDR	~9 hours	~2 days, 20 hours
SDDM	~7 hours	-
SEDD	~8 hours	-
USD3 (continuous version)	~8 hours	~2 days, 12 hours
USD3 (discrete version)	~8 hours	~2 days, 12 hours

表 8：图像生成质量Inception Score (IS) 和 Frechet Inception Dist。(FID) USD3* 在离散时间情况下无条件生成超过 5,000 个样本。 MCMC 校正器针对 100 个采样步骤的最后 10, 20 个时间步骤进行。

MCMC Configuration	IS (↑)	FID (↓)
Without MCMC	9.01	19.79
$\Delta n = 0.001, N = 2$, Start Steps:10	9.28	18.46
$\Delta n = 0.001, N = 2$, Start Steps:20	**9.59**	18.36
$\Delta n = 0.005, N = 2$, Start Steps:10	9.43	18.26
$\Delta n = 0.005, N = 2$, Start Steps:20	9.43	20.37
$\Delta n = 0.001, N = 5$, Start Steps:10	9.29	**18.02**
$\Delta n = 0.001, N = 5$, Start Steps:20	9.47	18.56
$\Delta n = 0.002, N = 5$, Start Steps:10	9.35	18.18
$\Delta n = 0.002, N = 5$, Start Steps:20	9.48	20.37

表 9：图像生成质量Inception Score (IS) 和 Frechet Inception Dist。(FID) USD3-CE 在连续时间情况下生成超过 5,000 个样本。 MCMC 校正器针对 100 个采样步骤的最后 10,20 个时间步骤进行。

MCMC Configuration	IS (↑)	FID (↓)
Without MCMC	8.98	19.19
$\Delta n = 0.001, N = 2$, Start Steps:10	9.12	17.62
$\Delta n = 0.001, N = 2$, Start Steps:20	**9.23**	17.35
$\Delta n = 0.005, N = 2$, Start Steps:10	9.01	17.71
$\Delta n = 0.005, N = 2$, Start Steps:20	**9.23**	17.58
$\Delta n = 0.001, N = 5$, Start Steps:10	9.03	**17.26**
$\Delta n = 0.001, N = 5$, Start Steps:20	9.00	17.42
$\Delta n = 0.002, N = 5$, Start Steps:10	9.16	17.98
$\Delta n = 0.002, N = 5$, Start Steps:20	8.97	17.83

A.19.7 采样时间离散时间和连续时间离散扩散

在表 10 中，我们计算了基线和模型在单个 A6000 GPU 上以 1000 个时间步长对 50,000 张 VQ 编码图像进行采样所需的时间。另外，我们还计算了USD3和τ-LDR在采样时需要MCMC校正器步骤(=5)所需的时间。我们在下表中比较我们的结果。在采样时间方面，RDM 由于其重新路由采样方案而产生最多的采样时间。由于τ跳跃方案，SEDD和τ-LDR在采样时间上相似。 USD3 不及 SDDM，与其他方法相比，SDDM 需要特殊的 NN 架构和不同的代码库 (jaxlib)。但在考虑 MCMC 采样时，USD3 比 τ-LDR 中的 tau-leaping 带来更快的采样时间，同时允许对连续时间和离散时间模型进行采样。

表 10：不同模型的采样时间和 MCMC 采样步骤。

Model	**Sample Time**	**MCMC Sampling Step=5**
RDM	∼23 hours	-
D3PM	∼8 hours	-
τ-LDR	∼9 hours	∼2 days, 20 hours
SDDM	∼7 hours	-
SEDD	∼8 hours	-
USD3 (continuous version)	∼8 hours	∼2 days, 12 hours
USD3 (discrete version)	∼8 hours	∼2 days, 12 hours